CONTENT

Topics

Chapter 1. Vector Algebra.....3 - 25

1. Scalars and Vectors – Introduction .....(3)
2. Types of Vectors .....(3)
3. Scalar Multiplication .....(4)
4. Addition Composition.....(5)
5. Parallelogram Law of Vector Addition.....(5)
6. Linear Combination.....(6)
7. Coplanar Vectors.....(6)
8. Vectors in Geometry .....(7)
9. Dot and Cross Product .....(8)
10. Vector Product .....(10)
11. Scalar Triple Product.....(11)
12. Vector Triple Product .....(12)
13. Application to Geometry .....(14)
14. Miscellaneous Examples .....(14)
15. Unsolved Exercise .....(20)

Chapter 2. Three Dimensional geometry.....26 - 51

1. Distance Formula .....(26)
2. Section Formula.....(26)
3. Direction Cosine .....(27)
4. Some Useful Results On DC's and DR's .....(28)
5. Angle Between Two Vectors .....(28)
6. Vector Equations of a Line .....(29)
7. Vector Equations of a Line Passing through Two Points .....(30)
8. Perpendicular Distance of a Point from a Line .....(31)
9. Angle Between Two Lines .....(32)
10. Shortest Distance Between Two-Skew-Lines .....(33)
11. Equation of a Plane .....(34)
12. Distance of a Point from a Plane .....(36)
13. Angle Between a Line and a Plane .....(37)
14. Angle Between Two Planes .....(38)
15. Equation of a Plane Passing Through the Intersection of Two Planes .....(39)
16. Planes Bisecting The Angles Between Two Planes .....(40)
17. Miscellaneous Examples .....(42)
18. Some Important Definitions and Results.....(45)
19. Unsolved Exercise .....(47)

Chapter 3. Three Dimensional geometry.....52 - 70

1. Introduction.....(52)
2. Some Definitions.....(52)
3. Probability.....(52)
4. Types Of Events.....(54)
5. Addition Theorems On Probability.....(54)
6. Conditional Probability.....(56)
7. Multiplication Theorems On Probability.....(56)
8. The Law Of Total Probability.....(57)
9. Baye's Rule.....(58)
10. Random Variable And Its Probability Distribution.....(58)
11. Binomial Distribution.....(59)
12. Miscellaneous Examples.....(61)
13. Unsolved Exercise.....(65)

Answer Key .....71 - 72

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CHAPTER 1

VECTOR ALGEBRA

1. SCALAR AND VECTORS

Before we take up vectors in detail, let's look at some of the terms associated with vectors.
A **Scalar** is a quantity which has magnitude but no direction.
A **Vector** on the other hand, is a quantity which has a magnitude as well as a direction.
Displacement, velocity, momentum are examples of vectors while distnace, speed, volume and mass are examples of scalars.
Geometrically a vector is represented by a directed line segment as shown in the accompanying figure. The direction is indicated by the arrow and magnitude by the length of the segment *AB*. *A* is called the initial point and *B* the terminal point $\overrightarrow{AB}$.



The line ℓ of which the segment *AB* forms a point is called the line of support. The sense of $\overrightarrow{AB}$ is from *A* to *B* and that of $\overrightarrow{BA}$ is from *B* to *A*.
Vectors which are fully characterised by the magnitude and direction only are called free vectors and those that are fully characterised by the magnitude, direction and also line of support are called line (or bound) vectors or **localised vectors**.
Displacement, velocity are free vectors, while force and moment of a force about a point are line vectors.

2. TYPES OF VECTORS

A. **Null or zero vector :** A vector whose magnitude is zero is called null or zero vector. it is a vector having same initial and terminal points. Thus, a vector having zero magnitude and arbitrary constant is known as zero vector.

B. **Unit vector :** A vector divided by its magnitude is a unit vector. Unit vector of $\vec{A}$ is $\hat{A}$ (read as *A* cap)

$$|\hat{A}| = 1$$

$$\vec{A} = |\vec{A}| \cdot \hat{A} \qquad \therefore \quad \hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

C. **Equal vectors :** Two vectors are said to be equal vectors if
(i) They have equal magnitude (ii) Same direction
If $\vec{A} = \vec{B}$, then *A* = *B* (not necessarity)

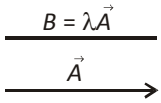
D. **Parallel vectors :** Two vectors $\vec{A}$ and $\vec{B}$ are said to be parallel if either both have same line of action/support.

OR

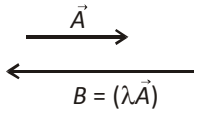
There exists a scalar λ such that

$\vec{A} = \lambda\vec{B}$

if $\lambda > 0$



if $\lambda < 0$



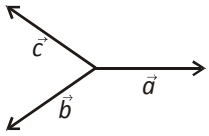
E. **Like vectors :** Vectors are said to like when they have same sense of direction i.e., angle between them is zero.

F. **Unlike vectors :** Vectors are said to be unlike when they have opposite sense of direction i.e., angle between them is π.

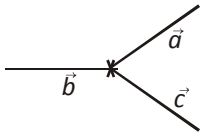
G. **Negative vector :** A vector is said to be negative of a given vector if its magnitude is the same as that of the given vector but direction is reversed. The negative vector of $\vec{a}$ is denoted by $-\vec{a}$.



H. **Co-initial vectors :** Co-initial vectors have the same initial points.



I. **Co-Terminus Vectors :** Co-terminus vectors have the same terminal point.



J. **Coplanar Vectors :** Vectors are said to be coplanar if they are in the same plane or parallel support of line.

Note : Any two vectors are always coplanar.

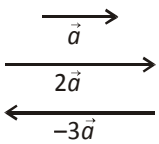
3. SCALAR MULTIPLICATION

It is simply the multiplication of vectors by scalar. Let $\vec{a}$ be any given vector and *m* be any given scalar. The symbol $m\vec{a}$ called the product of the vector $\vec{a}$, by the scalar *m*, is a vector such that

(i) The length of $m\vec{a}$ is given by $|m\vec{a}| = |m||\vec{a}|$, that is, the length of the vector is *m* times, or –*m* times that of $\vec{a}$, according as *m* is positive or negative. Thus $\vec{b} = m\vec{a} \Rightarrow |m| = \frac{|\vec{b}|}{|\vec{a}|}$.

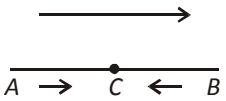
(ii) The support of $m\vec{a}$ is the same or parallel, to that of $\vec{a}$.

(iii) The sense of $m\vec{a}$ is the same or opposite to that of $\vec{a}$, according as *m* is positive or negative.

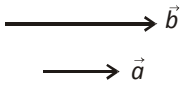


As a geometric example, let *C* be the mid-point of a line *AB*. Then we have

$$\overrightarrow{AB} = 2\overrightarrow{AB}, \overrightarrow{BC} = -\frac{1}{2}\overrightarrow{AB}$$



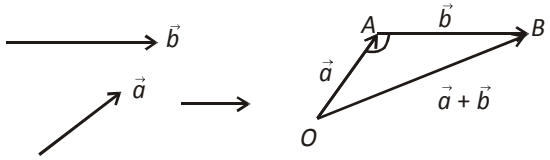
Two parallel vectors may also be throught of as collinear.



Note that $k_1(k_2\vec{a}) = k_2(k_1\vec{a}) = k_1k_2\vec{a}\ \forall k_1k_2 \in R$

4. ADDITION COMPOSITION

Let $\vec{a}, \vec{b}$ two given vecotrs. Take a point *O*.



Let $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$.

So that the terminal point of the vector $\vec{a}$ is the initial point of vector $\vec{b}$.

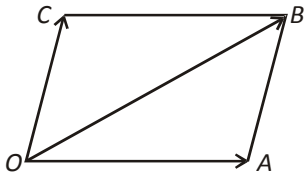
The vector $\overrightarrow{OB}$ is called the 'sum' of vectors $\vec{a}$ and $\vec{b}$ and is written as

$$\overrightarrow{OB} = \overrightarrow{OA} + \vec{a} + \vec{b}$$

We have $\vec{a} + 0 = \vec{a}$, for every vector $\vec{a}; \vec{0}$, being the zero vector.

5. PARALLELOGRAM LAW OF ADDITION OF VECTORS

Consider a parallelogram *OABC*. The sum of the vectors



$$\overrightarrow{OC} + \overrightarrow{CB} = \overrightarrow{OA} + \overrightarrow{OC} = \overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$$

The addition composition of vectors satisfies the following properties :

(i) Addition of vectors is commutative

$\vec{a} + \vec{b} = \vec{b} + \vec{a}$ for any pair of vectors $\vec{a}, \vec{b}$.

(ii) Addition of vectors is associative

$$\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$$

where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors

(a) $\vec{a} + 0 = \vec{a}$ (b) $\vec{a} + (-\vec{a}) = \vec{0}$

(c) $(k_1 + k_2)\vec{a} = k_1\vec{a} + k_2\vec{a}$ (d) $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$

***Illustration 1*:** If *ABCD* is a rhombus whose diagonals cut at the origin *O*, then $\vec{OA}+\vec{OB}+\vec{OC}+\vec{OD}$ equals

(a) $\vec{AB}+\vec{AC}$ (b) $\vec{0}$ (c) $2(\vec{AB}+\vec{BC})$ (d) $\vec{AC}+\vec{BD}$

Solution: **(b)** Since the diagonal of a rhombus bisect each other, therefore $\vec{OA}=-\vec{OC}$ and $\vec{OB}=-\vec{OD}$ and so $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}=\vec{0}$

***Illustration 2*:** If *A, B, C, D* be any four points and *E* and *F* be the middle points of *AC* and *BD* respectively, then $\overrightarrow{AB}+\overrightarrow{CB}+\overrightarrow{CD}+\overrightarrow{AD}$ is equal to

(a) $3\overrightarrow{EF}$ (b) $4\overrightarrow{EF}$ (c) $4\overrightarrow{FE}$ (d) $3\overrightarrow{FE}$

Solution: **(b)** Since *F* is the middle point of *BD*, therefore

$$\overrightarrow{AB}+\overrightarrow{AD}=2\overrightarrow{AF} \quad \text{....(i)}$$

Similarly, $\overrightarrow{CB}+\overrightarrow{CD}=2\overrightarrow{CF}$ ....(ii)

On adding eqs. (i) and (ii), we get

$$\overrightarrow{AB}+\overrightarrow{AD}+\overrightarrow{CB}+\overrightarrow{CD}=2(\overrightarrow{AB}+\overrightarrow{CF})=-2(\overrightarrow{FA}+\overrightarrow{FC})=-2(2\overrightarrow{AB}+\overrightarrow{AD}+\overrightarrow{CB}+\overrightarrow{CD}=2(\overrightarrow{AB}+\overrightarrow{CF})=-2(2\overrightarrow{FE})$$

[$\because$ *E* is the mid point of *AC*]

$$=4\overrightarrow{EF}$$

6. LINEAR COMBINATION

A vector $\vec{r}$ is said to be a linear combination of the vectors $\vec{a},\vec{b},\vec{c}$, if there exist scalars *x, y, z* such that $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$

Thus the vectors $2\vec{a}-\vec{b}+3\vec{c},\vec{a}-2\vec{b}+4\vec{c}$ are linear combinations of the vectors $\vec{a},\vec{b}$ and $\vec{c}$.

A linear combination of vectors, involves the two linear compositions of the addition of vectors and the multiplication of vectors by scalars.

If two vectors have same or parallel supports, then each is a linear combination of the other. Thus if $\vec{a}$ and $\vec{b}$ are two parallel vectors, then there exists a non-zero scalar *x* such that $\vec{b}=x\vec{a}$ or $\vec{a}=\frac{1}{x}\vec{b}$.

The vectors $\vec{a},\vec{b},\vec{c}$ are called linearly independent or linearly dependent according as

$$x\vec{a}+y\vec{b}+z\vec{c}=\vec{0}$$

implies *x* = *y* = *z* = 0 or there exist at least one pair of *x, y, z* in which at least one of *x, y, z* is not zero.

Expression as Linear Combination

7. COPLANAR VECTORS

If $\vec{a}$ and $\vec{b}$ be two given non-collinear vectors, then every $\vec{r}$, coplanar with $\vec{a}$ and $\vec{b}$ can be represented as a linear combination.

$$\vec{r}=x\vec{a}+y\vec{b}$$

x, y being some scalars. Moreover this representation is unique.

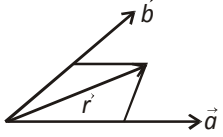


***Illustration 3*:** Let $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that

$\vec{r}_1 = \vec{a} - \vec{b} + \vec{c}, \vec{r}_2 = \vec{b} + \vec{c} - \vec{a}, \vec{r}_3 = \vec{c} + \vec{a} - \vec{b}$, $\vec{r} = 2\vec{a} - 2\vec{b} + 4\vec{c}$. If $\vec{r} = \lambda_1\vec{r}_1 + \lambda_2\vec{r}_2 + \lambda_3\vec{r}_3$, then

(a) $\lambda_1 = 7$ (b) $\lambda_1 + \lambda_3 = 3$ (c) $\lambda_1 + \lambda_2 + \lambda_3 = 4$ (d) $\lambda_3 + \lambda_2 = 2$

Solution: **(b, c)**

We have, $\vec{r} = \lambda_1\vec{r}_1 + \lambda_2\vec{r}_2 + \lambda_3\vec{r}_3$

$\Rightarrow \quad 2\vec{a} - 2\vec{b} + 4\vec{c} = (\lambda_1 - \lambda_2 + \lambda_3)\vec{a} + (-\lambda_1 + \lambda_2 - \lambda_3)\vec{b} + (\lambda_1 + \lambda_2 + \lambda_3)\vec{c}$

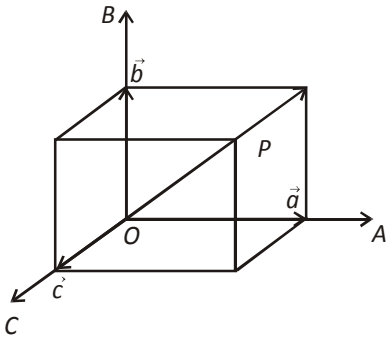
$\Rightarrow \quad \lambda_1 - \lambda_2 + \lambda_3 = 2, -\lambda_1 + \lambda_2 - \lambda_3 = -2, \lambda_1 + \lambda_2 + \lambda_3 = 4$ [$\because \vec{a}, \vec{b}, \vec{c}$ are non-coplanar]

$\lambda_1 + \lambda_2 = 3$ and $\lambda_1 + \lambda_2 + \lambda_3 = 4$

Arbitrary system of vectors : If $\vec{a}, \vec{b}, \vec{c}$ be three given non-coplanar vectors, then any $\vec{r}$ can be represented as a linear combination.

$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$

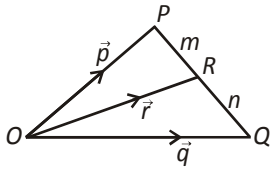
x, y, z being some scalar. Moreover, this represnetation is unique.



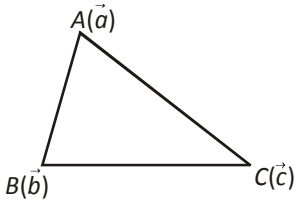
8. VECTORS IN GEOMETRY

Section Formula

If *R* divides *PQ* in the ratio *m* : *n*, then $\vec{r} = \dfrac{m\vec{q} + n\vec{p}}{m+n}$



If *R* be the mid-point of *PQ* then $\vec{r} = \dfrac{\vec{p} + \vec{q}}{2}$.



Again if *G* is the centroid of the triangle *ABC*, then $\vec{G} = \dfrac{\vec{a} + \vec{b} + \vec{c}}{3}$

***Illustration 4*:** Let *O* be the circumcentre, *G* be the centroid and *O'* be the orthocentre of a ΔABC.

Three vectors are taken through *O*, $\vec{a}=\overrightarrow{OA}, \vec{b}=\overrightarrow{OB}$ and $\vec{c}=\overrightarrow{OC}$, then $\vec{a}+\vec{b}+\vec{c}$ is

(a) $\overrightarrow{OG}$ (b) $2\overrightarrow{OG}$ (c) $\overrightarrow{OO'}$ (d) None of these

Solution: **(c)** Since centroid divides the line segment joining orthocentre and circumcentre in the ratio of 2 : 1.

$\therefore \quad \overrightarrow{OB}+\overrightarrow{OC}=2\overrightarrow{OD}$

$\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}=\overrightarrow{OA}+2\overrightarrow{OD}$

$\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}=\overrightarrow{OA}+2\overrightarrow{OD}=3\overrightarrow{OG}=\overrightarrow{OO'}$

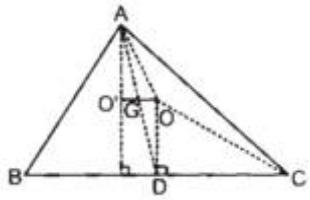


***Illustration 5*:** Let $\overrightarrow{OA}=\hat{i}+3\hat{j}-2\hat{k}$ and $\overrightarrow{OB}=3\hat{i}+\hat{j}-2\hat{k}$. The vector $\overrightarrow{OC}$ bisecting the angle *AOB* and *C* being a point on the line *AB*, is

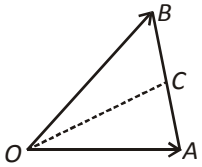
(a) $4(\hat{i}+\hat{j}-\hat{k})$ (b) $2(\hat{i}+\hat{j}-\hat{k})$ (c) $\hat{i}+\hat{j}-\hat{k}$ (d) None of these

Solution: **(b)** Taking *O* as the origin, the position vectors of *A* and *B* are $\vec{a}=\hat{i}+3\hat{j}-2\hat{k}$ and $\vec{b}=3\hat{i}+\hat{j}-2\hat{k}$ respectively

We have $|\vec{a}|=|\vec{b}|=\sqrt{14}$

So, the bisector *OC* of ΔAOB meets *AB* at its mid point *C*

$$\overrightarrow{OC}=\frac{1}{2}(\overrightarrow{OA}+\overrightarrow{OB})=2(\hat{i}+\hat{j}-2\hat{k})$$



9. DOT AND CROSS PRODUCTS

The scalar product of two vectors $\vec{a}$ and $\vec{b}$ is the scalar

$$\vec{a}.\vec{b}=|\vec{a}||\vec{b}|\cos\theta$$

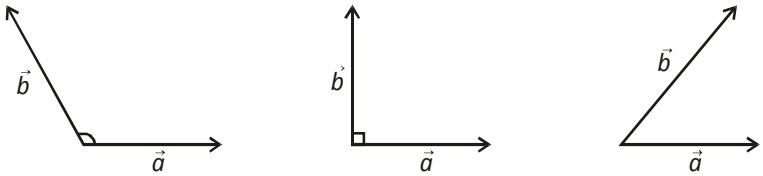
where θ is the angle between the vectors $\vec{a}$ and $\vec{b}$

The scalar product of the vectors $\vec{a}$ and $\vec{b}$ is denoted by the symbol $\vec{a}\cdot\vec{b}$ and due to this reason it is also called the dot product of the two vectors.

Thus, we have by definition

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$$

θ being angle between the vectors $\vec{a}$ and $\vec{b}$.



θ is acute $\Rightarrow \cos\theta > 0 \Rightarrow \vec{a}\cdot\vec{b}>0$

θ is right angle $\Rightarrow \cos\theta = 0 \Rightarrow \vec{a}\cdot\vec{b}=0$

θ is obtuse $\Rightarrow \cos\theta < 0 \Rightarrow \vec{a}\cdot\vec{b}<0$

If $\vec{a}$ be any vector then the scalar product $\vec{a}\cdot\vec{a}$ of $\vec{a}$ with itself is given by $\vec{a}\cdot\vec{a}=|\vec{a}|^2$

Angle between two vectors in terms of scalar product is given by $\theta=\cos^{-1}\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

(i) Scalar product is commutative $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{a}$

(ii) Scalar multiplication of vectors is distributive over the addition of numbers

$\vec{a}\cdot(\vec{b}+\vec{c})=\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}$

(iii) $(m\vec{a})\cdot(n\vec{b})=mn(\vec{a}\vec{b})$

(iv) If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k},\vec{b}=b_1\hat{i},b_2\hat{j}+b_3\hat{k}$, then $\vec{a}\cdot\vec{b}=a_1b_1+a_2b_2+a_3b_3$

(v) $\hat{i}.\hat{i}=\hat{j}.\hat{j}=\hat{k}.\hat{k}=1$

(vi) $\hat{i}.\hat{j}=\hat{j}.\hat{k}=\hat{k}.\hat{i}=0$

***Illustration 6*:** If $\vec{a},\vec{b},\vec{c}$ are three mutually perpendicular vectors each of magnitude unity, then $|\vec{a}+\vec{b}+\vec{c}|$ is equal to

(a) 3 (b) 1 (c) $\sqrt{3}$ (d) None of these

Solution: **(c)** We have $|\vec{a}|=|\vec{b}|=|\vec{c}|=1$ and $\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=0$

$\therefore\quad |\vec{a}+\vec{b}+\vec{c}|^2=(\vec{a}+\vec{b}+\vec{c}).(\vec{a}+\vec{b}+\vec{c})$

$=|\vec{a}|^2+|b|^2+|c|^2+2\vec{a}.\vec{b}+2\vec{b}.\vec{c}+2\vec{c}.\vec{a}=3$

$\Rightarrow\quad |\vec{a}+\vec{b}+\vec{c}|=\sqrt{3}$

***Illustration 7*:** The projection of the vector $\hat{i}-2\hat{j}+\hat{k}$ on the vector $4\hat{i}-4\hat{j}+7\hat{k}$ is

(a) $\dfrac{5\sqrt{6}}{10}$ (b) $\dfrac{19}{9}$ (c) $\dfrac{9}{19}$ (d) $\dfrac{\sqrt{6}}{19}$

Solution: **(b)**Let $\vec{a}=\hat{i}-2\hat{j}+\hat{k},\vec{b}=4\hat{i}-4\hat{j}+7\hat{k}$. Then, the projection of $\vec{a}$ on $\vec{b}$ is equal to

$$\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{4+8+7}{\sqrt{16+16+49}}=\frac{19}{9}$$

***Illustration 8*:** If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then $\left|\dfrac{\vec{a}-\vec{b}}{2}\right|$ is

(a) $\sin\dfrac{\theta}{2}$ (b) $\sin\theta$ (c) $2\sin\theta$ (d) $\sin 2\theta$

Solution: **(a)** We have, $|\vec{a}-\vec{b}|^2=|\vec{a}|^2+|\vec{b}|^2-2\vec{a}.\vec{b}$

$\Rightarrow\quad |\vec{a}-\vec{b}|^2=1+1-2|\vec{a}||\vec{b}|\sin\theta$

$\Rightarrow\quad |\vec{a}-\vec{b}|^2=2-2\cos\theta$

$\Rightarrow\quad |\vec{a}-\vec{b}|^2=4\sin^2\dfrac{\theta}{2}$

$\Rightarrow\quad \left|\dfrac{\vec{a}-\vec{b}}{2}\right|^2=\sin^2\dfrac{\theta}{2}$

$\Rightarrow\quad \left|\dfrac{\vec{a}-\vec{b}}{2}\right|=\sin\dfrac{\theta}{2}$

10. VECTOR PRODUCTS

The vector product denoted by $\vec{a}\times\vec{b}$ of two vectors, $\vec{a},\vec{b}$ taken in order, is the vector $\vec{c}$ where

$\vec{c}=|\vec{a}||\vec{b}|\sin\theta\hat{n}$

θ being the angle between vectors, $0^o \le \theta \le 180^o$.

The sense of $\vec{c}$ is such that the triad $\vec{a},\vec{b},\vec{c}$ forms a right-handed system.

(i) The vector product is not commutative. Indeed $\vec{a}\times\vec{b}=-\vec{b}\times\vec{a}$

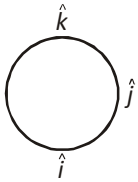
(ii) $-\vec{a}\times\vec{b}=-(\vec{a}\times\vec{b})$

$(-\vec{a})\times(-\vec{b})=\vec{a}\times\vec{b}$

(iii) $\hat{i}\times\hat{i}=\hat{j}\times\hat{j}=\hat{k}\times\hat{k}=0$

$\hat{i}\times\hat{j}=k,\ \hat{j}\times\hat{k}=\hat{i},\ \hat{k}\times\hat{i}=\hat{j}$

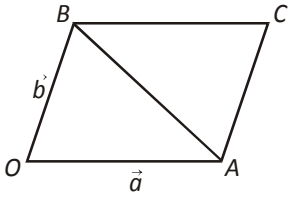
$\hat{j}\times\hat{i}=-\hat{k},\ \hat{k}\times\hat{j}=-\hat{i},\ \hat{i}\times\hat{k}=-\hat{j}$



In determinant form if $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k},\ \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

Then $\vec{a}\times\vec{b}$ is given by

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}=(a_2b_3-a_3b_2)\hat{i}-(a_1b_3-a_3b_1)\hat{j}+(a_2b_2-a_2b_1)\hat{k}$$



Area of triangle *OAB* = $\frac{1}{2}|\vec{a}\times\vec{b}|$

Area of parallelogram *OABC* = $|\vec{a}\times\vec{b}|$

where $\vec{a}$ and $\vec{b}$ are vectors corresponding to two adjacent sides.

***Illustration 9*:** If $\vec{a},\vec{b},\vec{c}$ are non-collinear vectors such that $\vec{a}+\vec{b}$ is parallel to $\vec{c}$ and $\vec{c}+\vec{a}$ is parallel to $\vec{b}$, then

(a) $\vec{a}+\vec{b}=\vec{c}$ (b) $\vec{a},\vec{b},\vec{c}$ taken in order form the sides of a triangle

(c) $\vec{b}+\vec{c}=\vec{a}$ (d) None of these

Solution: **(b)** It is given $(\vec{a}+\vec{b})||\vec{c}$ and $(\vec{c}+\vec{a})||\vec{b}$

$\Rightarrow$ $(\vec{a}+\vec{b})\times\vec{c}=0$ and $(\vec{c}+\vec{a})\times\vec{b}=0$

$\Rightarrow$ $\vec{a}\times\vec{c}+\vec{b}\times\vec{c}=0$ and $\vec{c}\times\vec{b}+\vec{a}\times\vec{b}=0$

$\Rightarrow$ $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

Hence, $\vec{a},\vec{b},\vec{c}$ are the sides of a triangle

***Illustration 10*:** If $\vec{a}\times\vec{b}=\vec{c}\times d$ and $\vec{a}\times\vec{c}=\vec{b}\times\vec{d}$, then

(a) $(\vec{a}-\vec{d})=\lambda(\vec{b}-\vec{c})$ (b) $\vec{a}+\vec{d}=\lambda(\vec{b}+\vec{c})$

(c) $(\vec{a}-\vec{d})=\lambda(\vec{c}+\vec{d})$ (d) None of these

Solution: **(a)** We have, $(\vec{a}-\vec{b})\times(\vec{b}-\vec{c})=\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{d}\times\vec{b}+\vec{d}\times\vec{c}$

$=\vec{c}\times\vec{d}-\vec{b}\times\vec{d}-\vec{d}\times\vec{b}+\vec{d}\times\vec{c}=0 \quad [\because \vec{a}\times\vec{b}=\vec{c}\times\vec{d}, \vec{a}\times\vec{c}=\vec{b}\times\vec{d}]$

$\Rightarrow (\vec{a}-\vec{d})||(\vec{b}-\vec{c})$

$\Rightarrow \vec{a}-\vec{d}=\lambda(\vec{b}-\vec{c})$

***Illustration 11*:** If $\vec{u}=\vec{a}-\vec{b}, \vec{v}=\vec{a}+\vec{b}$ and $|\vec{a}|=|\vec{b}|=2$, then $|\vec{u}\times\vec{v}|$ is

(a) $2\sqrt{16-(\vec{a}.\vec{b})^2}$ (b) $2\sqrt{4-(\vec{a}.\vec{b})^2}$ (c) $2\sqrt{16-(\vec{a}.\vec{b})^2}$ (d) $2\sqrt{4-(\vec{a}.\vec{b})^2}$

Solution: **(a)** We have, $\vec{u}=\vec{a}-\vec{b}, \vec{v}=\vec{a}+\vec{b}$

$\Rightarrow \vec{u}\times\vec{v}=(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})=2(\vec{a}\times\vec{b})$

$\Rightarrow |\vec{u}\times\vec{v}|=2|\vec{a}\times\vec{b}|=2\sqrt{|\vec{a}\times\vec{b}|^2}$

$=2\sqrt{|\vec{a}|^2|\vec{b}|^2-(\vec{a}.\vec{b})^2}$

$=2\sqrt{16-(\vec{a}.\vec{b})^2}$

11. SCALAR TRIPLE PRODUCTS

Let $\vec{a},\vec{b},\vec{c}$ be any three vectors. Consider the expression $\vec{a}\times\vec{b}\cdot\vec{c}$, which is scalar product of the vectors $\vec{a}\times\vec{b}$ and $\vec{c}$.

The expression $\vec{a}\times\vec{b}\cdot\vec{c}$ is called a scalar triple or mixed product of three vectors $\vec{a},\vec{b},\vec{c}$ taken in this order.

Based upon the permutation of three vectors in six different ways and each manner of writing down the three vectors, i.e. the position of dot and cross, we can have twelve scalar triple product.

$$\vec{a}\times\vec{b}.\vec{c} \quad \vec{b}\times\vec{c}.\vec{a} \quad \vec{c}\times\vec{a}.\vec{b} \quad \vec{a}.\vec{b}\times\vec{c} \quad \vec{b}.\vec{c}\times\vec{a} \quad \vec{c}.\vec{a}\times\vec{b}$$
$$\vec{a}\times\vec{c}.\vec{b} \quad \vec{b}\times\vec{a}.\vec{c} \quad \vec{c}\times\vec{b}.\vec{a} \quad \vec{a}.\vec{c}\times\vec{b} \quad \vec{b}.\vec{a}\times\vec{c} \quad \vec{c}.\vec{b}\times\vec{a}$$

The two important properties are

(i) A cyclic permutation of three vectors doesn't cahnge the value of the scalar triple product and an anti-cyclic permutation changes the value in sign but not in magnitude.

(ii) The positions of dot and cross can be interchanged without any change in the value of the scalar triple product.

In view of the above prokesties, we have a notation $\vec{a}\times\vec{b}.\vec{c}=[\vec{a}\,\vec{b}\,\vec{c}]$

This notation takes note of the cyclic order of the vectors and disregards the position of dot and cross which is not at all important

$[\vec{a}\,\vec{b}\,\vec{c}]=[\vec{b}\,\vec{c}\,\vec{a}]=[\vec{c}\,\vec{a}\,\vec{b}]$

$=-[\vec{a}\,\vec{c}\,\vec{b}]=-[\vec{b}\,a\,\vec{c}]=-[\vec{c}\,\vec{b}\,\vec{a}]$

A very remarkable fact is that the scalar triple product $[\vec{a}\,\vec{b}\,\vec{c}]$ is zero, if and only if, the three vectors $\vec{a},\vec{b},\vec{c}$ are coplanar, Also, the scalar triple product is zero if any two of the three vectors are parallel.

If $\hat{i},\hat{j},\hat{k}$ form orthonormal right-handed triad, then $[\hat{i},\hat{j},\hat{k}]=1$.

Scalar Tirple Product in Terms of Orthonormal Base

Consider

$$\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$$

$$\vec{b}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$$

$$\vec{c}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$$

We have $\vec{a}\times\vec{b}.\vec{c}=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

Geometric Interpretation : The volume of a tetrahedron *ABCD* is $\frac{1}{6}\left|\overrightarrow{AB}\times\overrightarrow{AC}.\overrightarrow{AD}\right|$.

Also the volume of the parallelepiped with $\overrightarrow{AB},\overrightarrow{AC}$ and $\overrightarrow{AD}$ as its co-terminous edges is nothing but the scalar triple product $\left|\overrightarrow{AB}\times\overrightarrow{AC}.\overrightarrow{AD}\right|$.

***Illustration 12*:** Which of the following expressions are meaningful?

(a) $\vec{u}.(\vec{v}\times\vec{w})$ (b) $(\vec{u}.\vec{v}).\vec{w}$ (c) $(\vec{u}.\vec{v})\vec{w}$ (d) $\vec{u}\times(\vec{v}.\vec{w})$

Solution: **(a, c)** Since $\vec{v}\times\vec{w}$ is a vector, therefore $\vec{u}.(\vec{v}\times\vec{w})$ is a scalar quantity.

$(\vec{u}.\vec{v})$ is a scalar, therefore, $(\vec{u}.\vec{v}).\vec{w}$ is not meaningful.

But $(\vec{u}.\vec{v})\vec{w}$ is scalar multiple of $\vec{w}$. So, it is meaningful.

Since, $\vec{v}.\vec{w}$ is scalar, so $\vec{u}\times(\vec{v}.\vec{w})$ is not meaningful.

***Illustration 13*:** If $\vec{r}.\vec{a}=\vec{r}.\vec{b}=\vec{r}.\vec{c}=0$ for some non-zero vector $\vec{r}$, then the value of $[\vec{a}\ \vec{b}\ \vec{c}]$ is

(a) 2 (b) 3 (c) 0 (d) None of these

Solution: **(c)** We have, $\vec{r}.\vec{a}=0 \Rightarrow \vec{r}\perp\vec{a}$

$\Rightarrow \vec{r}.\vec{b}=0$

$\Rightarrow \vec{r}\perp\vec{b}\quad \vec{r}.\vec{c}=0$

$\Rightarrow \vec{r}\perp\vec{c}$

$\vec{a},\vec{b},\vec{c}$ are coplanar $\Rightarrow[\vec{a}\ \vec{b}\ \vec{c}]=0$

12. VECTOR TRIPLE PRODUCT

Let $\vec{a},\vec{b},\vec{c}$ be any three vectors, then an expression of the form $\vec{a}(\vec{a}\times\vec{b})\times\vec{c}$, which is the vector product of $\vec{a}\times\vec{b}$ with $\vec{c}$ is known as vector triple product.

We have $\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}$

Evidently $\vec{a}\times(\vec{b}\times\vec{c})$ is linear combination of $\vec{b}$ and $\vec{c}$. Note that the vector $\vec{a}\times(\vec{b}\times\vec{c})$, being perpendicular $\vec{b}\times\vec{c}$, and as such it's a linear combination of $\vec{b}\times\vec{c}$.

As an aid to memory the formula for $\vec{a}\times(\vec{b}\times\vec{c})$ can be thought of as

(remote × adjacent) × outer = (outer.remote)adjacent – (outer . adjacent)remote

We have, two other expressions that extend the idea of product of vectors to four vectors.

Scalar Product of Four Vectors

$$(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d})=(\vec{a}.\vec{c})(\vec{b}.\vec{d})-(\vec{a}.\vec{d})(\vec{b}.\vec{c})$$

$$=\begin{vmatrix}\vec{a}.\vec{c} & \vec{a}.\vec{d}\\ \vec{b}.\vec{c} & \vec{b}.\vec{d}\end{vmatrix}$$

Vector Product of Four Vectors

$$(\vec{a}\times\vec{b})=[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}-[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}$$

$$=[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}-[\vec{b}\ \vec{c}\ \vec{d}]\vec{a}$$

Solving Vector equation

Sometimes one is given a vector equation in the unknown, say $\vec{r}$ and it is required to express $\vec{r}$ in terms of known vectors. There is no cut and derived formula to solve all such problems. Rather, the approach should be decided on the merits of the problem at hand. Often we need to take cross or dot product of the vector equation with some vector that will help isolate the unknown. Study the examples that follow carefully to learn the approach.

***Illustration 14*:** Let $\vec{a}=2\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ and a unit vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$, then $\vec{c}$ is equal to

(a) $\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$ (b) $\frac{1}{\sqrt{3}}(-\hat{i}-\hat{j}-\hat{k})$ (c) $\frac{1}{\sqrt{5}}(\hat{i}-2\hat{j})$ (d) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

Solution: **(a)** Required unit vector $=\frac{\vec{a}\times(\vec{a}\times\vec{b})}{|\vec{a}\times(\vec{a}\times\vec{b})|}$

Now, $\vec{a}\times(\vec{a}\times\vec{b})=(\vec{a}.\vec{b})\vec{a}-(\vec{a}.\vec{b})\vec{b}$

$=3(2\hat{i}+\hat{j}+\hat{k})-6(\hat{i}+2\hat{j}-\hat{k})\ =-9\hat{i}+9\hat{k}$

So, required unit vector $=\pm\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$

***Illustration 15*:** If $\hat{a},\hat{b},\hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a}\times(\hat{b}\times\hat{c})=\frac{1}{2}\hat{b}$, then the angle between $\hat{b}$ and $\hat{c}$ is

(a) 30° (b) 45° (c) 60° (d) 90°

Solution: **(c)** We have, $\hat{a}\times(\hat{b}\times\hat{c})=\frac{1}{2}\hat{b}$

$\Rightarrow\quad(\hat{a}.\hat{c})\hat{b}-(\hat{a}.\hat{b})\hat{c}=\frac{1}{2}\hat{b}$

$$\Rightarrow \left((\hat{a}.\hat{c})-\frac{1}{2}\right)\hat{b}-(\hat{a}.\hat{b})\hat{c}=0$$

$$\Rightarrow \hat{a}.\hat{c}-\frac{1}{2}=0 \text{ and } \hat{a}.\hat{b}=0$$

$$\Rightarrow \cos 60^0=\frac{1}{2}, \text{ where } 60^0 \text{ is the angle between } \hat{a} \text{ and } \hat{c}$$

$$\Rightarrow 60^0=\frac{\pi}{3}$$

***Illustration 16*:** Let $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=2\hat{i}-\hat{k}$, the point of intersection of the line $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$ and $\vec{r}\times b=\vec{a}\times\vec{b}$ is

(a) $\hat{i}+\hat{j}+\hat{k}$ (b) $3\hat{i}-\hat{j}+\hat{k}$ (c) $3\hat{i}+\hat{j}-\hat{k}$ (d) $\hat{i}-\hat{j}-\hat{k}$

Solution: **(c)** We have, $\vec{r}\times\vec{a}=\vec{b}\times\vec{a} \quad \Rightarrow(\vec{r}-\vec{b})\times\vec{a}=0$

$$\Rightarrow \vec{r}-\vec{b}\,||\,\vec{a} \quad \Rightarrow \quad \vec{r}-\vec{b}=\lambda\vec{a}$$

$$\Rightarrow \vec{r}=\vec{b}+\lambda\vec{a}$$

Similarly, the equation of the line $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ can be written as $\vec{r}=\vec{a}\times\mu\vec{b}$

For the point of intersection of the above two lines, we have $\vec{a}+\mu\vec{b}=\vec{b}+\lambda\vec{a} \Rightarrow\lambda=\mu=1$

Hence, the position vector fo the point of intersection of the given lines are $\vec{r}=\vec{a}+\vec{b}=3\hat{i}+\hat{j}-\hat{k}$

13. APPLICATION TO GEOMETRY

Many geometrical problems can easily be done using the language of vectors. Unlike the 2-dimensional or 3-dimensional co-ordinate geometry that uses two and three real numbers to specify a point, the vector algebra has the advantage that position of a point can be expressed in terms of a single vector, which has in it implicit the desired number of components. So we get the full-coordinate in one stroke rather than *x, y* or *z*-coordinate all one by one.

Again in a problem like this, one has the liberty to choose one's reference frame to suit one's purpose and this can result in much simplification.

MISCELLANEOUS EXAMPLES

Example 1: For any four points *P, Q, R, S* $\left|\overrightarrow{PQ}\times\overrightarrow{RS}-\overrightarrow{QR}\times\overrightarrow{PS}+\overrightarrow{RP}\times\overrightarrow{QS}\right|$ is equal to 4 times the area of the triangle

(a) *PQR* (b) *QRS*

(c) *PRS* (d) *PQS*

Solution: **(b)** Let $\vec{a},\vec{b},\vec{c},\vec{d}$ be the position vectors of points *P, Q, R* and *S* respectively. Then,

$$\overrightarrow{PQ}\times\overrightarrow{RS}=(\vec{b}-\vec{a})\times(\vec{d}-\vec{c})=\vec{b}\times\vec{d}-\vec{b}\times\vec{c}+\vec{d}\times\vec{a}-\vec{c}\times\vec{a}$$

$$\overrightarrow{QR}\times\overrightarrow{PS}=(\vec{c}-\vec{b})\times(\vec{d}-\vec{a})=\vec{c}\times\vec{d}-\vec{c}\times\vec{a}-\vec{b}\times d-\vec{a}\times\vec{b}$$

$|\overrightarrow{RP}\times\overrightarrow{QS}|=(\vec{a}-\vec{c})\times(\vec{d}-\vec{b}) \;=\vec{a}\times\vec{d}-\vec{a}\times\vec{b}-\vec{c}\times\vec{d}-\vec{b}\times\vec{c}$

$\therefore\ \overrightarrow{PQ}\times\overrightarrow{RS}-\overrightarrow{QR}\times\overrightarrow{PS}+\overrightarrow{RP}\times\overrightarrow{QS}$

$=\vec{b}\times\vec{d}-\vec{b}\times\vec{c}+\vec{d}\times\vec{a}-\vec{c}\times\vec{a}-\vec{c}\times d+\vec{c}\times\vec{a}+\vec{b}\times\vec{d}+\vec{a}\times\vec{b}+\vec{a}\times d-\vec{a}\times\vec{b}-\vec{c}\times\vec{d}-\vec{b}\times\vec{c}$

$=2(\vec{b}\times\vec{d})-2(\vec{b}\times\vec{c})-2(\vec{c}\times\vec{d}) \;=2\{\vec{b}\times\vec{d}+\vec{c}\times\vec{b}+\vec{d}\times\vec{c}\}$

$=\ 2(\overrightarrow{QS}\times\overrightarrow{QR})=4$ Area of the ΔQSR

Example 2: Let C_1, C_2, C_3 be the centroids of the triangular faces *OBC*, *OCA*, *OAB* of a tetrahedron *OABC*. If V_1 denote the volume of the tetrahedron *OABC* and V_2 that of the parallelopiped with OG_1, OG_2, OG_3 as three concurrent edges, then

(a) $4V_1 = 9V_2$ (b $9V_1 = 4V_2$

(c) $3V_1 = 2V_2$ (d) $3V_2 = 2V_1$

Solution: **(a)** Taking *O* as the origin, let the position vectors of *A, B* and *C* be $\vec{a}, \vec{b}$ and $\vec{c}$ respectively.

Then the position vectors G_1, G_2 and G_3 are $\frac{\vec{b}+\vec{c}}{3}, \frac{\vec{c}+\vec{a}}{3}$ and $\frac{\vec{a}+\vec{b}}{3}$ respectively

$\therefore \qquad V_1=\frac{1}{6}[\vec{a}\ \ \vec{b}\ \ \vec{c}]$ and $V_2=[\overrightarrow{OG_1}\ \overrightarrow{OG_2}\ \overrightarrow{OG_3}]$

Now, $\quad V_2=[\overrightarrow{OG_1}\ \overrightarrow{OG_2}\ \overrightarrow{OG_3}]$

$\Rightarrow \qquad V_2=\left[\frac{\vec{b}+\vec{c}}{3}\ \frac{\vec{c}+\vec{a}}{3}\ \frac{\vec{a}+\vec{b}}{3}\right] \qquad \Rightarrow\ V_2=\frac{1}{27}[\vec{b}+\vec{c}\ \ \vec{c}+\vec{a}\ \ \vec{a}+\vec{b}]=\frac{2}{27}[\vec{a}\ \ \vec{b}\ \ \vec{c}]$

$\Rightarrow \qquad V_2=\frac{2}{27}\times 6V_1 \qquad \Rightarrow 9V_2 = 4V_1$

Example 3: Let $\vec{a}=\hat{i}-\hat{k}, \vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}$ and $\vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$. Then, $[\vec{a}\ \ \vec{b}\ \ \vec{c}]$ depends on

(a) Only *x* (b) Only *y*

(c) Neither *x* nor *y* (d) Both *x* and *y*

Solution: **(c)** We have $[\vec{a}\ \ \vec{b}\ \ \vec{c}]=\begin{vmatrix}1 & 0 & -1\\ x & 1 & 1-x\\ y & x & 1+x-y\end{vmatrix}$

$\Rightarrow \qquad [\vec{a}\ \ \vec{b}\ \ \vec{c}]=\begin{vmatrix}1 & 0 & 0\\ x & 1 & 1\\ y & x & 1+x\end{vmatrix}$ [Using $C_3 \to C_3 + C_1$]

$\Rightarrow \qquad [\vec{a}\ \ \vec{b}\ \ \vec{c}]=1+x-x=1$

Clearly it does not depend on *x* and *y*.

Example 4: A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i}+\hat{j}$ and the plane determined by the vector $\hat{i}-\hat{j}, \hat{i}+\hat{k}$. The angle between $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$

(a) $\pi/4$ (b) $\pi/3$

(c) $\pi/6$ (d) $\pi/2$

Solution: (a) Let $\vec{n}_1$ and $\vec{n}_2$ be the vectors normal to the planes determined by $\hat{i}, \hat{i}+\hat{j}$ and $\hat{i}-\hat{j}, \hat{i}+\hat{k}$ respectively.

Then, $\vec{n}_1 = \hat{i}\times(\hat{i}+\hat{j}) = \hat{k}$ and $\vec{n}_2 = (\hat{i}-\hat{j})\times(\hat{i}+\hat{k}) = -\hat{i}-\hat{j}+\hat{k}$

Since, $\vec{a}$ is parallel to the line of intersection of the planes determined by $\hat{i}, \hat{i}+\hat{j}$ and $\hat{i}-\hat{j}, \hat{i}+\hat{k}$. Therefore, $\vec{a} \,||\, (\vec{n}_1 \times \vec{n}_2)$

$\Rightarrow \quad \vec{a} = \lambda(\vec{n}_1 \times \vec{n}_2)$

$$\Rightarrow \quad \vec{a} = \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -1 & -1 & 1 \end{vmatrix}$$

$\Rightarrow \quad \vec{a} = \lambda(\hat{i}-\hat{j})$

Let θ between angle $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$. Then, $\cos\theta = \dfrac{\lambda(\hat{i}-\hat{j}).(\hat{i}-2\hat{j}+2\hat{k})}{\sqrt{\lambda^2+\lambda^2}\sqrt{1+4+4}} = \dfrac{\lambda(1+2)}{\sqrt{2\lambda\times 3}} = \dfrac{1}{\sqrt{2}}$

$\therefore \quad \theta = \pi/4$

Example 5: Let $\vec{\alpha} = a\hat{i}+b\hat{j}+c\hat{k}, \vec{\beta} = b\hat{i}+c\hat{j}+a\hat{k}$ and $\vec{\gamma} = c\hat{i}+a\hat{j}+b\hat{k}$ be three coplanar vectors with $a \neq b$, and $\vec{v} = \hat{i}+\hat{j}+\hat{k}$. Then, $\vec{v}$ is perpendicular to

(a) $\vec{\alpha}$ (b) $\vec{\beta}$

(c) $\vec{\gamma}$ (d) None of these

Solution: **(a, b, c)** It is given that $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are coplanar vectors

$\therefore \quad [\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}] = 0 \quad \Rightarrow \quad \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$\Rightarrow \quad 3abc - a^3 - b^3 - c^3 = 0 \quad \Rightarrow \quad (a+b+c)(a^2+b^2+c^3-ab-bc-ca) = 0$

$a + b + c = 0 \quad [\because a^2+b^2+c^2-ab-bc-ca \neq 0]$

$\Rightarrow \quad \vec{v}.\vec{\alpha} = \vec{v}.\vec{\beta} = \vec{v}.\vec{\gamma} = 0$

$\Rightarrow \quad \vec{v}$ is perpendicular to $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$

Example 6: $\vec{a}$ and $\vec{c}$ are unit collinear vectors and $|\vec{b}| = 6$, then $\vec{b} - 3\vec{c} = \lambda\vec{a}$, if λ is

(a) –9, 3 (b) 9, 3

(c) 3, –3 (d) None of these

Solution: **(a)** We have, $\vec{b} - 3\vec{c} = \lambda\vec{a}$

Taking scalar product with $\vec{c}$, we have $(\vec{b}-3\vec{c}).\vec{c} = \lambda(\vec{a}.\vec{c})$

$\Rightarrow \quad \vec{b}.\vec{c} - 3(\vec{c}.\vec{c}) = \lambda(\vec{a}.\vec{c})$

$\Rightarrow \quad \vec{b}.\vec{c} - 3 = \lambda \quad [\because |\vec{a}| = |\vec{c}| = 1 \vec{a}$ and $\vec{c}$ are collinear vectors]

$\Rightarrow \quad \vec{b}.\vec{c} = 3 + \lambda \quad$ ....(i)

Again, $\vec{b}-3\vec{c}=\lambda\vec{a}$

$\Rightarrow \quad |\vec{b}-3\vec{c}|=|\lambda\vec{a}|$

$\Rightarrow \quad |\vec{b}-3\vec{c}|^2=\lambda^2|\vec{a}|^2$

$\Rightarrow \quad |\vec{b}|^2+9|\vec{c}|^2-6(\vec{b}.\vec{c})=\lambda^2|\vec{a}|^2$

$\Rightarrow \quad 36+9-6(3+\lambda)=\lambda^2$ [Using eq. (i)]

$\Rightarrow \quad 27-6\lambda=\lambda^2$

$\Rightarrow \quad \lambda^2+6\lambda-27=0 \quad \Rightarrow \lambda=-9, 3$

Example 7: If $\vec{a},\vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors, then the projection of the vector $\ell\frac{\vec{a}}{|\vec{a}|}+m\frac{\vec{b}}{|\vec{b}|}+n\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$ along the angle bisector of the vector $\vec{a}$ and $\vec{b}$, is

(a) $\frac{\ell^2+m^2}{\sqrt{\ell^2+m^2+n^2}}$ (b) $\sqrt{\ell^2+m^2+n^2}$

(c) $\frac{\sqrt{\ell^2+m^2}}{\sqrt{\ell^2+m^2+n^2}}$ (d) $\frac{\ell+m}{\sqrt{2}}$

Solution: **(d)** A vector to the bisector of the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|}+\frac{\vec{b}}{|\vec{b}|}=\hat{a}+\hat{b}$

$\therefore$ Unit vector along the bisector = $\frac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$

$$=\frac{1}{\sqrt{2}}(\hat{a}+\hat{b})$$

$[\because |\hat{a}+\hat{b}|^2=|\hat{a}|^2+|\hat{b}|^2+2\hat{a}.\hat{b} \Rightarrow |\hat{a}+\hat{b}|^2=1+1+0=2]$

$\therefore$ Required projection = $\left\{\ell\frac{a}{|\vec{a}|}+\frac{\vec{b}}{|\vec{b}|}+n\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}\right\}.\frac{1}{\sqrt{2}}(\hat{a}+\hat{b})$

$=\frac{1}{\sqrt{2}}(\ell+m)$ $[\because |\hat{a}|=|\hat{b}|=1$ and $\hat{a}.\hat{b}=\hat{a}.(\vec{a}\times\vec{b})=\hat{b}.(\hat{a}\times\hat{b})=0]$

Example 8: Let $\vec{u}=\hat{i}+\hat{j},\vec{v}=\hat{i}-\hat{j}$ and $\vec{w}=\hat{i}+2\hat{j}+3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u}.\hat{n}=-\vec{u}.\hat{n}=0$, then $|\vec{w}.\hat{n}|$ is equal to

(a) 3 (b) 0

(c) 1 (d) 2

Solution: We have, $\vec{u}.\hat{n}=0$ and $\vec{v}.\hat{n}=0$

$\Rightarrow \quad \hat{n}\perp\vec{u}$ and $\hat{n}\perp\vec{v}$ $\Rightarrow \quad \hat{n}=\pm\frac{\vec{u}\times\vec{v}}{|\vec{u}\times\vec{v}|}$

Now, $\vec{u}\times\vec{v}=(\hat{i}+\hat{j})\times(\hat{i}-\hat{j})=-2\hat{k}$

$\therefore \quad \hat{n}=\pm\hat{k}$

Hence, $|\vec{w}.\hat{n}|=|(\hat{i}+2\hat{j}+3\hat{k}).(\pm\hat{k})|=3$

Example 9: The value of 'a' so that volume of the parallelopiped formed by $\hat{i}+a\hat{j}+\hat{k}$, $\hat{j}+a\hat{k}$ and $a\hat{i}+\hat{k}$ becomes minimum, is

(a) 1/3 (b) 3

(c) $1/\sqrt{3}$ (d) $\sqrt{3}$

Solution: **(c)** Let V be the volume of the parallelopiped formed by the given vectors.

Let $\vec{\alpha}=\hat{i}+a\hat{j}+\hat{k}, \vec{\beta}=\hat{j}+a\hat{k}$ and $\vec{\gamma}=a\hat{i}+\hat{k}$

Then, $[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}]=\begin{vmatrix}1 & a & 1\\ 0 & 1 & a\\ a & 0 & 1\end{vmatrix}=1+a^3-a$

$\therefore \quad V=\left|[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}]\right|$

Clearly, V will be minimum when $\lambda = [\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}] = 1 + a^3 - a$ is minimum.

Now, $\lambda = 1 + a^3 - a$

$\Rightarrow \quad \frac{d\lambda}{da}=3a^2-1$ and $\frac{d^2\lambda}{da^2}=6a$

For maximum or minimum value of λ, we must have $\frac{d\lambda}{da}=0 \Rightarrow a=\pm\frac{1}{\sqrt{3}}$

Clearly, $\left(\frac{d^2\lambda}{da^2}\right)_{a=1/\sqrt{3}}=\frac{6}{\sqrt{3}}>0$

So, λ is minimum for $a = \frac{1}{\sqrt{3}}$ $\therefore$ V is minimum for $a = \frac{1}{\sqrt{3}}$

Example 10: Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors having magnitude 1, 1 and 2 respectively. If $\vec{a}\times(\vec{a}\times\vec{c})+\vec{b}=\vec{0}$, then the acute angle between $\vec{a}$ and $\vec{c}$ is

(a) $\pi/4$ (b) $\pi/6$

(c) $\pi/3$ (d) None of these

Solution: **(b)** We have, $|\vec{a}|=1, |\vec{b}|=1$ and $|\vec{c}|=1$

Also, $\vec{a}\times(\vec{a}\times\vec{b})+\vec{b}=0$

$\Rightarrow \quad (\vec{a}.\vec{c})\vec{a}-(\vec{a}.\vec{a})\vec{c}+\vec{b}=\vec{0}$

$\Rightarrow \quad (\vec{a}.\vec{c})\vec{a}.\vec{c}+\vec{b}=0 \qquad [\because \vec{a}.\vec{a}=|\vec{a}|^2=1]$

$\Rightarrow \quad (\vec{a}.\vec{c})\vec{a}-\vec{c}=-\vec{b} \qquad \Rightarrow \quad |(\vec{a}.\vec{c})\vec{a}-\vec{c}|=|-\vec{b}|$

$\Rightarrow \quad |(\vec{a}.\vec{c})\vec{a}-\vec{c}|^2=|-\vec{b}|^2 \qquad \Rightarrow \quad |(\vec{a}.\vec{c})\vec{a}|^2+|\vec{c}|^2-2\{(\vec{a}.\vec{c})\vec{a}.\vec{c}\}=|b|^2$

$\Rightarrow \quad (\vec{a}.\vec{c})^2|\vec{a}|^2+|\vec{c}|^2-2(\vec{a}.\vec{c})(\vec{a}.\vec{c})=|\vec{b}|^2 \qquad \Rightarrow \quad (\vec{a}.\vec{c})^2\{|\vec{a}|^2-2\}+|\vec{c}|^2=|\vec{b}|^2$

$\Rightarrow \quad -(\vec{a}.\vec{c})^2+4=1 \qquad \Rightarrow \quad (\vec{a}.\vec{c})^2=3$

$\Rightarrow \quad \vec{a}.\vec{c}=\pm\sqrt{3} \qquad \Rightarrow \quad |\vec{a}||\vec{c}|\cos\theta=\sqrt{3}$

where θ is an acute angle between $\vec{a}$ and $\vec{c}$

$\therefore \quad \cos\theta=\frac{\sqrt{3}}{2} \quad \Rightarrow \quad \theta=\frac{\pi}{6}$

Example 11: Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-zero vectors such that no two are colinear and $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$.

If θ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then sin θ is equal to

(a) $2\sqrt{2}/3$ (b) $\sqrt{2}/3$

(c) 2/3 (d) 1/3

Solution: **(a)** We have, $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$

$\Rightarrow \quad (\vec{a}.\vec{c})\vec{b} - (\vec{b}.\vec{c})\vec{a} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a} \quad \Rightarrow \quad (\vec{a}.\vec{c})\vec{b} - \left\{(\vec{b}.\vec{c}) + \frac{1}{3}|\vec{b}||\vec{c}|\right\}\vec{a} = \vec{0}$

$\Rightarrow \quad \vec{a}.\vec{c} = 0$ and $\vec{b}.\vec{c} + \frac{1}{3}|\vec{b}||\vec{c}| = 0$ [$\vec{a}, \vec{b}$ are non-collinear]

$\Rightarrow \quad |\vec{b}||\vec{c}|\cos\theta + \frac{1}{3}|\vec{b}||\vec{c}| = 0 \Rightarrow \cos\theta = -\frac{1}{3} \Rightarrow \sin\theta = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$

Example 12: $\vec{a}$ and $\vec{b}$ are mutually perpendicular unit vectors. $\vec{r}$ is a vector satisfying $\vec{r}.\vec{a} = 0, \vec{r}.\vec{b} = 1$ and $a[\vec{r}\ \ \vec{a}\ \ \vec{b}]$ is

(a) $\vec{a} + (\vec{a} \times \vec{b})$ (b) $\vec{b} + (\vec{a} \times \vec{b})$

(c) $\vec{a} + \vec{b}(\vec{a} \times \vec{b})$ (d) $\vec{a} - \vec{b} + (\vec{a} \times \vec{b})$

Solution: **(b)** Let $\vec{r} = x_1\vec{a} + x_2\vec{b} + x_3(\vec{a} \times \vec{b}) \Rightarrow \vec{r}.\vec{a} = x_1|\vec{a}|^2, \vec{r}.\vec{b} = x_2|\vec{b}|^2$

and $\vec{r}.\vec{a}(\vec{a} \times \vec{b}) = x_3|\vec{a} \times \vec{b}|^2$

$\Rightarrow \quad x_1 = 0, x_2 = 1, x_3 = 1 \quad \Rightarrow \quad \vec{r} = \vec{a} \times \vec{b} + \vec{b}$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Algebra of Vector

Q.1 Two $\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side *AD* is is rotated by an acute angle α in the plane of the parallelogram so that *AD* becomes *AD'*. If *AD'* makes a right angle with the side *AB*, then the cosine of the angle α is given by

(a) $\frac{8}{9}$ (b) $\frac{\sqrt{17}}{9}$

(c) $\frac{1}{9}$ (d) $\frac{4\sqrt{5}}{9}$

Q.2 Let *P, Q, R* and *S* be the point on the plane with position vector $-2\hat{i} - \hat{j}, 4\hat{i}, 3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral *PQRS* must be a

(a) Parallelogram, which is neither a rhombus nor a rectangle
(b) Square
(c) Rectangle, but not a square
(d) Rhombus, but not a square

Q.3 Let two non-collinear unit vector $\hat{a}$ and $\hat{b}$ form an acute angle. A point *P* moves so that at any time *t* the position vector $\overrightarrow{OP}$ (where *O* is the origin) is given by $\hat{a}\cos t + \hat{b}\sin t$. When *P* is farthest from origin *O*, let *M* be the length $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\hat{u}$. Then

(a) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a}.\hat{b})^{1/2}$

(b) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a}.\hat{b})^{1/2}$

(c) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a}.\hat{b})^{1/2}$

(d) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a}.\hat{b})^{1/2}$

Q.4 The position vectors of *P* and *Q* are respectively $\vec{a}$ and $\vec{b}$. If *R* is a point on $\overrightarrow{PQ}$ such that $\overrightarrow{PR} = 5\overrightarrow{PQ}$, then the position vector of *R* is

(a) $5\vec{b} - 4\vec{a}$ (b) $5\vec{b} + 4\vec{a}$

(c) $4\vec{b} - 5\vec{a}$ (d) $4\vec{b} + 5\vec{a}$

Q.5 The position vector of a point lying on the line joining the points whose position vectors are $\hat{i} + \hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is

(a) $\hat{j}$ (b) $\hat{i}$

(c) $\hat{k}$ (d) $\vec{0}$

2. Scalar Product of two & Vector Product of Two vector

Q.6 If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$, then the value of $[2\vec{a} + \vec{b}].[(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$ is

(a) 2 (b) 3
(c) 5 (d) 7

Q.7 If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are the unit vectors such that $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d}) = 1$ and $\vec{a}.\vec{c} = \frac{1}{2}$, then

(a) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar

(b) $\vec{a}, \vec{b}, \vec{d}$ are non-coplanar

(c) $\vec{b}, \vec{d}$ are non-parallel

(d) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel

Q.8 The edges of a parallelopiped are of unit length and are parallel to non-cpolanar unit vector $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a}.\hat{b} = \hat{b}.\hat{c} = \hat{c}.\hat{a} = \frac{1}{2}$. Then, the volume of the parallelopiped is

(a) $\frac{1}{\sqrt{2}}$ cu unit (b) $\frac{1}{2\sqrt{2}}$ cu unit

(c) $\frac{\sqrt{3}}{2}$ cu unit (d) $\frac{1}{\sqrt{3}}$ cu unit

Q.9 If $|\vec{a}|=3, |\vec{b}|=4$, then a value of λ for which $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{a}-\lambda\vec{b}$ is
(a) 9/16 (b) 3/4
(c) 3/2 (d) 4/3

Q.10 Let $\vec{a},\vec{b},\vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. Which one of the following is correct?
(a) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}=\vec{0}$
(b) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\neq\vec{0}$
(c) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{a}\times\vec{c}=\vec{0}$
(d) $\vec{a}\times\vec{b},\vec{b}\times\vec{c},\vec{c}\times\vec{a}$ are mutually perpendicular

Q.11 $\vec{a}=\hat{i}+2\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k},\vec{c}=\hat{i}+\hat{j}-\hat{k}$. A vector coplanar to $\vec{a}$ and $\vec{b}$ has a projection along $\vec{c}$ of magnitude $\frac{1}{\sqrt{3}}$, then the vector is
(a) $4\hat{i}-\hat{j}+4\hat{k}$ (b) $4\hat{i}+\hat{j}-4\hat{k}$
(c) $2\hat{i}+\hat{j}+\hat{k}$ (d) None of these

Q.12 Let $\vec{a}=\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$. Then, the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}=0$ and $\vec{a}.\vec{b}=3$, is
(A) $\hat{i}+\hat{j}-2\hat{k}$ (B) $2\hat{i}-\hat{j}+2\hat{k}$
(C) $\hat{i}-\hat{j}-2\hat{k}$ (D) $\hat{i}+\hat{j}-2\hat{k}$

Q.13 The non-zero vectors $\vec{a},\vec{b}$ and $\vec{c}$ are related by $\vec{a}=8\vec{b}$ and $\vec{c}=-7\vec{b}$. Then, the angle between $\vec{a}$ and $\vec{c}$ is
(a) 0 (b) $\pi/4$
(c) $\pi/2$ (d) π

Q.14 If $\hat{u}$ and $\hat{v}$ are unit vectors and θ is the acute angle between them, then $2\hat{u}\times3\hat{v}$ is a unit vector for
(a) Exactly two values of θ
(b) More than two values of θ
(c) No values of θ
(d) Exactly one value of θ

Q.15 the value of *a*, for which the points *A, B, C* with position vectors $2\hat{i}-\hat{j}+\hat{k},\hat{i}-3\hat{j}-5\hat{k}$ and $a\hat{i}-3\hat{j}+\hat{k}$ respectively are the vertices of a right angled triangle with $C = \pi/2$ are
(a) –2 and –1 (b) –2 and 1
(c) 2 and –1 (d) 2 and 1

Q.16 If the position vectors of *A, B* and *C* are respectively $2\hat{i}-\hat{j}+\hat{k},\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$, then $\cos^2 A$ is equal to
(a) 0 (b) 6/41
(c) 35/41 (d) 1

Q.17 The projection of the vector $2\hat{i}+\hat{j}-3\hat{k}$ on the vector $\hat{i}-2\hat{j}+\hat{k}$ is
(a) $-\frac{3}{\sqrt{14}}$ (b) $\frac{3}{\sqrt{14}}$
(c) $-\sqrt{\frac{3}{2}}$ (d) $\frac{3}{\sqrt{2}}$

Q.18 If $\vec{a},\vec{b}$ and $\vec{c}$ are mutually perpendicular unit vector, then $|\vec{a}+\vec{b}+\vec{c}|$ is equal to
(a) 3 (b) $\sqrt{3}$
(c) $\frac{(\sqrt{a^2+b^2+c^2})}{3}$ (d) 1

Q.19 If the vectors $\overrightarrow{AB}=-3\hat{i}+4\hat{k}$ and $\overrightarrow{AC}=5\hat{i}-2\hat{j}+4\hat{k}$ are the sides of a ΔABC. Then, the length of the medium through *A* is
(a) $\sqrt{14}$ (b) $\sqrt{18}$
(c) $\sqrt{29}$ (d) 4

3. Scalar Triple Product, Dot Product, Mixed Product, Vector Triple Product & Solving Equations and Peciprocal of Vector

Q.20 The number of distinct real values of λ, for which the vectors $-\lambda^2\hat{i}+\hat{j}+\hat{k},\hat{i}-\lambda^2\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}-\lambda^2\hat{k}$ are coplanar, is
(a) Zero (b) One
(c) Two (d) Three

Q.21 If the vectors $\vec{a}=\hat{i}-\hat{j}+2\hat{k},\vec{b}=2\hat{i}+4\hat{j}+\hat{k}$ and $\vec{c}=\lambda\hat{i}+\hat{j}+\mu\hat{k}$ are mutually orthogonal, then (λ, μ) is equal to
(a) (–3, 2) (b) (2, –3)
(c) (–2, 3) (d) (3, –2)

Q.22 If $\vec{u},\vec{v},\vec{w}$ are non-coplanar vectors and *p, q* are real numbers, then the equality. $[3\vec{u}\ p\vec{v}\ p\vec{w}] - [p\vec{v}\ \vec{w}\ q\vec{u}] - [2\vec{w}\ 2q\vec{v}\ q\vec{u}] = 0$ holds for

(a) Exactly two values of (*p, q*)
(b) More than two but not all values of (*p, q*)
(c) All values of (*p, q*)
(d) Exactly one value of (*p, q*)

Q.23 The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then, which one of the following gives possible values of α and β?

(a) $\alpha = 2, \beta = 2$ (b) $\alpha = 1, \beta = 2$
(c) $\alpha = 2, \beta = 1$ (d) $\alpha = 1, \beta = 1$

Q.24 Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = x\hat{i} + (x-2)\hat{j} - \hat{k}$, if the vector $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then *x* equals

(a) 0 (b) 1
(c) –4 (d) –2

Q.25 If $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}.\vec{b} \neq 0, \vec{b}.\vec{c} \neq 0$, then $\vec{a}$ and $\vec{c}$ are

(a) Inclined at an angle of $\pi/6$ between them
(b) Perpendicular
(c) Parallel
(d) Inclined at an angle of $\pi/3$ between them

Q.26 Let $\vec{a}$ be a unit vector, $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + 3\hat{k}$. Then, maximum value of $[\vec{a}\ \vec{b}\ \vec{c}]$ is

(a) –1 (b) $\sqrt{10} + \sqrt{6}$
(c) $\sqrt{10} - \sqrt{6}$ (d) $\sqrt{59}$

Q.27 The unit vector which is orthogonal to the vector $3\hat{i} + 2\hat{j} + 6\hat{k}$ and is coplanar with the vectors $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is

(a) $\frac{2\hat{i} - 6\hat{j} + \hat{k}}{\sqrt{41}}$ (b) $\frac{2\hat{i} - 3\hat{j}}{\sqrt{13}}$
(c) $\frac{3\hat{j} - \hat{k}}{\sqrt{10}}$ (d) $\frac{4\hat{i} + 3\hat{j} - 3\hat{k}}{\sqrt{34}}$

Q.28 Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and let $\vec{p}, \vec{q}$ and $\vec{r}$ be vectors defined by the relation $\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]}, \vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]}$ and $\vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$.

Then, the value of the expression $(\vec{a} + \vec{b}).\vec{p} + (\vec{b} + \vec{c}).\vec{q} + (\vec{c} + \vec{a}).\vec{r}$ is equal to

(a) 0 (b) 1
(c) 2 (d) 3

Q.29 The magnitude of cross product of two vectors is $\sqrt{3}$ times the dot product. The angle between the vectors is

(a) $\pi/6$ (b) $\pi/3$
(c) $\pi/2$ (d) $\pi/4$

Q.30 If the volume of the parallelopiped with $\vec{a}, \vec{b}$ and $\vec{c}$ as conterminous edges is 40 cu unit, then the volume of the parallelopiped having $\vec{b} + \vec{c}, \vec{c} + \vec{a}$ and $\vec{a} + \vec{b}$ as coterminous edges in cubic unit is

(a) 80 (b) 120
(c) 160 (d) 40

Section-B
(Previous Year Question)

Q.1 If $\left[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}\right] = \lambda\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2$, then λ is equal to

(a) 3 (b) 0
(c) 1 (d) 2

Q.2 If the vectors $\vec{AB} = 3\hat{i} + 4\hat{k}$ and $\vec{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is

(a) $\sqrt{45}$ (b) $\sqrt{18}$
(c) $\sqrt{72}$ (d) $\sqrt{33}$

Q.3 Let $\hat{a}$ and $\hat{b}$ be two unit vetors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} + 4\hat{b}$ are peeprendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

Q.4 Let ABCD be a parallelogram such that $\vec{AB} = \vec{q}, \vec{AD} = \vec{p}$ and $\angle BAD$ be an acute angle $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by

(a) $\vec{r}=\vec{q}-\left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$

(b) $\vec{r}=-3\vec{q}+\frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(c) $\vec{r}=3\vec{q}-\frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(d) $\vec{r}=-\vec{q}+\left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$

Q.5 If $\vec{a}=\frac{1}{\sqrt{10}}\left(3\hat{i}+\hat{k}\right)$ and $\vec{b}=\frac{1}{7}\left(2\hat{i}+3\hat{j}-6\hat{k}\right)$, then the value of $\left(2\vec{a}-\vec{b}\right).\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}+2\vec{b}\right)\right]$ is

(a) 5 (b) 3
(c) -5 (d) -3

Q.6 The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying : $\vec{b}\times\vec{c}=\vec{b}\times\vec{d}$ and $\vec{a}.\vec{d}=0$. Then the vector $\vec{d}$ is equal to

(a) $\vec{b}+\left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (b) $\vec{c}-\left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

(c) $\vec{b}-\left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (d) $\vec{c}+\left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

Q.7 Let $\vec{a}=\vec{j}-\vec{k}$ and $\vec{c}=\hat{i}-\vec{j}-\vec{k}$. Then the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}+\vec{c}=0$ and $\vec{a}.\vec{b}=3$ is

(a) $-\hat{i}+\hat{j}-2\hat{k}$ (b) $2\hat{i}-\hat{j}+2\hat{k}$
(c) $\hat{i}-\hat{j}-2\hat{k}$ (d) $\hat{i}+\hat{j}-2\hat{k}$

Q.8 If the vectors $\hat{a}=\hat{i}-\hat{j}+2\hat{k},\vec{b}=2\hat{i}+4\hat{j}+4\hat{k}$ and $\vec{c}=\lambda\hat{i}+\hat{j}+\mu\hat{k}$ are mutually orthogonal, then (λ,μ) =

(a) (-3, 2) (b) (2,-3)
(c) (-2, 3) (d) (3, -2)

Q.9 If $\vec{u},\vec{v},\vec{w}$ are non-coplanar vectors and p,q are real numbers, then the equality $\left[3\vec{u},p\vec{v},p\vec{w}\right]-\left[p\vec{v}\,\vec{w}\,q\vec{u}\right]-\left[2\vec{w}\,q\vec{v}\,q\vec{u}\right]=0$ holds for

(a) exactly two volumes of (p,q)
(b) more than two but not all values of (p,q)
(c) all values of (p,q)
(d) exactly one value of (p,q)

Q.10 The non-zero vectors $\vec{a},\vec{b}$ and $\vec{c}$ are realted by $\vec{a}=8\vec{b}$ and $\vec{c}=-7\vec{b}$. Then the angle between $\vec{a}$ and $\vec{c}$ is

(a) π (b) 0
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

Q.11 The vector $\vec{a}=\alpha\hat{i}+2\hat{j}+\beta\hat{k}$ lies in the plane of the vectors $\vec{b}=\hat{i}+\hat{j}$ and $\vec{c}=\hat{j}+\hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β ?

(a) $\alpha=1,\beta=1$ (b) $\alpha=2,\beta=2$
(c) $\alpha=1,\beta=2$ (d) $\alpha=2,\beta=1$

Q.12 If $\hat{u}$ and $\hat{v}$ are unit vectors and θ is the acute angle between them, then $2\hat{u}\times3\hat{v}$ is a unit vector for

(a) no value of θ
(b) exactaly one value of θ
(c) exactly two values of θ
(d) more than two values of θ

Q.13 The values of a, for which the points A,B,C with position vectors $2\hat{i}-\hat{j}+\hat{k},\hat{i}-3\hat{j}-5\hat{k}$ and $a\hat{i}-3\hat{j}+\hat{k}$ respectvely are the vertices of a right-angled triangle at C are

(a) 2 and 1 (b) -2 and -1
(c) -2 and 1 (d) 2 and -1

Q.14 If $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\vec{a}\times\left(\vec{b}\times\vec{c}\right)$, where $\vec{a},\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}.\vec{b}\neq0,\vec{b}.\vec{c}\neq0$, then $\vec{a}$ and $\vec{c}$ are

(a) inclined at an angle of $\pi/3$
(b) inclined at an angle of $\pi/6$ between them
(c) perpendicualr
(d) parallel

Q.15 If $\vec{a},\vec{b},\vec{c}$ are non-coplanar vector and λ is a real number then $\left[\lambda\left(\vec{a}+\vec{b}\right)\lambda^2\vec{b}\ \lambda\vec{c}\right]=\left[\vec{a}\ \ \vec{b}+\vec{c}\ \ \vec{b}\right]$ for
(a) no value of λ
(b) exactly one value of λ
(c)exactly two values of λ
(d) exactly three values of λ

Q.16 Let a,b and c be distinct non-negative numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}+c\hat{k},\hat{i}+\hat{k}$ and $C\hat{i}+c\hat{j}+b\hat{k}$ lie in a plane, then c is
(a) the arithmetic mean of a and b
(b) the geometric mean of a and b
(c) the harmoic mean of a and b
(d) equal to zero

Q.17 Let $\vec{a}=\hat{i}-\hat{k},\vec{b}=x\hat{i}+\hat{j}+\left(1-x\right)\hat{k}$ and $\vec{c}=y\hat{i}+x\hat{j}+\left(1+x-y\right)\hat{k}$. Then $\left[\vec{a},\ \vec{b},\vec{c}\right]$ depends on
(a) only x (b) only y
(c) neither x nor y (d) both x and y

Q.18 For any vector $\vec{a}$, the value of $\left(\vec{a}\times\hat{i}\right)^2+\left(\vec{a}\times\hat{j}\right)^2+\left(\vec{a}\times\vec{k}\right)^2$ is equal
(a) $\vec{a}^2$ (b) $3\vec{a}^2$
(c) $4\vec{a}^2$ (d) $2\vec{a}^2$

Q.19 If C is the mid point of AB and P is any point outside AB, then
(a) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=0$ (b) $\overrightarrow{PA}+\overrightarrow{PB}+2\overrightarrow{PC}=\vec{0}$
(c) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$ (d) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$

Q.20 Let $\vec{a},\vec{b}$ and $\vec{c}$ be non-zero vectors such that $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\frac{1}{3}\left|\vec{b}\right|\left|\vec{c}\right|\vec{a}$. If θ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin\theta$ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin\theta$ equals
(a) $\frac{2}{3}$ (b) $\frac{\sqrt{2}}{3}$
(c) $\frac{1}{3}$ (d) $\frac{2\sqrt{2}}{3}$

Q.21 Let $\vec{u},\vec{v},\vec{w}$ be such that $|\vec{u}|=1,|\vec{v}|=2,|\vec{w}|=3.$ If the projection $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v}$, $\vec{w}$ are perpendicualr to each other then $|\vec{u}-\vec{v}+\vec{w}|$ equals
(a) $\sqrt{14}$ (b) $\sqrt{7}$
(c)2 (d) 14

Q.22 If $\vec{a},\vec{b},\vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors $\vec{a}+2\vec{b}+3\vec{c},\lambda\vec{b}+4\vec{c}$ and $\left(2\lambda-1\right)\vec{c}$ are non-coplanar for
(a) all except two values of λ
(b) all except one value of λ
(c) all values of λ
(d) no value of λ

Q.23 A particle is acted upon by constant forces $4\hat{i}+\hat{j}-3\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ which displace it form a point $\hat{i}+2\hat{j}+3\hat{k}$ the the point $5\hat{i}+4\hat{j}+\hat{k}$. The work doen in standard units by the forces is given by
(a) 25 (b) 30
(c) 40 (d) 15

Q.24 Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}+6\vec{c}$ equals
(a) $\lambda\vec{c}$ (b) $\lambda\vec{b}$
(c) $\lambda\vec{a}$ (d) 0

Q.25 $\vec{a},\vec{b},\vec{c}$ are three vectors, such that $\vec{a}+\vec{b}+\vec{c}=0$, $|\vec{b}|=2,|\vec{c}|=3,$ then $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ is equal to
(a) -7 (b) 7
(c) 1 (d) 0

Q.26 The vectors $\overrightarrow{AB}=3\hat{i}+4\hat{k}$ and $\overrightarrow{AC}=5\hat{i}-2\hat{j}+4\hat{k}$ are the sides of a triangle ABC. The length of the median through A is
(a) $\sqrt{72}$ (b) $\sqrt{33}$
(c) $\sqrt{288}$ (d) $\sqrt{18}$

Q.27 Let $\vec{u}=\hat{i}+\hat{j},\vec{v}=\hat{i}-\hat{j}$ and $\vec{w}=\hat{i}+2\hat{j}+3\hat{k}.$ If $\hat{n}$ is a unit vector such taht $\vec{u}.\hat{n}=0$ and $\vec{v}.\hat{n}=0$, then $|\vec{w}.\hat{n}|$ is equal to
(a) 1 (b) 2
(c) 3 (d) 0

Q.28 If $\vec{u},\vec{v}$ and $\vec{w}$ are three non-coplanar vectors, then $(\vec{u}+\vec{v}-\vec{w}).(\vec{u}-\vec{v})\times(\vec{v}-\vec{w})$ equals
(a) $\vec{u}.\vec{v}\times\vec{w}$ (b) $\vec{u}.\vec{w}\times\vec{v}$
(c) $3u.\vec{u}\times\vec{w}$ (d) 0

Q.29 Consider A, B, C and D with position vectors $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ a n d $5\hat{i}-\hat{j}+5\hat{k}$ respectively. then ABCD is a
(a) rhombus
(b) rectangle
(c) paralleogram but not a rhombus
(d) square

Q.30 If $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$ then $\vec{a}+\vec{b}+\vec{c}=$
(a) abc (b) -1
(c) 0 (d) 2

Section-C
(Board pattern)

Q.1 Write a unit vector in the direction of $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$.

Q.2 If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 9\hat{k}$, find a unit vector parallel to $\vec{a} + \vec{b}$.

Q.3 Write a unit vector in the direction of $\vec{b} = 2\hat{i} + \hat{j} + 2\hat{k}$.

Q.4 Dot product of a vector with $\hat{i} + \hat{j} - 3\hat{k}, \hat{i} + 3\hat{j} - 2\hat{k}$ and $2\hat{i} + \hat{j} + 4\hat{k}$ are 0, 5 and 8 respectively. Find the vector.

Q.5 Express the vector $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ as the sum of two vectors such that one is parallel to the vector $\vec{b} = 3\hat{i} + \hat{k}$ and other is perpendicular to $\vec{b}$.

Q.6 If $\vec{a}.\vec{a} = 0$ and $\vec{a}.\vec{b} = 0$, what can you conclude about the vector $\vec{b}$?

Q.7 If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are orthogonal.

Q.8 What is the angle between vectors $\vec{a}$ and $\vec{b}$ with magnitudes 2 and Ö3 respectively? Given $\vec{a}.\vec{b} = \sqrt{3}$.

Q.9 Find the angle between the vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$.

Q.10 For what value of l are the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ per perpendicualr to each other.

Q.11 Find the projection of $\vec{a}$ on $\vec{b}$ if $\vec{a}.\vec{b} = 8$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$.

Q.12 Write the value of *p* for which $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} + p\hat{j} + 3\hat{k}$ are parallel vectors.

CHAPTER 2

THREE DIMENSIONAL GEOMETRY

1. DISTANCE FORMULAE

The distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2. SECTION FORMULAE

(i) The coordinates of the point dividing the line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m_1 : m_2$ internally, are

$$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right)$$

(ii) The coordinates of the point dividing the line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio of $m_1 : m_2$ extrenally, are

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)$$

(iii) The coordinates of the mid point of the line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, are

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$$

Illustration 1: The ratio in which the line joining (1, 2, 3) and (–3, 4, –5) is divided by *xy*-plane, is
(a) 5 : 3 (b) 3 : 5 (c) 2 : 3 (d) None of these

Solution : **(b)** Let $A \equiv (1, 2, 3)$ and $B \equiv (-3, 4, -5)$. Let the point P where the line AB meets the *xy*-plane divides it in the ratio $k : 1$.

$\therefore$ The coordinates of P are $\left(\frac{-3k+1}{k+1}, \frac{4k+2}{k+1}, \frac{-5k+3}{k+1}\right)$

Since, the point P lies upon *xy*-plane.

$\therefore$ *z*-coordinate of P must be zero.

$$\therefore \quad \frac{-5k+3}{k+1} = 0$$

$$\Rightarrow \quad -5k + 3 = 0$$

$$\Rightarrow \quad k = 3/5$$

Therefore, required ratio is $\frac{5}{3}$: 1 ie, 3 : 5 internally.

Centroid of a Triangle : The coordinates of the centroid of the ΔABC, whose vertices are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$

3. DIRECTION COSINES

(i) If a vector makes angles α, β, γ with the positive directions of *x*-axis, *y*-axis and *z*-axis respectively then cos α, cos β, cos γ are called its direction cosines.
The direction cosines are generally denoted as ℓ, *m, n* ie, ℓ = cos α, *m* = cos β, *n* = cos γ.
The angles α, β, γ are known as direction angles.

(ii) Three numbers α, β, γ proportional to the direction cosines ℓ, *m, n* of a vectors are known as the direction ratios of the vector.

Thus, *a, b, c* are the direction ratios of a vector, provided $\frac{\ell}{a}=\frac{m}{b}=\frac{n}{c}$.

***Illustration 2*:** *A*(3, 2, 0), *B*(5, 3, 2), *C*(–9, 6, –3) are three points forming a triangle. If *AD*, the bisector of $\angle BAC$ meets *BC* in *D*, then coordinates of *D* is

(a) $\left(-\frac{19}{8},\frac{57}{16},\frac{17}{16}\right)$ (b) $\left(\frac{19}{8},-\frac{57}{16},\frac{17}{16}\right)$ (c) $\left(\frac{19}{8},\frac{57}{16},-\frac{17}{16}\right)$ (d) $\left(\frac{19}{8},\frac{57}{16},\frac{17}{16}\right)$

Solution : **(d)** Since, *AD* is the bisector of $\angle BAC$

$\therefore \quad \frac{BD}{DC}=\frac{AB}{AC}$

Now, $AB=\sqrt{(5-3)^2+(3-2)^2+(2-0)^2}$

$=\sqrt{4+1+4}\ =\sqrt{9}=3$

and $AC=\sqrt{(-9-3)^3+(6-2)^2+(-3-0)^2}=13$

$\therefore \quad \frac{BD}{DC}=\frac{3}{13}$

$\Rightarrow$ *D* divides *BC* in the ratio 3 : 13.

$\therefore$ The coordinates of *D* are $\left(\frac{3-(9)+13(5)}{3+13},\frac{3(6)+13(3)}{3+13},\frac{3(-3)+13(2)}{3+13}\right)$

$=\left(\frac{19}{8},\frac{57}{16},\frac{17}{16}\right)$

***Illustration 3*:** If a line *OP* makes with *x*-axis an angle of measure 120° and with *y*-axis of measure 60°, then the angle made with the line with *z*-axis is

(a) 45° (b) 120° (c) 60° (d) None of these

Solution : **(a)** We have, α = 120° and β = 60°

$\therefore \quad \cos\alpha = \cos 120^\circ = -\frac{1}{2}$ and $\cos\beta = \cos 60^\circ = \frac{1}{2}$

But $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\therefore \quad \left(-\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2+\cos^2\lambda=1$

$\cos^2\lambda=1-\frac{1}{4}-\frac{1}{4}=\frac{1}{2}$

$\Rightarrow \quad \cos\lambda = \pm\frac{1}{\sqrt{2}}$

$\therefore \quad \lambda = 45^\circ$ or 135°

4. SOME USEFUL RESULTS ON DIRECTION COSINES & DIRECTION RATIOS

If $P(x, y, z)$ is a point in space such that $\vec{r} = OP$ has direction cosines ℓ, m, n, then

(a) $x = \ell|\vec{r}|, y = m|\vec{r}|, z = n|\vec{r}|$

(b) $\ell|\vec{r}|$, $m|\vec{r}|$, $n|\vec{r}|$ are projections of $\vec{r}$ on *OX, OY, OZ* respectively.

(c) $\vec{r} = |\vec{r}|(\ell\hat{i} + m\hat{j} + n\hat{k})$ and $\hat{r} = \ell\hat{i} + m\hat{j} + n\hat{k}$

(d) $\ell^2 + m^2 + n^2 = 1$

(e) If $\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$, then

(i) *a, b, c* are the direection ratios of $\vec{r}$.

(ii) Direction cosines of $\vec{r}$ are given by $\ell = \frac{a}{\sqrt{a^2+b^2+c^2}}, m = \frac{b}{\sqrt{a^2+b^2+c^2}}, n = \frac{c}{\sqrt{a^2+b^2+c^2}}$

(f) Direction ratios of the line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ and $x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$ and its direction cosines are

$$\frac{x_2 - x_1}{|\overrightarrow{PQ}|}, \frac{y_2 - y_1}{|\overrightarrow{PQ}|}, \frac{z_2 - z_1}{|\overrightarrow{PQ}|}$$

(g) The direction cosines of

(i) $\overrightarrow{OX}$ are (1, 0, 0) (ii) $\overrightarrow{OY}$ are (0, 1, 0) (iii) $\overrightarrow{OZ}$ are (0, 0, 1)

Note :- Direction cosines of a line are unique but direction ratios of a line are not unique and can be infinite.

5. ANGLE BETWEEN TWO VECTORS

If θ is the angle between two vectors whose direction cosines are ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2, then

$\cos\theta = \ell_1\ell_2 + m_1m_2 + n_1n_2$ and

$\sin\theta = \sqrt{(m_1n_2 - m_2n_1)^2 + (n_1\ell_2 - n_2\ell_1)^2 + (\ell_1m_2 - \ell_2m_1)^2}$

Note :-

(i) $\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$, then two vectors $\vec{r}_1$ and $\vec{r}_2$ having direction cosines ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 are orthogonal.

(ii) If $\frac{\ell_1}{\ell_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$, then two vectors are parallel.

(i) Angle in terms of direction ratios : If θ is the angle between two vectors having direction ratios a_1, b_1, c_1 and a_2, b_2, c_2, then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

The two vectors are orthogonal, where

$a_1a_2 + b_1b_2 + c_1c_2 = 0$

and parallel, when $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(ii) Projection of the join of two points on a line : If $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are two points, then the length of projection of PQ on a line whose direction cosines are ℓ, m, n, is given by $(x_2 - x_1)\ell + (y_2 - y_1)m + (z_2 - z_1)n$

***Illustration 4*:** The acute angle between the lines whose direction cosines are proportional to (2, 3, –6) and (3, –4, 5), is

(a) $\cos^{-1}\left(\frac{36}{7\sqrt{48}}\right)$ (b) $\cos^{-1}\left(\frac{36}{7\sqrt{50}}\right)$ (c) $\cos^{-1}\left(\frac{18}{7\sqrt{50}}\right)$ (d) None of these

Solution : **(b)** The actual DC's of the lines are $\left(\frac{2}{7}, \frac{3}{7}, -\frac{6}{7}\right)\left(\frac{3}{\sqrt{(50)}}, -\frac{4}{\sqrt{(50)}}, \frac{5}{\sqrt{(500)}}\right)$

If θ is the angle between them, then $\cos\theta = \pm \frac{2.3 + 3(-4) + (-6)5}{7 \times \sqrt{(50)}} = \pm\frac{36}{7\sqrt{50}}$

$\therefore$ The acute angle between the lines = $\cos^{-1}\left(\frac{18}{7\sqrt{50}}\right)$

***Illustration 5*:** The acute angle between two lines whose direction ratios are 2, 3, 6 and 1, 2, 2, is

(a) $\cos^{-1}(20/21)$ (b) $\cos^{-1}(18/21)$ (c) $\cos^{-1}(17/21)$ (d) None of these

Solution : **(a)** We have, $a_1 = 2, b_1 = 3, c_1 = 6, a_2 = 1, b_2 = 2, c_2 = 2$

If θ be the angle between two lines whose DR's are given, then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}} = \frac{2\times1 + 3\times2 + 6\times2}{\sqrt{2^2 + 3^2 + 6^2}\sqrt{1^2 + 2^2 + 3^2}}$$

$$= \frac{2\times1 + 3\times2 + 6\times2}{\sqrt{2^2 + 3^2 + 6^2}\sqrt{1^2 + 2^2 + 3^2}} = \frac{2 + 6 + 12}{7\times3} = \frac{20}{21}$$

$\therefore$ $\theta = \cos^{-1}(20/21)$

6. VECTOR EQUATIONS OF A LINE

Parametric Form

The vector equation of a line passing through a point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is

$\vec{r} = \vec{a} + \lambda\vec{b}$, where λ is a scalar.

Proof : Let O be the origin and let A be the point with position vector $\vec{a}$. Then, $\overrightarrow{OA} = \vec{a}$. Let $\vec{r}$ be the position vector of an arbitrary point P on the line through A and parallel to $\vec{b}$ as shown in figure. Then,

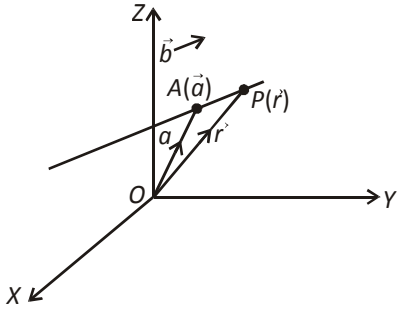
$\overrightarrow{OP} = \vec{r}$

Clearly, $\overrightarrow{AP} \,||\, \vec{b}$

$\therefore$ $\overrightarrow{AP} = \lambda\vec{b}$ for some scalar λ.

$\Rightarrow$ $\vec{r} - \vec{a} = \lambda\vec{b}$

$\Rightarrow$ $\vec{r} = \vec{a} + \lambda\vec{b}$



Since, every point on the line passing through *A* and parallel to $\vec{b}$ satisfied the above equation and for each value of λ the above equation gives the position vector of a point *P* on the line. Hence, the vector equation of the line is $\vec{r} = \vec{a} + \lambda\vec{b}$.

Non-Parametric Form

The vector equation of a line passing through a point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$ or $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$.

Proof : We have, $\overrightarrow{AP} \,||\, \vec{b}$

$\Rightarrow \quad \overrightarrow{AP} \times \vec{b} = \vec{0}$

$\Rightarrow \quad (\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$

$\Rightarrow \quad \vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

Hence, the vector equation of the line in non-parametric form is $(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$ or $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

7. VECTOR EQUATIONS OF A LINE PASSING THROUGH TWO POINTS

Parametric From

The vector equation of a line passing through two given points having position vector $\vec{a}$ and $\vec{b}$ is

$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$ or $\vec{r} = \vec{b} + \lambda(\vec{b} - \vec{a})$

Proof : Let *O* be the origin and let *A* and *B* be the points with position vectors $\vec{a}$ and $\vec{b}$ respectively. Then,

$\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$

Let $\vec{r}$ be the position vector of any point *P* on the line through *A* and *B* as shown in figure. Then

$\overrightarrow{OP} = \vec{r}$

Clearly, $\overrightarrow{AP}$ and $\overrightarrow{AB}$ are collinear vectors.

$\therefore \quad \overrightarrow{AP} = \lambda\overrightarrow{AB}$ for some scalar λ.

$\Rightarrow \quad \vec{r} - \vec{a} = \lambda(\vec{b} - \vec{a})$

$\Rightarrow \quad \vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

Since, every pont on the line satisfies the above equation for each value of λ and the above equation gives the position vector of a point *P* on the line. Hence, the vector equation of the line is

$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

Non-Parametric Form

The vector equation of a line passing through two given points with position vectors $\vec{a}$ and $\vec{b}$ is

$(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = \vec{0}$ or $\vec{r} \times (\vec{b} - \vec{a}) = \vec{a} \times \vec{b}$

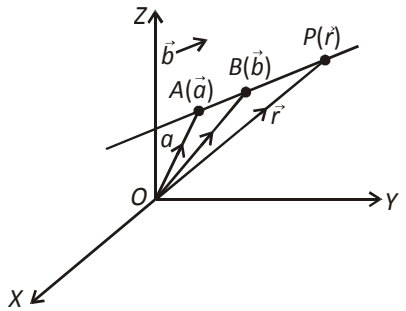
Proof : In the given figure, we have

$\overrightarrow{AP} = \overrightarrow{AB}$

$\Rightarrow \quad \overrightarrow{AP} \times \overrightarrow{AB} = \vec{0}$

$\Rightarrow \quad (\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = 0$

$\Rightarrow \quad \vec{r} \times (\vec{b} - \vec{a}) = \vec{a} \times \vec{b}$



Cartesian Form

The equation of a line passing through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$

8. PREPENDICULAR DISTANCE OF A POINT FROM A LINE

The length of the perpendicular from a point $P(\vec{a})$ on the line $\vec{r}=\vec{b}+\mu\vec{c}$ is given by

$$\sqrt{|\vec{a}-\vec{b}|^2-\{(\vec{a}-\vec{b}).\hat{c}\}^2}=\sqrt{|\vec{a}-\vec{b}|^2-\left\{\frac{(\vec{a}-\vec{b}).\vec{c}}{|\vec{c}|}\right\}^2}$$

Proof : The line $\vec{r}=\vec{b}+\mu\vec{c}$ passes through point $A(\vec{b})$ and is parallel to $\vec{b}$. Let *PM* be perpendicular from *P* on the given line.

Clearly, *AM* = Projection of $\overrightarrow{AP}$ on $\vec{c}$

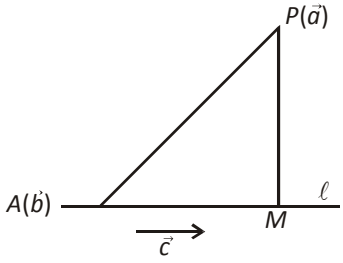
$$=\left(\overrightarrow{AP}.\hat{c}\right)=(\vec{a}-\vec{b}).\hat{c}$$

In right angled triangle *AMP*, we have

$AP^2 = AM^2 + PM^2$

$$\Rightarrow \quad PM=\sqrt{AP^2-AM^2}$$

$$\Rightarrow \quad \sqrt{|\vec{a}-\vec{b}|^2-\{(\vec{a}-\vec{b}).\hat{c}\}^2}=\sqrt{|\vec{a}-\vec{b}|^2-\left\{\frac{(\vec{a}-\vec{b}).\vec{c}}{|\vec{c}|}\right\}^2}$$



Note :- PM is also equal to $\left|\frac{(\vec{b}-\vec{a})\times\vec{c}}{|\vec{c}|}\right|$.

***Illustration 6*:** The perpendicular distance of a corner of a unit cube from a diagonal not passing through it, is

(a) $\sqrt{\frac{2}{3}}$ (b) $\frac{2}{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{3}{2}$

Solution : **(a)** Let the edges of *OA, OB, OC* of the unit cube, be along *OX, OY* and *OZ* respectively. Since,

OA = *OB* = *OC* = 1 unit. Therefore,

$$\overrightarrow{OA}=\hat{i},\overrightarrow{OB}=\hat{j},\overrightarrow{OC}=\hat{k}$$

Let *CM* be the perpendicular from the corner *C* on the diagonal *OP*.

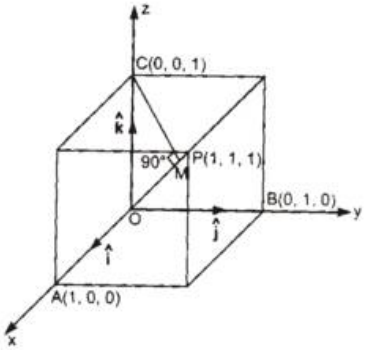
The vector equation of *OP* is $\vec{r}=\lambda(\hat{i}+\hat{j}+\hat{k})$.

$\therefore$ OM = Projection of $\overrightarrow{OC}$ on $\overrightarrow{OP}$

$$=\overrightarrow{OC}\,.\,\overrightarrow{OP}$$

$$=\hat{k}.\frac{(\hat{i}+\hat{j}+\hat{k})}{\sqrt{3}}=\frac{1}{\sqrt{3}}$$

Now, $OC^2 = OM^2 + CM^2$



$$\Rightarrow \quad CM^2 = OC^2 - OM^2 = |\overrightarrow{OC}|^2 - OM^2 = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\Rightarrow \quad CM = \sqrt{\frac{2}{3}}$$

***Illustration 7*:** The perpendicular distance of $P(1, 2, 3)$ from the lines $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ is

(a) 7 (b) 5 (c) 0 (d) 10

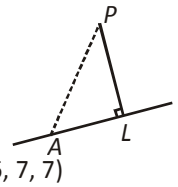
Solution : **(a)** The ponit $A(6, 7, 7)$ is on the line. Let the perpendicular from P meet the line in L. Then,

$AP^2 = (6 - 1)^2 + (7 - 2)^2 + (7 - 3)^2 = 56$

Also, AL = projection of AP on line

$$\left(\text{actual DC's are } \frac{3}{\sqrt{17}}, \frac{2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}\right)$$

$$= (6-1).\frac{3}{\sqrt{17}} + (7-2).\frac{2}{\sqrt{17}} + (7-3).\left(\frac{-2}{\sqrt{17}}\right) = \sqrt{17}$$



$\therefore$ Perpendicular distance d of P from the line is given by

$d^2 = AP^2 - AL^2 = 66 - 17 = 49 \quad \Rightarrow \quad d = 7$

9. ANGLE BETWEEN TWO LINES

Vector Form

The angle between the two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ is given by $\cos\theta = \frac{\vec{b}_1.\vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$

Cartesian Form

The angle between the two lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ is given by

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

***Illustration 8*:** The angle between the straight lines $\frac{x+1}{2} = \frac{y-2}{5} = \frac{z+3}{4}$ and $\frac{x-1}{1} = \frac{y+2}{2} = \frac{z-3}{-3}$ is

(a) 45° (b) 30° (c) 60° (d) 90°

Solution : **(d)** The angle between the given lines is $\cos\theta = \frac{(2)(1)+(5)(2)+(4)(-3)}{\sqrt{4+25+16}\sqrt{1+4+9}} = 0$

$\therefore \quad \theta = 90^o$

Condition of coplanarity of Two lines

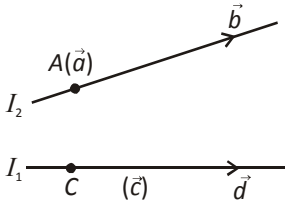
Two lines $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{c} + \mu\vec{d}$ are coplanar or intersecting, if

$$(\vec{a} - \vec{c}).(\vec{b} \times \vec{d}) = 0 \qquad \Rightarrow \qquad [\vec{a}\ \ \vec{b}\ \ \vec{c}] = [\vec{c}\ \ \vec{b}\ \ \vec{d}]$$

Proof : The line $\vec{r} = \vec{a} + \lambda\vec{b}$, passes through the point $A(\vec{a})$ and is parallel to $\vec{b}$.

The line $\vec{r} = \vec{c} + \mu\vec{d}$ passes through the point $C(\vec{c})$ and is parellel to $\vec{d}$.

If these two lines lie in the same plane, then $\overrightarrow{AC}$ lies in the same plane and $\vec{b} \times \vec{d}$ is normal to the plane. Therefore



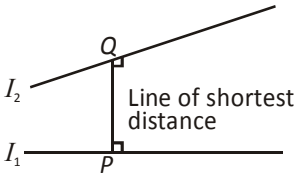
$$\overrightarrow{AC}.(\vec{b} \times \vec{d}) = 0$$

$$\Rightarrow \quad (\vec{c} - \vec{a}).(\vec{b} \times \vec{d}) = 0$$

$$\Rightarrow \quad [\vec{c}\ \ \vec{b}\ \ \vec{d}] = [\vec{a}\ \ \vec{b}\ \ \vec{d}]$$

10. SHORTEST DISTANCE BETWEEN TWO SKEW-LINES

Skew-Lines : Two straight lines in space which are neither parallel nor intersecting are called skew-lines.
Thus, skew-lines are those lines which do not lie in the same plane.



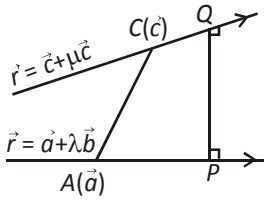
Line of shortest distance : If ℓ_1 and ℓ_2 are two skew-lines, then there is one and only one line perpendicular to each of the lines ℓ_1 and ℓ_2 which is known as the line of shortest distance.

Shortest distance : The shortest distance between two lines ℓ_1 and ℓ_2 is the distance *PQ* between the points *P* and *Q* where the line of shortest distance intersects the two given lines.

The shortest distance between two skew-lines $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{c} + \mu\vec{d}$ is given by

$$SD = \left| \frac{(\vec{c} \times \vec{a}).(\vec{b} \times \vec{d})}{|\vec{b} \times \vec{d}|} \right|$$

Proof : Let *PQ* be the shortest distance between the given lines. Then, *PQ* is perpendiuclar to both the lines. So, it is parallel to $\vec{b} \times \vec{d}$.



Clearly, *PQ* is the projection of $\overrightarrow{AC}$ on $\overrightarrow{PQ}$. Therefore,

$$PQ = \overrightarrow{AC}.\overrightarrow{PQ}$$

$$= \left| (\vec{c} - \vec{a}).\frac{(\vec{b} \times \vec{d})}{|\vec{b} \times \vec{d}|} \right| = \left| \frac{(\vec{c} - \vec{a}).(\vec{b} \times \vec{d})}{|\vec{b} \times \vec{d}|} \right|$$

***Illustration 9*:** The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y-2}{-4} = \frac{z-3}{-6}$ is

(a) $\sqrt{30}$ (b) $2\sqrt{30}$ (c) $5\sqrt{30}$ (d) $3\sqrt{30}$

Solution : **(d)** Given lines can be written as $\vec{r} = \vec{a}_1 + t\vec{b}_1$ and $\vec{r} = \vec{a}_2 + s\vec{b}_2$

where, $\vec{a}_1 = 3\hat{i} + 8\hat{j} + 3\hat{k}, \vec{a}_2 = -3\hat{i} - 7\hat{j} + 6\hat{k}, \vec{b}_1 = 3\hat{i} - \hat{j} + \hat{k}$ and $\vec{b}_2 = -3\hat{i} + 2\hat{j} + 4\hat{k}$

Required *SD* = $\frac{|(\vec{a}_2 - \vec{a}_1).(\vec{b}_1, \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

11. EQUATION OF A PLANE

Plane : It is a surface such that if any two points on it are taken, then every point of the line joining them lies on it.

General equation of a plane : The general equation of a plane is $ax + by + cz + d = 0$, where a, b, c are not all zero.

Theorem : The vector equation of a plane passing through a point having position vector $\vec{a}$ and normal to the vector $\vec{n}$ is $(\vec{r} - \vec{a}).\vec{n} = \vec{0}$ or, $\vec{r}.\vec{n} = \vec{a}.\vec{n}$

Proof : Let $P(\vec{r})$ be an arbitrary point on the plane. The plane passes through a point $A(\vec{a})$.

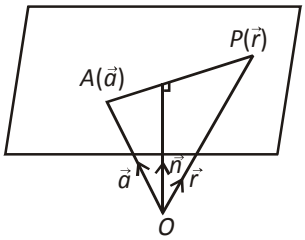
$\therefore \quad \overrightarrow{AP} = \vec{r} - \vec{a}$

Since, the plane is perpendicular to $\vec{n}$. Therefore,

$\overrightarrow{AP} \perp \vec{n} \quad \Rightarrow \quad \overrightarrow{AP}.\vec{n} = 0$

$\Rightarrow \quad (\vec{r} - \vec{a}).\vec{n} = 0 \quad \Rightarrow \quad \vec{r}.\vec{n} = \vec{a}.\vec{n}$

This is the required equation of the plane.



Equation of a plane in intercept form : If a plane makes intercepts of lengths a, b, c with x-axis, y-axis and z-axis respectively, then the equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

***Illustration 10*:** A plane meets the coordinate axes at A, B, C such the centroid of the ΔABC is the point (a, b, c). The equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = k$, where k is equal to

(a) 1 (b) 2 (c) 3 (d) None of these

Solution : **(c)** Let the equation of the plane be $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$. Then, points A, B, C are $(\alpha, 0, 0)$, $(0, \beta, 0)$ and $(0, 0, \gamma)$ respectively. Now, (a, b, c) are the coordinates of centroid of ΔABC.

$\therefore \quad a = \frac{\alpha + 0 + 0}{3}, b = \frac{0 + \beta + 0}{3}, c = \frac{0 + 0 + \gamma}{3}$

$\Rightarrow \quad \alpha = 3a, \beta = 3b, \gamma = 3c$

$$\frac{x}{3a} + \frac{y}{3b} + \frac{z}{3c} = 1$$

$$\Rightarrow \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$$

$\Rightarrow \quad k = 3$

Equation of a plane passing through three point

Vector Form : The equation of a plane passing through three points having position vectors $\vec{r}_1, \vec{r}_2$ and $\vec{r}_3$ is $[\vec{r} - \vec{r}_1 \ \ \vec{r} - \vec{r}_2 \ \ \vec{r} - \vec{r}_3] = 0$.

Cartesian form : The equation of a plane passing through three given points $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ is $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x - x_2 & y - y_2 & z - z_2 \\ x - x_3 & y - y_3 & z - z_3 \end{vmatrix} = 0$

***Illustration 11*:** The equation of the plane through the points (1, 1, 0), (–2, 2, –1) and (1, 2, 1) is
(a) $2x + 3y + 3z = 5$ (b) $2x – 3y + 3z = 5$ (c) $2x + 3y – 3z = 3$ (d) $2x + 3y – 3z = 5$

Solution : **(d)** A plane through (1, 1, 0) is $a(x – 1) + b(y – 1) + cz = 0$
It passes through (–2, 2, –1) and (1, 2, 1), hence $–3a + b – c = 0$ and $a + b + c = 0$

$$\Rightarrow \quad \frac{a}{1+1}=\frac{b}{3}=\frac{c}{-3}$$

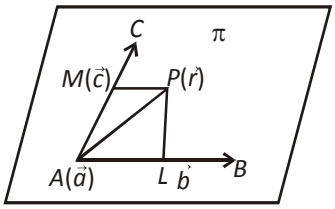
$$\Rightarrow \quad \frac{a}{2}=\frac{b}{3}=\frac{c}{-3}$$

Therefore, the plane is $2(x – 1) + 3(y – 1) – 3z = 0$
$\Rightarrow \quad 2x + 3y – 3z = 5$

Equation of plane passing through a given point and parallel to two given vectors

Parametric form : The vector equation of a plane passing through a point having position vector $\vec{a}$ and parallel to vector $\vec{b}$ and $\vec{c}$ is $\vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$, where λ and μ are scalar.

Proof : Let *O* be the origin and π be a plane passing through *A* point *A* having position vector $\vec{a}$. Let the plane π be parallel to $\vec{b}$ and $\vec{c}$ respectively. through the point *A* draw two vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ such that $\overrightarrow{AB}=\vec{b}$ and $\overrightarrow{AC}=\vec{c}$. Let *P* be an arbitrary point on the plane. Let the position vector of *P* be $\vec{r}$. Complete the parallelogram *ALPM*. Since, $\overrightarrow{AL}$ and $\overrightarrow{AM}$ are parallel to $\vec{b}$ and $\vec{c}$ respectively. Therfore, $\overrightarrow{AL}=\lambda\vec{b}$ and $\overrightarrow{AM}=\mu\vec{c}$



In ΔAPL, we have

$$\overrightarrow{AP}=\overrightarrow{AL}+\overrightarrow{LP}$$

$$\Rightarrow \quad \vec{r}-\vec{a}=\lambda\vec{b}+\mu\vec{c}$$

$$\Rightarrow \quad \vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$$

Since, $\vec{r}$ denotes the positive vector of any point *P* on the plane. Therefore, the equation of the plane is $\vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$, where λ and μ are scalar.

Note :- In the above equation λ amd μ are variable scalar, because for different points on the plane the values of λ and μ are different.

That is why, it is called the parametric form of the plane passing through a given point and parallel to given vectors.

Non-parametric form : The equation of the plane passing through a point having position vector $\vec{a}$ and parallel to vectors $\vec{b}$ and $\vec{c}$ is $(\vec{r}-\vec{a}).(\vec{b}\times\vec{c})=0$ or $\vec{r}.(\vec{b}\times\vec{c})=\vec{a}.(\vec{b}\times\vec{c})$

or $\quad [\vec{r}\ \ \vec{b}\ \ \vec{c}]=[\vec{a}\ \ \vec{b}\ \ \vec{c}]$

Proof : From previous figure, the plane is parallel to vectors $\overrightarrow{AB}=\vec{b}$ and $\overrightarrow{AC}=\vec{c}$. Therefore, it is perpendicular to the vector $\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\vec{b}\times\vec{c}$.

Since, the plane passes through a point *A* having position vector $\vec{a}$. Therefore, the equation of the plane in scalar product form is

$$(\vec{r}-\vec{a}).(\vec{b}\times\vec{c})=0 \qquad [\text{Using } (\vec{r}\times\vec{a})\vec{n}=0]$$

$$\Rightarrow \quad \vec{r}.(\vec{b}\times\vec{c})-\vec{a}.(\vec{b}\times\vec{c})=0 \qquad \Rightarrow \quad \vec{r}.(\vec{b}\times\vec{c})=\vec{a}.(\vec{b}\times\vec{c})$$

$$\Rightarrow \quad [\vec{r}\ \ \vec{b}\ \ \vec{c}]=[\vec{a}\ \ \vec{b}\ \ \vec{c}]$$

***Illustration 12*:** The equation of the plane through the points (2, 3, 1) and (4, –5, 3) and parallel to *x*-axis, is

(a) $x - z - 1 = 0$ (b) $4x + y - 11 = 0$ (c) $y + 4z - 7 = 0$ (d) None of these

Solution : **(c)** Any plane parallel to *x*-axis is $y + 4z - 7 = 0$. If it passes through (2, 3, 1) and (4, –5, 3), then

$3b + c + d = 0$ and $-5b + 3c + d = 0$

$$\Rightarrow \quad \frac{b}{1-3} = \frac{c}{-5-3} = \frac{d}{9+5}$$

$$\Rightarrow \quad \frac{b}{1} = \frac{c}{4} = \frac{d}{-7}$$

Therefore, the plane parallel to *x*-axis is $y + 4z - 7 = 0$.

12. DISTANCE OF A POINT FROM A PLANE

Vector form : The length of the perpendicular from a point having position vector $\vec{a}$ to the plane $\vec{r}.\vec{n} = d$ is given by

$$p = \frac{|\vec{a}.\vec{n} - d|}{|\vec{n}|}$$

Proof : Let π be the given plane of $P(\vec{a})$ be the given point. Let *PM* be the length of the perpendicular from *P* on the plane π. Since, line *PM* passes through $P(\vec{a})$ and is parallel to the vector $\vec{n}$ which is normal to the plane π. So vector equation of line *PM* is

$\vec{r} = \vec{a} + \lambda\vec{n}$, where λ is a scalar ....(i)

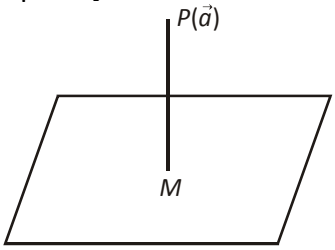
Point *M* is the intersection of line (i) and the given plane. Therefore, for point *M*

$(\vec{a} + \lambda\vec{n}).\vec{n} = d$ [On substituting $\vec{r} = \vec{a} + \lambda\vec{n}$ in the equation of the plane]

$$\Rightarrow \quad \vec{a}.\vec{n} + \lambda\vec{n}.\vec{n} = d$$

$$\lambda = \frac{d - (\vec{a}.\vec{n})}{\vec{n}.\vec{n}}$$

$$\Rightarrow \quad \lambda = \frac{d - (\vec{a}.\vec{n})}{|\vec{n}|}$$



On putting this value of λ in Eq. (i), we obtain the position vector of *M* given by

$$\vec{r} = \vec{a} + \left\{\frac{d - (\vec{a}.\hat{n})}{|\vec{n}|}\right\}\vec{n}$$

$$\Rightarrow \quad \overrightarrow{PM} = PV \text{ of } M - PV \text{ of } P$$

$$\Rightarrow \quad \overrightarrow{PM} = \vec{a} + \left\{\frac{d(\vec{a}.\vec{n})}{|\vec{n}|^2}\right\}\vec{n} - \vec{a}$$

$$\Rightarrow \quad \overrightarrow{PM} = \frac{\{\vec{d} - (\vec{a}.\vec{n})\vec{n}\}}{|\vec{n}|^2}$$

$$\Rightarrow \quad \overrightarrow{PM} = \frac{|\vec{d} - (\vec{a}.\vec{n})|\,|\vec{n}|}{|\vec{n}|^2} = \frac{|(\vec{a}.\vec{n}) - d|}{|\vec{n}|}$$

***Illustration 13*:** The distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ and the palne $\vec{a}.(\hat{i} + 5\hat{j} + \hat{k}) = 5$ is

(a) $\frac{10}{9}$ (b) $\frac{10}{3\sqrt{3}}$ (c) $\frac{10}{3}$ (d) None of these

Solution : **(b)** The given line is $\vec{r} = \vec{a} + t\vec{b}$, where $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}, \vec{b} = \hat{i} - \hat{j} + 4\hat{k}$ and given planes is $\vec{r}.\vec{n} = \vec{d}$, where $\vec{n} = \hat{i} + 5\hat{j} + \hat{k}, d = 5$.

Since, $\vec{b}.\vec{n} = 1 - 5 + 4 = 0$

$\therefore$ Given line is parallel to the given plane.

$\therefore$ The distance between the line and the plane is equal to length of the prependicular from the point $\vec{a} = 2\hat{i} - 2\hat{j} + 3\hat{k}$ on the line to given plane.

$\therefore$ Required distance = $\left|\frac{(2\hat{i} - 2\hat{j} + 3\hat{k}).(\hat{i} + 5\hat{j} + \hat{k}) - 5}{\sqrt{1 + 25 + 1}}\right|$

$$= \left|\frac{2 - 10 + 3 - 5}{\sqrt{27}}\right| = \frac{10}{3\sqrt{3}}$$

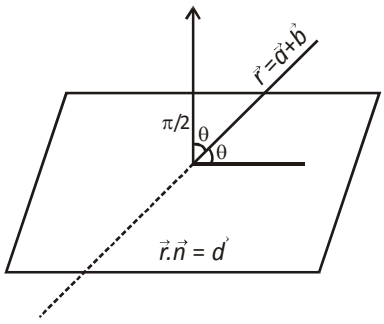
13. ANGLE BETWEEN A LINE AND A PLANE

The angle between a line and a plane is the complement of the angle between the line and normal to the plane.

Vector form : If θ is the angle between a line $\vec{r} = (\vec{a} + \lambda\vec{b})$ and the plane $\vec{r}.\vec{n} = d$, then

$$\sin\theta = \frac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$$

Proof : The line $\vec{r} = \vec{a} + \lambda\vec{b}$ is parallel to the vector $\vec{b}$ and the plane is normal to the vector $\vec{n}$. Therefore, if θ is the angle between the given line and given plane, then the angle between $\vec{b}$ and $\vec{a}$ is $\left(\frac{\pi}{2} - \theta\right)$



So, $\cos\left(\frac{\pi}{2} - \theta\right) = \frac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$

$\Rightarrow$ $\sin\theta = \frac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$

Condition of Perpendicularity : If the line is perpendicular to the plane, then it is parallel to the normal to the plane. Therefore, $\vec{b}$ and $\vec{n}$ are parallel.

So, $\vec{b} \times \vec{n} = 0$ or $\vec{b} = \lambda\vec{n}$ for some scalar λ.

Condition of parallelism : If the line is parallel to the plane, then it is perpendicular to the normal to the plane. Therefore, $\vec{b}$ and $\vec{n}$ are perpendicular. So, $\vec{b}.\vec{n} = 0$.

14. ANGLE BETWEEN TWO PLANES

The angle between two planes $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$ is defined as the angle between their normals. Thus, if θ is the angle between these two planes, then

$$\cos\theta = \frac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

If the planes are perpendicular, then $\vec{n}_1.\vec{n}_2 = 0$. If the planes are parallel, then $\vec{n}_1$ and $\vec{n}_2$ are parallel.

***Illustration 14*:** The angle between the planes $\vec{r}.(3\hat{i}+\hat{j}-\hat{k})$ = 1 and $\vec{r}.(\hat{i}+4\hat{j}-2\hat{k})$ = 2 is

(a) $\cos^{-1}\left(\frac{9}{\sqrt{231}}\right)$ (b) $\cos^{-1}\left(\frac{4}{\sqrt{231}}\right)$ (c) $\cos^{-1}\left(\frac{11}{\sqrt{231}}\right)$ (d) None of these

Solution : **(a)** Equations of planes are

$\vec{r}.\vec{n}_1 = 1$ where $\vec{n}_1 = 3\hat{i}+\hat{j}-\hat{k}$

and $\vec{r}.\vec{n}_2 = 2$ where $\vec{n}_2 = \hat{i}+4\hat{j}-2\hat{k}$

If θ is the angle between the planes, then

$$\cos\theta = \frac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$= \frac{(3)(1)+(1)(4)+(-1)(-2)}{\sqrt{3^2+1^2+(-1)^2}\sqrt{(1)^2+4^2+(-2)^2}} = \frac{3+4+2}{\sqrt{11}\sqrt{21}} = \frac{9}{\sqrt{239}}$$

$$\therefore \quad \theta = \cos^{-1}\left(\frac{9}{\sqrt{231}}\right)$$

***Illustration 15*:** If the planes $\vec{r}.(2\hat{i}+\lambda\hat{j}-3\hat{k}) = 0$ and $\vec{r}.(\lambda\hat{i}+3\hat{j}+\hat{k}) = 5$ are perpendicular, then λ is equal to

(a) 2 (b) –2 (c) 3 (d) –3

Solution : **(d)** When two planes $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$ are perpendicular, then

$$\vec{n}_1.\vec{n}_2 = 0$$

Here, $\vec{n}_1 = 2\hat{i}+\lambda\hat{j}-3\hat{k}$ and $\vec{n}_2 = \lambda\hat{i}-3\hat{j}+\hat{k}$

$\therefore \quad (2\hat{i}+\lambda\hat{j}-3\hat{k}).(\lambda\hat{i}-3\hat{j}+\hat{k}) = 0$

$\Rightarrow$ $2\lambda + \lambda(-3) + (-3)(1) = 0$

$\Rightarrow$ $2\lambda - 3\lambda - 3 = 0 \Rightarrow -\lambda - 3 = 0$

$\Rightarrow$ $\lambda = -3$

***Illustration 16*:** If the planes $\vec{r}.(2\hat{i}-\lambda\hat{j}+\hat{k}) = 3$ and $\vec{r}.(4\hat{i}+\hat{j}-\mu\hat{k}) = 5$ are parallel, then the values of λ and μ are

(a) 1/2, –2 (b) –1/2, 2 (c) –1/2, –2 (d) 1/2, 2

Solution : **(c)**

Two planes $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$ are parallel, and $\vec{n}_1$ and $\vec{n}_2$ are parallel, ie, if $\vec{n}_1 = t\vec{n}_2$

$\Rightarrow$ $2\hat{i}-\lambda\hat{j}+\hat{k} = t(4\hat{i}+\hat{j}-\mu\hat{k})$

On comparing the coeficients of $\hat{i}$, $\hat{j}$ and $\hat{k}$, we get $2 = 4t$, $-\lambda = t$ and $1 = -\mu t$

Which given $t = \frac{1}{2}$, $\lambda = -\frac{1}{2}$ and $\mu = -2$

Therefore, $\lambda = -\frac{1}{2}$ and $\mu = -2$

15. EQUATION OF A PLANE PASSING THROUGH THE INTERSECTION OF TWO PLANES

The intersection of two planes is always a straight lines. For example *xy*-plane and *xz*-plane intersect to form *x*-axis. The plane containing the line of intersection of two given planes is known as the plane passing through the intersection of two given planes. In the following discussion we will obtain the equation of family of planes passing through the intersection of two given planes.

Vector Form : The equation of a plane passing through the intersection of the planes $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$ is

$$(\vec{r}.\vec{n}_1 - \vec{d}_1) + \lambda(\vec{r}.\vec{n}_2 - \vec{d}_2) = 0$$

or $\vec{r}.(\vec{n}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$ where λ is an arbitrary constant.

Proof : The equation $\vec{r}.(\vec{n}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$ is of the form $\vec{r}.\vec{n} = d$, so it represents a plane. In order to prove that it represents a plane, it is sufficient to show that every point on the lines of intersection of the plane $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$ lies on $\vec{r}.(\vec{n}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$.

Let $\vec{r}_1$ be the position vector of any point on the line of intersection of $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$. then,

$\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$

$\Rightarrow$ $\vec{r}_1.\vec{n}_1 = 0$ and $\vec{r}_1.\vec{n}_2 - d_2 = 0$

$\Rightarrow$ $(\vec{r}_1.\vec{n}_1 - d_1) + \lambda(\vec{r}_1.\vec{n}_2 - d_2) = 0$

$\Rightarrow$ $\vec{r}_1.(\vec{n}_1 + \lambda\vec{n}_2) = \vec{d}_1 + \lambda d_2$

$\Rightarrow$ $\vec{r}_1$ lies on the plane $\vec{r}.(\vec{n}_1 + \lambda\vec{n}_2) = \vec{d}_1 + \lambda d_2$

Thus $\vec{r}.(\vec{r}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$ represents a plane passing through the intersection of $\vec{r}.\vec{n}_1 = d_1$ and $\vec{r}.\vec{n}_2 = d_2$

***Illustration 17*:** The vector equation of the line intersection of the planes, $\vec{r} = \vec{b} + \lambda_1(\vec{b} - \vec{a}) + \mu_1(\vec{a} + \vec{c})$ and $\vec{r} = \vec{c} + \lambda_2(\vec{b} - \vec{c}) + \mu_2(\vec{a} + \vec{b})$, $\vec{a}, \vec{b}, \vec{c}$ being non coplanar vectors, is

(a) $\vec{r} = \vec{b} + \mu_1(\vec{a} + \vec{c})$ (b) $\vec{r} = \vec{b} + \lambda_1(\vec{a} - \vec{c})$

(c) $\vec{r} = 2\vec{b} + \lambda_2(\vec{a} - \vec{c})$ (d) None of these

Solution : **(a)** At the point of intersection of the two given planes, we have

$\vec{b} + \lambda(\vec{b} - \vec{a}) + \mu_1(\vec{a} + \vec{c}) = \vec{c} + \lambda_2(\vec{b} - \vec{c}) + \mu_2(\vec{a} + \vec{b})$

$\Rightarrow$ $(-\lambda_1 + \mu_1 - \mu_2)\vec{a} + (1 + -\lambda_1 + \lambda_2 - \mu_2)\vec{b} + (-\mu_1 + 1 + \lambda_2)\vec{c} = \vec{0}$

$\Rightarrow$ $-\lambda_1 + \mu_1 - \mu_2 = 0$, $1 + \lambda_2 - \lambda_2 - \mu_2 = 0$,

and $\mu_1 - 1 - \lambda_2 = 0$ $[\because \vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors]

From the last two equations, we get $\lambda_1 + \mu_1 - \mu_2 = 0$
On solving this equation with $-\lambda_1 + \mu_1 - \mu_2 = 0$, we get $\mu_1 = \mu_2$
and $\lambda_1 = 0$
$\therefore \quad \lambda_1 = 0, \mu_1 = \mu_2$ and $\lambda_2 = 1 - \mu_1$
On substituting these values in either of the given equations, we obtain

$$\vec{r} = \vec{b} + \mu_1(\vec{a} + \vec{c})$$

as the required line of intersection of the given planes.

16. PLANE BISECTING THE ANGLES BETWEEN TWO PLANES

Cartesian form : The equations of the planes bisecting the angles between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are

$$\frac{(a_1x + b_1y + c_1z + d_1)}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{(a_2x + b_2y + c_2z + d_2)}{\sqrt{a_2^2 + b_2^2 + c_2^2}} \qquad \text{....(i)}$$

Bisector of acute / obtuse angle :

(a) Write the equations of the given planes such that their constant terms are positive.
(b) If $a_1a_2 + b_1b_2 + c_1c_2 > 0$, then origin lies in obtuse angle and hence positive sign in eq. (i) gives the bisector of the obtuse angle.
(c) If $a_1a_2 + b_1b_2 + c_1c_2 < 0$, then origin lies in acute angle and hence positive sign in eq. (i) gives the bisector of the acute angle.

***Illustration 18*:** The equation of the plane through (1, 1, 1) and passing through the line of intersection of the planes $x + 2y - z + 1 = 0$ and $3x - y - 4z + 3 = 0$ is

(a) $8x + 5y - 11z + 8 = 0$ (b) $8x + 5y + 11z + 8 = 0$
(c) $8x - 5y - 11z + 8 = 0$ (d) None of these

Solution : **(c)** Any plane through the line is $(x + 2y - z + 1) + \lambda(3x - y - 4z + 3) = 0$
It passes through (1, 1, 1).
$\therefore \quad (1 + 2 - 1 + 1) + \lambda(3 - 1 - 4 + 3) = 0$
$\Rightarrow \quad \lambda = -3$
Therefore, the required plane is $8x - 5y - 11z + 8 = 0$

***Illustration 19*:** The equation of the plane passing through the line of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z = 8$, is

(a) $2x + 17y + 9z = 0$ (b) $28x - 17y + 9z = 0$
(c) $2x + 17y - 9z = 0$ (d) None of these

Solution : **(b)** Any plane through the line is $(2x - y) + \lambda(3z - y) = 0$ ....(i)
The given plane is $4x + 5y - 3z - 8 = 0$ ....(ii)
Since, eq. (i) and (ii) are perpendicular to each other, we have $2.\ 4 - (1 + \lambda).5 + 3\lambda(-3) = 0$
$\Rightarrow \quad \lambda = 3/14$
On substituting the value of λ in eq. (i), we get
$14(2x - y) + 3(3z - y) = 0$
$\Rightarrow \quad 28x - 17y + 9z = 0$
which is the required plane.

***Illustration 20*:** The sine of the angle between the st. line $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ and the plane 2x – 2y + z = 5 is

(a) $\frac{10}{6\sqrt{5}}$ (b) $\frac{4}{5\sqrt{2}}$ (c) $\frac{\sqrt{2}}{10}$ (d) $\frac{2\sqrt{3}}{5}$

Solution : **(c)** Direction numbers of the given lines are <3, 4, 5>. Direction cosines of the normal to the plane 2x – 2y + z – 5 = 0 are proporational to 2, –2, 1.

If θ is the angle between the two, then

$$\sin\theta=\frac{\left|3\times2+4(-2)+5\times1\right|}{\sqrt{3^2+4^2+5^2}\sqrt{2^2+(-2)^2+1^2}}=\frac{3}{\sqrt{50}\sqrt{9}}=\frac{1}{5\sqrt{2}}=\frac{\sqrt{2}}{10}$$

***Illustration 21*:** The equation of the plane which bisects the line joining (2, 3, 4) and (6, 7, 8) is

(a) x – y – z – 15 = 0 (b) x – y + z – 15 = 0

(c) x + y + z – 15 = 0 (d) x + y + z + 15 = 0

Solution : **(c)** Mid pt. of AB where A is (2, 3, 4), B is (6, 7, 8) is (4, 5, 6) Clearly only (c) satisfies it.

***Illustration 22*:** The equation of the plane through the point (1, 2, 3), (–1, 4, 2) and (3, 1, 1) is

(a) 5x + y + 12z – 23 = 0 (b) 5x + 6y + 2z – 23 = 0

(c) x + 6y + 2z – 13 = 0 (d) x + y + z – 13 = 0

Solution : **(b)** Clearly (1, 2, 3) lies on 5x + 6y + 2z – 23 = 0

Other two point only lie on it.

$\therefore$ required plane is 5x + 6y + 2z – 23 = 0

***Illustration 23*:** The equation of the plane thro' the line of intersection of the planes x + y + z – 6 = 0, 2x + 3y + 4z + 5 = 0 and the point (1, 1, 1) is

(a) 7x – 9y + 8z = 0 (b) 7x + y + 8z = 0

(c) 2x – 2y – 3z = 14 (d) 20x + 23y + 26z – 69 = 0.

Solution : **(d)** Any plane thro' the given two planes is (x + y + z – 6) + K (2x + 3y + 4z + 5) = 0

It passes thro' (1,1,1)

$\therefore$ 1 + 1 + 1 – 6 + K (2 + 3 + 4 + 5) = 0

$$K=\frac{3}{14}$$

$\therefore$ reqd. plane is

$$x+y+z-6+\frac{3}{14}(2x+3y+4z+5)=0$$

$\Rightarrow$ 14x + 14y + 14z – 84 + 6x + 9y + 12z + 15 = 0

$\Rightarrow$ 20x + 23y + 26z – 69 = 0

***Illustration 24*:** The equation of the plane that contains the point (1, –1, 2) and is prependicular to each of the planer 2x + 3y – 2z = 5 and x + 2y – 3z = 8 is

(a) 5x + 4y – z = 7 (b) 5x – 4y + z = 7

(c) –5x + 4y – z = 7 (d) 5x – 4y – z = 7

Solution : **(d)** Any plane thro' (1, –1, 2) is

a(x – 1) + b(y + 1) + c(z – 2) = 0 ... (1)

(1) is $\perp$ to the given planes

$\therefore$ 2a + 3b – 2c = 0 ... (2)

and a + 2b – 2c = 0 ... (3)

$$\therefore\frac{a}{-9+4}=\frac{b}{-2+6}=\frac{c}{4-3}$$

$$\Rightarrow\frac{a}{-5}=\frac{b}{4}=\frac{c}{1}$$

$\therefore$ reqd. plane is –5 (x – 1) + 4 (y + 1) + 1 (z – 2) = 0

$\Rightarrow$ 5x – 4y – z – 7 = 0 i.e., 5x – 4y – z = 7

MISCELLANEOUS EXAMPLES

Example 1: The points $\hat{i}-\hat{j}+3\hat{k}$ and $3\hat{i}+3\hat{j}+3\hat{k}$ are equidistant from the plane $\vec{r}.(5\hat{i}+2\hat{j}-7\hat{k})=0$, then the are

(a) On the same sides of the plane (b) Parallel of the plane

(c) On the opposite sides of the plane (d) None of these

Answer: **(c)** The given plane is $\vec{r}.(5\hat{i}+2\hat{j}-7\hat{k})=-9$

Length of the perpendicular from $\hat{i}-\hat{j}+3\hat{k}$ to it is $\dfrac{-9-(\hat{i}-\hat{j}+3\hat{k}).(5\hat{i}+2\hat{j}-7\hat{k})}{\sqrt{5+4+49}}$

$$=\frac{-9-5+2+21}{\sqrt{78}}=\frac{9}{\sqrt{78}}$$

Length of the perpendicular from $3\hat{i}+3\hat{j}+3\hat{k}$

$$=\frac{-9-(3\hat{i}+3\hat{j}+3\hat{k}).(5\hat{i}+2\hat{j}-7\hat{k})}{\sqrt{78}}=\frac{-9-15-6+21}{\sqrt{78}}=\frac{9}{\sqrt{78}}$$

Thus, the length of the two perpendicular are equal in magnitude but opposite in sign. Hence, they are located on opposite side of the plane.

Example 2: A line L_1 passes through the point $3\hat{i}$ and is parallel to the vector $-\hat{i}+\hat{j}+\hat{k}$ and another line L_2 passes through the point $\hat{i}+\hat{j}$ and is parallel to the vector $\hat{i}+\hat{k}$, then point of intersection of he lines is

(a) $\hat{i}+2\hat{j}+\hat{k}$ (b) $2\hat{i}+\hat{j}+\hat{k}$

(c) $\hat{i}-2\hat{j}-\hat{k}$ (d) $\hat{i}-2\hat{j}+\hat{k}$

Answer: **(b)** Equation of L_1 is $\vec{r}=3\hat{i}+\lambda(-\hat{i}+\hat{j}+\hat{k})=(3-\lambda)\hat{i}+\lambda\hat{j}+\lambda\hat{k}$

Equation of L_2 is $\vec{r}=(\hat{i}+\hat{j})+\mu(\hat{i}+\hat{k})=(1+\mu)\hat{i}+\hat{j}+\mu\hat{k}$

For point of intersection, we get $\lambda = \mu = 1$

$\Rightarrow$ $\vec{r}=2\hat{i}+\hat{j}+\hat{k}$

Example 3: The equation of a plane containing the lines $\vec{r}=\vec{a}_1+t\vec{b}_1$ and $\vec{r}=\vec{a}_2+t\vec{b}_2$, where $[(\vec{a}_2-\vec{a}_1).(\vec{b}_1\times\vec{b}_2)]=1$, is

(a) $[(\vec{r}+\vec{a}_1).(\vec{b}_1\times\vec{b}_2)]=1$ (b) $[(\vec{r}-\vec{b}_1).\vec{a}.(\vec{b}_1\times\vec{b}_2)]=0$

(c) $[(\vec{r}-\vec{a}_1).(\vec{b}_1\times\vec{b}_2)]=0$ (d) None of these

Answer: **(c)** The given line passes through the points with position vectors $\vec{a}_1$ and $\vec{a}_2$ and normal to both the line is $\vec{b}_1\times\vec{b}_2$ (the given condition $[(\vec{a}_2-\vec{a}_1)\vec{b}_1\vec{b}_2]=0$ ensure that the lines are either parallel or intersecting)

Thus, the required equation of the plane is $\vec{r}.(\vec{b}_1\times\vec{b}_2)=\vec{a}_1.(\vec{b}_1\times\vec{b}_2)$

or $[\vec{r}.(\vec{r}-\vec{a}_1).(\vec{b}_1\times\vec{b}_2)]=0$

Example 4: The direction ratio of the line $x - y + z - 5 = 0 = x - 3y - 6$ are

(a) 3, 1, –2 (b) 2, –4, 1

(c) $\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$ (d) $\frac{2}{\sqrt{21}}, \frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}$

Answer: **(a, c)** Let the DR's of line are a, b, c.

As the line is perpendicular to both the planes

$\Rightarrow$ $a - b + c = 0$

$a - 3b + 0 \cdot c = 0$

$$\frac{a}{3} = \frac{b}{1} = \frac{c}{-2}$$

Example 5: The equation of the line $x + y + z - 1 = 0, 4x + 3y - 2z + 2 = 0$ written in the symmetrical form is

(a) $\frac{x+1}{1} = \frac{y-2}{-2} = \frac{z-0}{1}$ (b) $\frac{x}{1} = \frac{x}{-2} = \frac{z-1}{1}$

(c) $\frac{x+1/2}{1} = \frac{y-1}{-2} = \frac{z-1/2}{1}$ (d) $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-2}{2}$

Answer: **(a, b, c, d)** $x + y + z - 1 = 0$

$4x + y - 2z + 2 = 0$

$\therefore$ Direction ratios of the line are (–3, 6, –3).

ie, <1, –2, 1>

Let $z = k$, then $x = k - 1$, $y = 1 - 2k$

ie, $(k - 1, 2 - 2k, k)$ is any point on the line

$\therefore$ (–1, 2, 0), (0, 0, 1), $\left(-\frac{1}{2}, 1, \frac{1}{2}\right)$ and (1, –2, 2) are points on the line.

Example 6: If the plane $x - cy - bz = 0$, $cx - y + az = 0$ and $bx + ay - z = 0$ are through a line, then find the value of $a^2 + b^2 + c^2 + 2abc$.

Answer: Given, planes are

$x - cy - bz = 0$ ....(i)

$cx - y + az = 0$ ....(ii)

$bx + ay - z = 0$ ....(iii)

Equation of plane passing through the line of intersection of the planes (i) and (ii) may be taken as

$(x - cy - bz) + \lambda(cx - y + az) = 0$ ....(iv)

Now, planes (iii) and (iv) are same

$\therefore$ $$\frac{1 + c\lambda}{b} = \frac{-(c + \lambda)}{a} = \frac{-b + a\lambda}{-1}$$

By eliminating λ, we get $a^2 + b^2 + c^2 + 2abc = 1$

Example 7: If the line $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies exactly on the plane $2x - 4y + z = 7$, the value of k is ____.

Answer: The point $(4, 2, k)$ must satisfy the plane.

So, $8 - 8 - k = 0$

$\Rightarrow$ $k = 7$

Here, also $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Example 8: If the shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$ is $\lambda\sqrt{30}$ unit, then the value of λ is _____.

Answer: Given, lines are $\vec{r}=(3,8,3)+\lambda(3,-1,1)$ and $\vec{r}=(-3,-7,6)+\mu(-3,2,4)$

Where λ, μ are parameters.

$$\text{Shortest distance} = \frac{(-3-3,-7-8,6-3).[(3,-1,1)\times(-3,2,4)]}{[(3,-1,1)\times(-3,2,4)]}$$

$$= \frac{(-6,-15,3).(-6,-15,3)}{\sqrt{36+225+9}}$$

$$= \sqrt{270} = 3\sqrt{30} \text{ unit}$$

Example 9: If the perpendicular distance of the point (6, 5, 8) from the y-axis is 5λ unit, then λ is equal to ___.

Answer: Foot of perpendicular from (6, 5, 8) on *y*-axis is (0, 5, 0).

$$\text{Required distance} = \sqrt{(6-0)^2+(5-5)^2+(8-0)^2} = 10 \text{ unit}$$

$\Rightarrow \quad 5\lambda = 10$

$\Rightarrow \quad \lambda = \frac{10}{5} = 2$

Example 10: A parallelopiped is formed by planes drawn through the points (2, 4, 5) and (5, 9, 7) parallel to the coordinate planes. The length of the diagonal of the parallelopiped is ____.

Answer: The length of the edges are given by $a = 5 - 2 = 3$, $b = 9 - 3 = 6$, $c = 7 - 5 = 2$, so length of the diagonal

$$= \sqrt{a^2+b^2+c^2}$$

$$= \sqrt{9+36+4} = 7 \text{ unit}$$

SOME IMPORTANT DEFINITIONS AND RESULTS

Equation of a line in space :

(i) **Vector equation** of a line passing through a point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$.

(ii) **Cartesian equation** of a line passing through a point (x_1, y_1, z_1) and having direction ratios $<a, b, c>$ is $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

(iii) **Vector equation** of a line passing through two points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

(iv) **Cartesian equation** of a line passing through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

Angle between two lines : Let θ be angle between two given lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ and let $<a_1, b_1, c_1>$ be the direction ratios of two given lines then, $\cos\theta = \frac{\vec{b}_1.\vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$ or

$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

(i) Two lines are perpendicular iff $\vec{b}_1.\vec{b}_2 = 0$ or $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

(ii) Two lines are parallel iff $\vec{b}_1 \times \vec{b}_2 = 0$ or $\vec{b}_1 = \lambda\vec{b}_2$ or $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Shortest distance between two skew lines :

(i) Let $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ be two skew lines then shortest distance is given by $d = \left|\frac{(\vec{b}_1 \times \vec{b}_2)(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$

(ii) The shortest distance between two lines ℓ_1 : $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$; $\ell_2 = \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ is given by $d = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_1 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}$

(iii) Two lines intersect (or co-planar) iff $(\vec{b}_1 \times \vec{b}_2).(\vec{a}_2 - \vec{a}_1) = 0$ or $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$

Shortest distance between two parallel lines : Consider two parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}$ then shortest distance between the lines is given by $d = \left|\frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|$.

Equation of a plane :

(i) General equation of the plane in vector form is $\vec{r}.\vec{n} = d$, where $\vec{n}$ is a normal vector to the plane.

(ii) General equation of the plane in Cartesian form is $ax + by + cz + d = 0$, where $<a, b, c>$ are direction ratios of the normal to plane.

(iii) **Vector equation** of the plane passing through a point with position $\vec{a}$ perpendicular to the vector $\vec{n}$ is $(\vec{r}-\vec{a}).\vec{n}=0$.

(iv) **Cartesian equation** of the plane passing through a point (x_1, y_1, z_1) is $A(x-x_1)+B(y-y_1)+C(z-z_1)=0$, where $<A, B, C>$ are the direction ratios of the normal to plane.

(v) **Vector equation** of the plane passing through three points with position vectors $\vec{a},\vec{b}$ and $\vec{c}$ is given by $(\vec{r}-\vec{a}).[(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})]=0$.

(vi) **Cartesian equation** of plane passing through the points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) is given by $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0$.

(vii) **Intercept form :** Equation of a plane which makes intercepts a,b and c with x, y and z axes is given by

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

(viii) **Vector equation** of plane in normal form is $\vec{r}.\hat{n}=p$, where $\hat{n}$ is unit vector along perpendicular from origin and p is distance of plane from origin.

(ix) **Cartesian equation** of plane is normal form is $\ell x + my + nz = p$, where $<\ell, m, n>$ are direction cosines of perpendicular from origin.

(x) **Vector equation** of plane passing through the intersection of two planes $\vec{r}.\vec{n}_1=p_1$ and $\vec{r}.\vec{n}_2=p_2$ is given by $\vec{r}.(\vec{n}_1+\lambda\vec{n}_2)=p_1+\lambda p_2$

(ix) **Cartesian equation** of plane passing through the intersection of two planes $A_1x + B_1y + C_1z+D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$ is given by $A_1x + B_1y + C_1z + D_1 + k(A_2x + B_2y + C_2z + D_2) = 0$, where k is real can be calculated from a given condition.

Angle between two planes :

(i) Angle θ between two planes $\vec{r}.\vec{n}_1=p_1$ and $\vec{r}.\vec{n}_2=p_2$ is given by $\cos\theta = \dfrac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$

(ii) Angle θ between two planes $A_1x + B_1y + C_1z+D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$ is given by

$$\cos\theta = \frac{A_1A_2+B_1B_2+C_1C_2}{\sqrt{A_1^2+B_1^2+C_1^2}\sqrt{A_2^2+B_2^2+C_2^2}}$$

Angle between the line and plane :

(i) Angle θ between the line $\vec{r}=a+\lambda\vec{b}$ and $\vec{r}.\vec{n}=p$ given by $\sin\theta = \dfrac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$

(ii) Angle θ between the line $\dfrac{x-x_1}{a}=\dfrac{y-y_1}{b}=\dfrac{z-z_1}{c}$ and the plane $Ax + By + Cz + D = 0$ is given by

$$\sin\theta=\frac{Aa+Bb+Cc}{\sqrt{A^2+B^2+C^2}\sqrt{a^2+b^2+c^2}}$$

Distance of a point from the plane : The perpendicualr distance of point (x_1, y_1, z_1) from the plane $Ax + By + Cz + D = 0$ is given by $d = \dfrac{|Ax_1+By_1+Cz_1+D|}{\sqrt{A^2+B^2+C^2}}$

General equation of any plane parallel to plane $Ax + By + Cz + D = 0$ is $Ax + By + Cz + k = 0$, where k is real

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Direction Cosines and Direction Ratios

Q.1 A line AB is three-dimensional space makes angles 45° and 120° with the positive x-axis and the positive y-axis respectively. If AB makes an acute angle θ with the positive z-axis, then θ equals
(a) 30° (b) 45°
(c) 60° (d) 75°

Q.2 The projection of a vector on the three coordinate axes are 6, –3, 2 respectively. The direction of the vector are
(a) 6, –3, 2 (b) $\frac{6}{5},-\frac{3}{5},\frac{2}{5}$
(c) $\frac{6}{7},-\frac{3}{7},\frac{2}{7}$ (d) $-\frac{6}{7},-\frac{3}{7},\frac{2}{7}$

Q.3 If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is
(a) $\pi/6$ (b) $\pi/3$
(c) $\pi/4$ (d) $\pi/2$

Q.4 The angle between the lines whose direction cosines are $\left(\frac{\sqrt{3}}{4},\frac{1}{4},\frac{\sqrt{3}}{2}\right)$ and $\left(\frac{\sqrt{3}}{4},\frac{1}{4},\frac{-\sqrt{3}}{2}\right)$ is
(a) π (b) $\pi/2$
(c) $\pi/3$ (d) $\pi/4$

Q.5 The shortest distance between the straight lines through the points $A_1 = (6, 2, 2)$ and $A_2 = (–4, 0, –1)$, tin the directions of (1, –2, 2) and (3, –2, –2) is
(a) 6 (b) 8
(c) 12 (d) 9

Q.6 The perimeter of the triangle with vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1) is
(a) 3 (b) 2
(c) $2\sqrt{2}$ (d) $3\sqrt{2}$

2. Straight line in space and Shortest Distance

Q.7 Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+1}{2}$ lies in the plane $x + 2y + \alpha z + \beta = 0$. Then, (α, β) equals
(a) (6, –17) (b) (–6, 7)
(c) (5, –15) (d) (–5, 15)

Q.8 The line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point $\left(0,\frac{17}{2},-\frac{13}{2}\right)$. Then
(a) $a = 2, b = 8$ (b) $a = 4, b = 6$
(c) $a = 6, b = 4$ (d) $a = 8, b = 2$

Q.9 If the straight lines $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$ inctersect at a point, then the integer k is equal to
(a) –5 (b) 5
(c) 2 (d) –2

Q.10 The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ are perpendicular to each other, if
(a) $aa' + cc' = 1$ (b) $\frac{a}{a'}+\frac{c}{c'}=-1$
(c) $\frac{a}{a'}+\frac{c}{c'}=1$ (d) $aa' + cc' = –1$

Q.11 In DABC the mid points of the sides AB, BC and CA are respectively $(\ell, 0, 0)$, $(0, m, 0)$ and $(0, 0, n)$. Then, $\frac{AB^2+BC^2+CA^2}{\ell^2+m^2+n^2}$ is equal to
(a) 2 (b) 4
(c) 8 (d) 16

Q.12 The equation of the plane containing then line $\frac{x+1}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$ and the point (0, 7, –7) is
(a) $x + y + z = 1$ (b) $x + y + 2 = 0$
(c) $x + y + z = 0$ (d) None of these

3. Equation of a Plan

Q.13 Equation of the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicular to the plane containng the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$ is
(a) $x + 2y – 2z = 0$ (b) $3x + 2y – 2z = 0$
(c) $x – 2y + 2z = 0$ (d) $5x + 2y – 4z = 0$

Q.14 If the disttance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$ is 5, then the foot of the perpendicular from P to the plane is

(a) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ (b) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$

(c) $\left(\frac{1}{3}, \frac{2}{3}, -\frac{10}{3}\right)$ (d) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

Q.15 If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and

$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is

(a) 3 (b) 4
(c) 5 (d) f6

Q.16 Let $P(3, 2, 6)$ be a point in space and Q be a point on the line. Then, the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is

(a) 1/4 (b) –1/4
(c) 1/8 (d) –1/8

Q.17 A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals

(a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

Q.18 A plane passes through (1, –2, 1) and is perpendicular to two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$. Then, the distance of the plane from the point (1, 2, 2) is

(a) 0 (b) 1
(c) $\sqrt{2}$ (d) $2\sqrt{2}$

Q.19 Let L be the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$. If L makes an angle α with the positive x-axis, then cos α equals

(a) $1/\sqrt{3}$ (b) 1/2
(c) 1 (d) $1/\sqrt{2}$

Q.20 The image of the point (–1, 3, 4) in the plane $x - 2y = 0$ is

(a) (15, 11, 4) (b) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$

(c) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$ (d) None of these

Q.21 The points (5, –4, 2) (4, –3, 1), (7, –6, 4) and (8, –7, 5) are the vertices of

(a) A rectangle (b) A square
(c) A parallelogram (d) None of these

Q.22 The image of the piont (1, 2, 3) by the plane, $x + y + z + 3 = 0$ is

(a) (–5, 4, –3) (b) (–5, –4, –3)
(c) (5, –4, 3) (d) (5, 4, 3)

Q.23 If a plane meets the coordinate axes at A, B and C in such a way that the centroid of ΔABC is at the point (1, 2, 3), then equation of the plane is

(a) $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$ (b) $\frac{x}{3} + \frac{y}{6} + \frac{z}{9} = 1$

(c) $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = \frac{1}{3}$ (d) None of these

Q.24 The $\vec{r}^2 - \vec{r}.\vec{c} + h = 0, |\vec{c}| > \sqrt{h}$, represents

(a) Circle (b) Ellipse
(c) Cone (d) Sphere

Section-B
(Previous Year Question)

Q.1 The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line]

(a) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$ (b) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$

(c) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$ (d) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$

Q.2 The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 + m^2 + n^2$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$

Q.3 If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar, then k can have
(a) exactly three values
(b) any value
(c) exactly one value
(d) exactly two values

Q.4 Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z=0$ is
(a) $\frac{5}{2}$ (b) $\frac{7}{2}$
(c) $\frac{9}{2}$ (d) $\frac{3}{2}$

Q.5 An equation of a plane parallel to the plane $x-2y+2z-5=0$ and at a unit distance from the origin is
(a) $x-2y+2z-1=0$ (b) $x-2y+2z+5=0$
(c) $x-2y+2z-3=0$ (d) $x-2y+2z+1=0$

Q.6 If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y+k}{2}=\frac{z}{1}$ intersect, then k is equal to
(a) 9/ 2 (b) 0
(c) -1 (d) 2/ 9

Q.7 If the angle between the line $x=\frac{y-1}{2}=\frac{z-3}{\lambda}$ and the plane $x+2y+3z=4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$ then λ equals
(a) 2/ 5 (b) 5 / 3
(c) 2 / 3 (d) 3/ 2

Q.8 A line AB in three-dimensional space makes angles 45^0 and 120^0 with the positive x-axis and the positive y-axis respectively. If AB makes an acute angle θ with the positive z-axis, then θ equals
(a) 30^0 (b) 45^0
(c) 60^0 (d) 75^0

Q.9 Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lie in the plane $x+3y-\alpha z+\beta=0$. Then (α,β) equals
(a) $(-6,7)$ (b) $(5,-15)$
(c) $(-5,5)$ (d) $(6,-17)$

Q.10 The projections of a vector on the three coordinate axis are 6, -3, 2 respectively. the direction cosines of the vector are
(a) $\frac{6}{5},\frac{-3}{5},\frac{2}{5}$ (b) $\frac{6}{7},\frac{-3}{7},\frac{2}{7}$
(c) $\frac{-6}{7},\frac{-3}{7},\frac{2}{7}$ (d) $6,-3,2$

Q.11 If the straight lines $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-2}{k}=\frac{z-1}{2}$ intersect at a point, then the integer k is equal to
(a) -2 (b) -5
(c) 5 (d) 2

Q.12 the line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point $\left(0,\frac{17}{2},\frac{-13}{2}\right)$.Then
(a) $a=8, b=2$ (b) $a=2, b=8$
(c) $a=4, b=6$ (d) $a=6, b=4$

Q.13 Let L be the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$. If L makes an angle α with the positive x-axis, then cos α equals
(a) 1 (b) $\frac{1}{\sqrt{2}}$
(b) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

Q.14 Let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=x\hat{i}+(x-2)\hat{j}-\hat{k}$. If the vectors $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then x equals
(a) -4 (b) -2
(c) 0 (d) 1

Q.15 If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$

Q.16 The image of the point $(-1,3,4)$ in the 3 plane $x-2y=0$ is
(a) $\left(-\frac{17}{3},-\frac{19}{3},4\right)$ (b) $(15, 11, 4)$
(c) $\left(-\frac{17}{3},-\frac{19}{3},1\right)$ (d) $\left(\frac{9}{5},-\frac{13}{5},4\right)$

Q.17 The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ are perpendicular to each other if

(a) $aa' + cc' = -1$ (b) $aa' + cc' = 1$

(c) $\frac{a}{a'} + \frac{c}{c'} = -1$ (d) $\frac{a}{a'} + \frac{c}{c'} = 1$

Q.18 The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is

(a) 90^0 (b) 0^0

(c) 30^0 (d) 45^0

Q.19 If the angle θ between the lien $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}z + 4 = 0$ is such that $\sin\theta = \frac{1}{3}$, the value of λ is

(a) $-\frac{3}{5}$ (b) $\frac{5}{3}$

(c) $\frac{-4}{3}$ (d) $\frac{3}{4}$

Q.20 The distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda\left(\hat{i} - \hat{j} + 4\hat{k}\right)$ and the plane $\vec{r}.\left(\hat{i} + 5\hat{j} + \hat{k}\right) = 5$ is

(a) $\frac{10}{3\sqrt{3}}$ (b) $\frac{10}{9}$

(c) $\frac{10}{3}$ (d) $\frac{3}{10}$

Q.21 A line with direction consines proportional to 2,1,2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. the co-ordinates of each of the points of intersection are given by

(a) $(3a, 2a, 3a), (a, a, 2a)$

(b) $(3a, 2a, 3a), (a, a, a)$

(c) $(3a, 3a, 3a), (a, a, a)$

(d) $(2a, 3a, 3a), (2a, a, a)$

Q.22 Distance between two parallel planes $3x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is

(a) 7 / 2 (b) 5 / 2

(c) 3 / 2 (d) 9 / 2

Q.23 A line makes the same angle θ, with each of the x and z axis. If the angle β, which it makes with y-axis, is such that $\sin^2\beta = 3\sin^2\theta$, then $\cos^2\theta$ equals

(a) 3/ 5 (b) 1/ 5

(c) 2 / 3 (d) 2 / 5

Q.24 If the straight line $x = 1 + s, y = -3 - \lambda s,\ z = 1 + \lambda s$ and $x = \frac{t}{2}, y = 1 + t, z = 2 - t$, with paramenters s and t respectivley, are coplanar, then λ equals

(a) -1 / 2 (b) -1

(c) -2 (d) 0

Q.25 The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if

(a) $k = 1$ *or* -1 (b) $k = 0$ *or* -3

(c) $k = 3$ *or* -3 (d) $k = 0$ *or* -1

Q.26 The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d$ will be perpendicualr, if and only if

(a) $aa' + bb' + cc' = 0$

(b) $(a + a')(b + b') + (c + c') = 0$

(c) $aa' + cc' + 1 = 0$

(d) $aa' + bb' + cc' + 1 = 0$

Section-C
(Board pattern)

Q.1 Find the coordinates of the foot of the perpendicular drawn from the point *A*(1, 2, 1) to the line joining *B*(1, 4, 6) and *C*(5, 4, 4).

Q.2 The cartesian equations of a line are $6x - 2 = 3y + 1 = 2z - 2$. Find its direction ratios and also find vector equation of the line.

Q.3 Find the point on the line $\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2}$ at a distance of $3\sqrt{2}$ from the point (1, 2, 3).

Q.4 Find the cartesian and vector equations of a line which passes through the point (1, 2, 3) and is parallel to the line $\frac{-x-2}{1} = \frac{y+3}{7} = \frac{2z-6}{3}$.

Q.5 The cartesian equations of a line are $3x + 1 = 6y - 2 = 1 - z$. Find the fixed point through which it passes, its direction ratios and also its vector equation.

Q.6 Find the equation of the line passing through the point (–1, 3, –2) and perpendicular to the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ and $\frac{x+2}{-3}=\frac{y-1}{2}=\frac{z+1}{5}$.

Q.7 If the lines $\frac{x-1}{-3}=\frac{y-2}{2\lambda}=\frac{z-3}{2}$ and $\frac{x-1}{3\lambda}=\frac{y-1}{1}=\frac{z-6}{-5}$ are perpendicular, then find the value of λ.

Q.8 Find the value of λ so that the following lines are perpendicular to each other.

$$\frac{x-5}{5\lambda+2}=\frac{2-y}{5}=\frac{1-z}{-1}, \frac{x}{1}=\frac{2y+1}{4\lambda}=\frac{1-3}{-3}$$

Q.9 Show that the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{5}=\frac{y-1}{2}=z$ intersect. Find their point of intersection.

Q.10 Show that the lines $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-1}{5}$ and $\frac{x+2}{4}=\frac{y-1}{3}=\frac{z+1}{-2}$ do not intersect.

Q.11 Find the perpendicular distance of the point (1, 0, 0) from the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$. Also, find the coordinates of the foot of the perpendicular.

Q.12 Find the shortest distance between the pairs of lines whose Cartesian equation are

$$\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1} \text{ and } \frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$$

Q.13 A plane meets the coordinate axes in *A, B, C* such that the centroid of triangle *ABC* is the point (*p, q, r*). Show that the equation of the plane is $\frac{x}{p}+\frac{y}{q}=\frac{z}{r}=3$.

Q.14 Find the equation of the plane passing through the point (1, 2, 1) and perpendicular to the line joining the points (1, 4, 2) and (2, 3, 5). Find also the perpendicular distance of the origin from this plane.

Q.15 Find the equation of the plane passing through the point (1, 1, –1) and perpendicualr to the planes $x + 2y + 3z - 7 = 0$ and $2x - 3y + 4z = 0$.

Q.16 Find the equation of a plane passing through the point (–1, –1, 2) and perpendicular to the planes $3x + 2y - 3z = 1$ and $5x - 4y + z = 5$.

Q.17 Obtain the equation of the plane passing through the point (1, –3, –2) and perpendicular to the planes $x + 2y + 2z = 5$ and $5x - 4y + 2z = 8$.

Q.18 Find the equation of the plane passing through the point (1, –1, 2) and (2, –2, 2) and which is perpendicular to the plane $6x - 2y + 2z = 9$.

Q.19 Find the equation of the plane which is perpendicualr to the plane $5x + 3y + 6z + 8 = 0$ and which contains the line of intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$.

Q.20 Find the equation of the plane passing through the intersection of the plane $\vec{r}.(2\hat{i}+\hat{j}+3\hat{k})=7$, $\vec{r}.(2\hat{i}+5\hat{j}+3\hat{k})=9$ and the point (2, 1, 3).

Q.21 Find the equation of the plane passing through the points (3, 4, 1) and (0, 1, 0) and parallel to the line $\frac{x+3}{2}=\frac{y-3}{7}=\frac{z-2}{5}$.

Q.22 Find the distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured along a plane parallel to $\frac{x}{2}=\frac{y}{3}=\frac{z}{-6}$.

Q.23 Find the length and the foot of the perpendicular from the point (1, 1, 2) to the plane $\vec{r}.(\hat{i}-2\hat{j}+4\hat{k})+5=0$

Q.24 Find the image of the two points (1, 6, 3) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{3-2}{3}$.

CHAPTER 3

PROBABILITY

1. INTRODUCTION

There are two approaches to probability viz. (i) classical approach and (ii) axiomatic approach

Deterministic experiments : Those experimetns which when repeated under identical conditions produce the same result or outcome are known as deterministic experiments. When experiments in science or engineering are repeated under identical conditions, we get almost the same result everytime.

Probabilistic or random experiment : If an experiment, when repeated under identical conditions, do not produce the same outcome every time but the outcome in a trial is one of the several possible outcomes, then such an experiment is known as a probabilistic experiment or a random experiment.

2. SOME DEFINITION

Elementary events : If a random experiment is performed, then each of its outcomes is known as an elmeentary event.

Sample sapce : The set of all possible outcomes of a random experiment is called the sample space associated with it and it is generally denoted by S.

eg. Let there be a bag containing 3 white and 2 black balls. Let the white balls be denoted by W_1, W_2, W_3 and black balls be denoted by B_1 and B_2. If we draw two balls from the bag, then there are ${}^3C_2$ elementary events associated to this experiment.

These elementary are B_1W_1, B_1W_2, B_1W_3, B_2W_1, B_2W_2, B_2W_3, W_1W_2, W_1,W_3, W_2W_3 and B_1B_2. The set of all these elementary events is the samle space associated to a random experiment is defined as compound event, if it is disjoint union of single element subsets of the sample space.

Occurence of an event : An event A associated to a random experiment is said to occur, if any one of the elementary events associated to it is an outcome.

Negation of an event : Corresponding to every event A associated with a random experiment we define an event "not A" which occurs when and only when A does not occur.

Favourable elementary events : Let S be the sample space associated with a random experiment and A be an event associated to the experiment. Then elementary events belonging to A are known as favourable elementary events to the event A.

3. PROBABILITY

If there are n elementary events associated with a random expermient and m of them are favourable to an event A, then the probability of happening or occurrence of A is denoted by $P(A)$ and is defined as the ratio $\frac{m}{n}$.

Thus, $P(A)=\frac{m}{n}$

Clearly, $0 \le m \le n$

Therefore, $0 \le \frac{m}{n} \le 1 \quad \Rightarrow \quad 0 \le P(A) \le 1$

***Illustration 1*:** Five boys and three girls are seated at random in a row. The probability that no boy sits between two girls, is

(a) 1/56 (b) 1/8 (c) 3/28 (d) None of these

Solution: **(c)** $n(S) = 8!$

$n(E)$ = The number of arrangements of 5 boys and 3 girls, when the 3 girls are consecutive = $6! \times 3!$

$\therefore$ The required probability = $\frac{6! \times 3!}{8!} = \frac{3}{28}$

***Illustration 2*:** In a convex hexagon two diagonals are drawn at random. The probability that the diagonals intersect at an interior point of the hexagon, is

(a) 5/12 (b) 7/12 (c) 2/5 (d) None of these

Solution: **(a)** $n(S)$ = total number of selections of two diagonals = ${}^{9}C_{2} = \frac{9 \times 8}{2} = 36$

$n(E)$ = the number of selections of two diagonals which intersect at an interior point
= the total number of selections of four vertices
= ${}^{6}C_{4} = 15$

$\therefore$ The required probability = $\frac{15}{36} = \frac{5}{12}$

***Illustration 3*:** Two cards are drawn at random from a pack of 52 cards. The probability of getting at least a spade and an ace, is

(a) 1/34 (b) 8/221 (c) 1/26 (d) 2/51

Solution: **(c)** $n(S) = {}^{52}C_{2}$

$n(E)$ = The number of selections of 1 spade, 1 ace from 3 aces or selections of the ace of spade and 1 other spade

$= {}^{13}C_{1} \times {}^{3}C_{1} + {}^{12}C_{2} \times {}^{1}C_{1} = 51$

$\therefore \quad P(E) = \frac{51}{{}^{52}C_{2}} = \frac{1}{26}$

***Illustration 4*:** If the letters of the word ATTEMPT are written down at random, the chance that all T's are consecutive, is

(a) 1/42 (b) 6/7 (c) 1/7 (d) None of these

Solution: **(c)** $n(S) = \frac{7!}{3!}$, $n(E) = 5!$ So, $P(E) = \frac{5!}{7!/3!} = \frac{1}{7}$

***Illustration 5*:** The probability that out of 10 persons, all born in April, at least two have the same birthday, is

(a) $\frac{{}^{30}C_{10}}{(30)^{10}}$ (b) $1 - \frac{{}^{30}C_{10}}{30!}$ (c) $\frac{30^{10} - {}^{30}C_{10}}{(30)^{10}}$ (d) None of these

Solution: **(c)** There are 30 days in April

$n(S)$ = The number of ways in which 10 persons can have birthdays in the month of April = $30 \times 30 \times$ ..... to 10 times = 30^{10} (because each person can have birthday in 30 ways)

$n(E)$ = $n(S)$ – the number of ways in which 10 persons can have different birthdays

= $30^{10} - {}^{30}C_{10}$

$\therefore \quad P(E) = \frac{30^{10} - {}^{30}C_{10}}{30^{10}}$

4. TYPES OF EVENTS

Mutually exclusive events : Two or more events associated to a random experiment are mutually exclusive, if the occurrence of one of them prevents or denies the ocurrence all others.
Thus, if two events A and B are mutually exclusive, then $P(A \cap B) = 0$
Similarly, if A, B and C are mutually exclusive events, then $P(A \cap B \cap C) = 0$
Exhaustive events : Two or more events associated to a random experiments are exhaustive, if their union is the sample space. ie, events $A_1, A_2, \ldots\ldots, A_n$ associated to a random experiment with sample sapace S are exhaustive, if $A_1 \cup A_2 \cup \ldots\ldots \cup A_n = S$
Independent events : Two events A and B associated to a random experiment are independent, if the probability of occurrence or non occurrence of A is not affected by the occurrence or non-occurrence of B.

5. ADDITION THEOREMS ON PROBABILITY

(i) **Addition theorem for two events :** If A and B are two events associated with a random experiment, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(ii) **Addition theorem for three events :** If A, B, C are three events associated with a random experiment, then $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

(iii) **Generalized addition theorem :** If $A_1, A_2, \ldots\ldots, A_n$ are n events associated to a random experiment, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum\sum_{1\le i<j\le n} P(A_i \cap A_j) + \sum_{1\le i}^{n}\sum_{<j}^{n}\sum_{<k\le n}^{n} P(A_i A_j A_k) + \ldots\ldots + (-1)^{n-1} P(A_1 \cap A_2 \cap \ldots\ldots \cap A_n)$$

(iv) Let A and B be two events associated to a random experiment. Then,

(a) $P(\overline{A} \cap B) = P(B) - P(A \cap B)$

(b) $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

(c) $P((A \cap \overline{B}) \cup (\overline{A} \cap B)) = P(A) + P(B) - 2P(A \cap B)$

(v) For any two events A and B; $P(A \cap B) \le P(A) \le P(A \cup B) \le P(A) + P(B)$

(vi) For any two events A and B, the probability that exactly one of A, B occurs is given by
$P(A) + P(B) - 2P(A \cap B) = P(A \cup B) - P(A \cap B)$

(vii) If A, B, C are three vents, then

(a) P(at least two of A, B, C occur) $= P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$

(b) P(Exactly two of A, B, C occur) $= P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$

(c) P(Exactly one of A, B, C occur) $= P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(B \cap C) - 2P(A \cap C) + 3P(A \cap B \cap C)$

***Illustration 6*:** A and B are two events where $P(A) = 0.25$ and $P(B) = 0.5$. The probability of both happening together is 0.14. The probability of both A and B not happening is
(a) 0.39 (b) 0.25 (c) 0.11 (d) None of these

Solution: **(a)** $P(\overline{A} \cup \overline{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$
$= 1 - \{P(A) + P(B) - P(A \cap B)\}$
$= 1 - \{0.25 + 0.50 - 0.14\} = 0.39$

Illustration 7: Let A and B be two independent events such that $P(A) = \frac{1}{5}$, $P(A \cup B) = \frac{7}{10}$. Then $P(\bar{B})$ is equal to

(a) 3/8 (b) 2/7 (c) 7/9 (d) None of these

Solution: **(a)** Let $P(\bar{B}) = x$. Then $P(B) = 1 - x$

$\therefore \quad P(A \cap B) = P(A) \cdot P(B) = \frac{1}{5}(1 - x)$

But $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad \frac{7}{10} = \frac{1}{5} + (1 - x) - \frac{1}{5}(1 - x)$

or $\quad \frac{1}{2} = \frac{4}{5}(1 - x)$

$\Rightarrow \quad 1 - x = \frac{5}{8}$

$\therefore \quad x = \frac{3}{8}$

***Illustration 8*:** The probability that at least one of the events A and B occurs is 3/5. If A and B occur simultaneously with probability 1/5, then $P(\bar{A}) + P(\bar{B})$ is

(a) 2/5 (b) 4/5 (c) 6/5 (d) 7/5

Solution: **(c)** Here, $P(A \cup B) = \frac{3}{5}$ and $P(A \cap B) = \frac{1}{5}$.

So, from the addition theorem,

$$\frac{3}{5} = P(A) + P(B) - \frac{1}{5}$$

or $\quad \frac{4}{5} = 1 - P(\bar{A}) + 1 - P(\bar{B})$

$\therefore \quad P(\bar{A}) + P(\bar{B}) = 2 - \frac{4}{5} = \frac{6}{5}$

***Illustration 9*:** If $P(A) = \frac{3}{5}$ and $P(B) = \frac{2}{3}$, then

(a) $P(A \cup B) \geq \frac{2}{3}$ (b) $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$ (c) $\frac{2}{5} \leq P\left(\frac{A}{B}\right) \leq \frac{9}{10}$ (d) $P(A \cap \bar{B}) \leq \frac{1}{3}$

Solution: **(a, b, c, d)** $B \subseteq A \cup B$. Therefore, $P(B) \leq P(A \cup B) \Rightarrow P(A \cup B) \geq \frac{2}{3}$

So, alternative (a) is the correct answer.

Now, $\quad P(A \cap B) \geq \frac{3}{5} + \frac{2}{3} - 1 = \frac{4}{15}$

Now, $A \cup B \subseteq A \Rightarrow P(A \cap B) \le P(A)$

$\Rightarrow \quad P(A \cap B) \le \frac{3}{5}$

$\therefore \quad \frac{4}{15} \le P(A \cap B) \le \frac{3}{5}$

So, alternative (b) is the correct answer.

Now, $\frac{2}{5} \le P\left(\frac{A}{B}\right) \le \frac{9}{10}$

$\Rightarrow \quad \frac{4}{15} \le \frac{2}{3} P\left(\frac{A}{B}\right) \le \frac{3}{5} \qquad \Rightarrow \quad \frac{2}{5} \le P\left(\frac{A}{B}\right) \le \frac{9}{10}$

So, alternative (c) is the correct answer.

Finally, $P(A \cap \bar{B}) = P(A) - P(A \cap B) \le \frac{3}{5} - \frac{4}{15} = \frac{1}{3}$

So, alternative (d) is also the correct answer.

6. CONDITIONAL PROBABILITY

Let A and B be two events associated with a random experiment. Then, the probability of occurrence of event A under the condition that B has already occurred and $P(B) \neq 0$, is called the conditional probability and it is denoted by $P(A/B)$. Thus, we have

$P(A/B)$ = Probability of uccurrence of A given that B has already occurred.

Infact, the meanings of symbols $P(A/B)$ and $P(B/A)$ depends on the nature of the events A and B and also on the nature of the random experiment. These two symbols have the following meaning also.

$P(A/B)$ = Probability of occurrence of A when B occurs
Probability of occurrence of A with respect to B.

and $P(B/A)$ = Probability of occurrence of B when A occurs
Probability of occurrence of B with respect to A.

7. MULTIPLICATION THEOREMS ON PROBABILITY

(i) If A and B are two events associated with a random experiment, then

$P(A \cap B) = P(A)P(B/A)$, if $P(A) \neq 0$

or $P(A \cap B) = P(B)P(A/B)$, if $P(B) \neq 0$

or $P(A \cap B) = P(B)P(A/B)$, if $P(B) \neq 0$

(ii) **Extension of multiplication theorem :** If $A_1, A_2, ...A_n$ are n events associated with a random experiment, then

$P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap .....\cap A_n) = P(A_1)P(A_2/A_1)\, P(A_3/A_1 \cap A_2).....P(A_n/A_1 \cap A_2 \cap ....\cap A_{n-1})$

where $P(A_i/A_1 \cap A_2 \cap .....\cap A_{i-1})$ represents the conditional probability of the occurrence of event A_i, given that the events $A_1, A_2, ...A_{i-1}$ have already occured.

***Illustration 10*:** A draws two cards at random from a pack of 52 cards .After returning them to the pack and shuffling it. B draws two cards at random. The probability that their draws contain exactly one common card, is

(a) 25/546 (b) 50/663 (c) 25/663 (d) None of these

Solution: **(b)** The probability of both drawing the common card x
= $P(x)$ = (Probability of A drawing the card x and any other card y) × (probability of B drawing the card x and a card other than y)

$\therefore \quad P(x) = \frac{^{51}C_1}{^{52}C_2} \times \frac{^{50}C_1}{^{52}C_2}$ for all x, where x has 52 values.

$\therefore$ The required probability = $\Sigma P(x) = 52 \times \frac{51 \times 50 \times 4}{52 \times 51 \times 52 \times 51} = \frac{50}{663}$

***Illustration 11*:** A man draws a card from a pack of 52 cards and then replaces it. After shuffling the pack, he again draws a card. This he repeats a number of times. The probability that he will draw a heart for the first time in the third draw, is

(a) $\frac{9}{64}$ (b) $\frac{27}{64}$ (c) $\frac{1}{4} \times \frac{^{39}C_2}{^{52}C_2}$ (d) None of these

Solution: **(a)** Let E = The event of drawing a heart. Clearly, $P(E) = \frac{^{13}C_1}{^{52}C_1} = \frac{1}{4}$

$\therefore$ The required probability = $P(\bar{E}\ \bar{E}\ E) = P(\bar{E}).P(\bar{E}).P(E) = \frac{3}{4}.\frac{3}{4}.\frac{1}{4} = \frac{9}{64}$

***Illustration 12*:** A card is drawn from a pack. The card is replaced and the pack is reshuffled. If this is done six times, the probability that 2 hearts, 2 diamonds and 2 black cards are drawn, is

(a) $90.\left(\frac{1}{4}\right)^6$ (b) $\frac{45}{2}.\left(\frac{3}{4}\right)^6$ (c) $\frac{90}{2^{10}}$ (d) None of these

Solution: **(c)** The probability of getting a heart in one draw = $\frac{13}{52} = \frac{1}{4}$

Similarly for a diamond, the probability = $\frac{1}{4}$

The probability of getting a black card in one draw = $\frac{26}{52} = \frac{1}{2}$

$\therefore$ The required probability = $^6C_2 \times {}^4C_2 \times \left(\frac{1}{4}\right)^2 \times \left(\frac{1}{4}\right)^2 \times \left(\frac{1}{2}\right)^2 = 90.\left(\frac{1}{2}\right)^2$.

because in 6 draws, 2 draws will be for hearts, 2 for diamonds and 2 for black card and this selection can be done in $^6C_2 \times {}^4C_2 \times {}^2C_2$ ways.

8. THE LAW OF TOTAL PROBABILITY

Let S be the sample space and let $E_1, E_2, \ldots, E_n$ be n mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ..... or E_n, then

$P(A) = P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + \ldots + P(E_n)P(A/E_n)$

$= \sum_{r=1}^{n} P(E_r)P(A/E_r)$

9. BAYER'S RUNE

Let S be the sample space and let $E_1, E_2, ..., E_n$ be n mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ..... or E_n, then

$$P(E_i / A) = \frac{P(E_i)P(A / E_i)}{\sum_{i=1}^{n} P(E_i)P(A / E_i)}, i = 1,2,....n$$

***Illustration 13*:** The probability of a number n showing in a throw of a dice marked 1 to 6 is proportional to n. Then, the probability of the number 3 showing in a throw, is

(a) 1/2 (b) 1/6 (c) 1/7 (d) 1/21

Solution: **(d)** $P(E_n)$ = probability of the dice showing $n = kn$

Clearly, $P(E_1) + P(E_2) + ... + P(E_6) = 1$

$\Rightarrow \quad k(1 + 2 + 3 + 4 + 5 + 6) = 1$

$$\therefore \quad k = \frac{1}{6.7/2} = \frac{1}{21}$$

***Illustration 14*:** There are 4 white and 3 black balls in a box. In another box there are 3 white and 4 black balls. An unbiased dice is rolled. If it shows a number less than or equal to 3, then a ball is drawn from the Second box. If the ball drawn is black, then the probability that the ball was drawn from the first box, is

(a) 1/5 (b) 6/7 (c) 4/7 (d) 3/7

Solution: **(d)** E_1 = The event of drawing from the first box.

E_2 = The event of drawing from the second box.

$$P(E_1) = \frac{3}{6} = \frac{1}{2} \text{ and } P(E_2) = \frac{3}{2} = \frac{1}{2}$$

Clearly, E_1 and E_2 are mutually exclusive and exhaustive.

Now, $\quad P\left(\frac{B}{E_1}\right) = \frac{3}{7}$ and $P\left(\frac{B}{E_2}\right) = \frac{4}{7}$

We have to find $P\left(\frac{E_1}{B}\right)$

By Bayes' theorem,

$$P\left(\frac{E_1}{B}\right) = \frac{P(E_1).P(B / E_1)}{P(E_1).P(B / E_1) + P(E_2).P(B / E_2)} = \frac{1/2 \cdot 3/7}{1/2 \cdot 3/7 + 1/2 \cdot 4/7} = \frac{3}{7}$$

10. RANDOM VARIABLE AND ITS PROBABILITY DISTRIBUTION

Definition : Let S be the sample space associated with a given random experiment. Then, a real valued function X which assigns to each event $w \in S$ to a unique real number $X(w)$ is called a random variable.

Probability distribution : If a random variable X takes values $x_1, x_2,.....,x_n$ with respective probabilities $p_1, p_2, p_3,.....p_n$, then

X	:	x_1	x_2	x_3	.....	x_n
$P(X)$	:	p_1	p_2	p_3	.....	p_n

is known as the probability distribution of X.

This, a tabular description giving the values of the random variable along with the corresponding probabilities is called the probability distribution.

Note :-

The probability distribution of a random variable X is defined only when we have the various value of the random variable eg,

$x_1, x_2,.....,x_n$ together with respective probabilities $p_1, p_2, ...,p_n$ satisfying $\sum_{i=1}^{n} p_i = 1$.

Mean and variance of a random variable

Mean : If X is a discrete random variable which assumes values $x_1, x_2, x_3,.....x_n$ with respective probabilities $p_1, p_2, p_3,.....,p_n$, then the mean $\bar{X}$ of X is defined as

$$\bar{X} = p_1x_1 + p_2x_2 + ... + p_nx_n \qquad \text{or} \quad \bar{X} = \sum_{i=1}^{n} p_ix_i$$

Variance : If X be a discrete random variable which assumes values $x_1, x_2, x_3,.....x_n$ with the respective probabilities $p_1, p_2, p_3,.....,p_n$, then variance of X is defined as

$$\text{Var}(X) = p_1(x_1 - \bar{X})^2 + p_2(x_2 - \bar{X})^2 + ... + p_n(x_n - \bar{X})^2$$

$$= \sum_{i=1}^{n} p_i(x_i - \bar{X})^2 \text{ , where } \bar{X} = \sum_{i=1}^{n} p_ix_i \text{ is the mean of } X$$

Now, $$\text{Var}(X) = \sum_{i=1}^{n} p_i(x_i - \bar{X})^2 = \sum_{i=1}^{n} p_i(x_i^2 - 2x_i\bar{X} + \bar{X}^2)$$

$$= \sum_{i=1}^{n} p_ix_i^2 - 2\bar{X}\left(\sum_{i=1}^{n} p_ix_i\right) + \bar{X}^2\left(\sum_{i=1}^{n} p_i\right)$$

$$= \sum_{i=1}^{n} p_ix_i^2 - 2\bar{X}.\bar{X} + \bar{X}^2 \qquad \left[\because \sum_{i=1}^{n} p_i = 1\right]$$

$$= \sum_{i=1}^{n} p_ix_i^2 - 2\bar{X}^2 + \bar{X}^2$$

$$= \sum_{i=1}^{n} p_ix_i^2 - \bar{X}^2 = \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} p_1x_i\right)^2$$

Thus, $$\text{Var}(X) = \sum_{i=1}^{n} p_ix_i^2 - \left(\sum_{i=1}^{n} p_ix_i\right)^2$$

11. BINOMIAL DISTRIBTION

A random variable X which takes values 0, 1, 2, ...n, is said to follow binomial distribution, if its probability distribution function is given by

$P(X = r) = {}^nC_r p^r q^{n-r}$, $r = 0, 1, 2, ..., n$

where $p, q > 0$ such that $p + q = 1$

The two constants n and p in the distribution are known as the parameters of the distribution.

The notation $X \sim B(n, p)$ is generally used to denote that the random variable X follows binomial

distribution with parameters n and p.

We have, $P(X = 0) + P(X = 1) + ... + P(X = n)$

$= {}^nP_0p^0q^{n-0} + {}^nC_1p^1q^{n-1} + ... + {}^nC_np^nq^{n-n} = (q + p)^n = 1^n = 1$

Thus, the assignment of probabilities to the random variable X is permissible.

***Illustration 15*:** A coin is tossed 7 times. The probability that he wins the toss on more occasions, is

(a) 1/4 (b) 5/8 (c) 1/2 (d) None of these

Solution:(c)

The man has to win at least 4 times.

$\therefore$ The required probability

$$= {}^7C_4\left(\frac{1}{2}\right)^4.\left(\frac{1}{2}\right)^3 + {}^7C_5.\left(\frac{1}{2}\right)^5.\left(\frac{1}{2}\right)^2 + {}^7C_6\left(\frac{1}{2}\right)^6.\frac{1}{2} + {}^7C_7\left(\frac{1}{2}\right)^7$$

$$= ({}^7C_4 + {}^7C_5 + {}^7C_6 + {}^7C_7).\frac{1}{2^7} = \frac{64}{2^7} = \frac{1}{2}$$

***Illustration 16*:** From a box containing 20 tickets of value 1 to 20, four tickets are drawn one by one. After each draw, the ticket is replaced. The probability that the largest value of tickets drawn is 15, is

(a) $(3/4)^4$ (b) 27/320 (c) 27/1280 (d) None of these

Solution: **(b)** The probability of drawing a number less than or equal to 15 in a draw = $\frac{15}{20} = \frac{3}{4}$

The probability of drawing the ticket of value 15 in a draw = $\frac{1}{20}$

$\therefore$ The required probability = ${}^4C_1 . \frac{1}{20} . \left(\frac{3}{4}\right)^3$

***Illustration 17*:** A dice is thrown $2n + 1$ times, $n \in N$. The probability that faces with even numbers show odd numbers of times, is

(a) $\frac{2n+1}{4n+3}$ (b) Less than 1/2 (c) Greater than 1/2 (d) None of these

Solution: **(d)** The probability of showing an even number in a throw = $\frac{3}{6} = \frac{1}{2}$

$\therefore$ The required probability

$$= {}^{2n+1}C_1.\frac{1}{2}.\left(\frac{1}{2}\right)^{2n} + {}^{2n+1}C_3.\left(\frac{1}{2}\right)^3.\left(\frac{1}{2}\right)^{2n-2} + ... + {}^{2n+1}C_{2n+1}\left(\frac{1}{2}\right)^{2n+1}$$

$$= \left({}^{2n+1}C_1 + {}^{2n+1}C_3 + ... + {}^{2n+1}C_{2n+1}\right)\left(\frac{1}{2}\right)^{2n+1}$$

$$= 2^{2n}.\left(\frac{1}{2}\right)^{2n+1} = \frac{1}{2}$$

MISCELLANEOUS EXAMPLES

Example 1: An urn contains *m* white and *n* black balls. A ball is drawn at random and is put back into the urn along with *k* additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. Then the probability that the ball drawn now is white, is

(a) $\frac{n}{m+n}$ (b) $\frac{m}{m+n}$

(c) $\frac{2n}{m+n}$ (d) None of these

Solution: **(b)** Let w_1 = ball drawn in the first draw is white, b_1 = ball drawn in the first draw is black w_2 = ball drawn in second draw is white

Then, $P(w_2) = P(w_1)\,.\,P(w_2/w_1) + P(b_1)P(w_2/b_1)$

$$=\left(\frac{m}{m+n}\right)\left(\frac{m+k}{m+n+k}\right)+\left(\frac{n}{m+n}\right)\left(\frac{m}{m+n+k}\right)=\frac{m}{m+n}$$

Example 2: A box contains *N* coins, *m* of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is 1/2, while it is 2/3 when a bised coin is tossed. A coin is drawn from the box at random and is tossed twice. Then the probability that the coin drawn is fair, is

(a) $\frac{9m}{8N+m}$ (b) $\frac{9m}{8N-m}$

(c) $\frac{9m}{8m-N}$ (d) $\frac{9m}{8m+N}$

Solution: **(a)** E_1 : coin is fair, E_2 : coin is biased, A second toss shows tail.

$$P\left(\frac{E_1}{A}\right)=\frac{P(A/E_1)P(E_1)}{P(A/E_1)P(E_1)+P(A/E_1)P(E_2)}$$

$$=\frac{\frac{m}{N}.\frac{1}{2}.\frac{1}{2}}{\frac{m}{N}.\frac{1}{2}.\frac{1}{2}+\frac{N-m}{N}.\frac{2}{3}.\frac{1}{3}}$$

$$=\frac{9m}{9m+8N-8m}=\frac{9m}{8N+m}$$

Example 3: *A* is targetting to *B*, *B* and *C* are targetting to *A*. Probability of hitting the target by *A*, *B* and *C* are $\frac{2}{3},\frac{1}{2}$ and $\frac{1}{3}$ respectively. If *A* is hit then, the probability that *B* hits the target and *C* does not, is

(a) 1/2 (b) 3/4

(c) 2/3 (d) None of these

Solution: **(a)** Let $P(A)$ = Probability that *A* will hit *B; P*(*B*) = Probability that *B* will hit *A*

$P(C)$ = Probability that *C* will hit *A; P*(*E*) = Probability that *A* will be hit

Then, $P(E)=1-P(\bar{B}\cap\bar{C})=1-P(\bar{B}).P(\bar{C})=1-\frac{1}{2}.\frac{2}{3}=\frac{2}{3}$

$$\Rightarrow \qquad P\left(\frac{B\cap\overline{C}}{E}\right)=\frac{P(B).P(\overline{C})}{P(E)}=\frac{\frac{1}{2}\cdot\frac{2}{3}}{\frac{2}{3}}=\frac{1}{2}$$

Example 4: Four cards are drawn from a pack of 52 playing cards. Then the probability of drawing at least one pair, is

(a) $1-\frac{4^3\times{}^{13}C_4}{{}^{52}C_4}$ (b) $1-\frac{{}^{13}C_4}{{}^{52}C_4}$

(c) $1-\frac{4^4\times 13!}{52!}$ (d) None of these

Solution: **(a)** $n(S) = {}^{52}C_4$

there are exactly 13 types of cards

$\Rightarrow \qquad n(E) = {}^{52}C_4 - {}^{13}C_4 \times {}^4C_1 \times {}^4C_1 \times {}^4C_1 \times {}^4C_1$

$$\Rightarrow \qquad P(E)=1-\frac{4^4\times{}^{13}C_4}{{}^{52}C_4}$$

Example 5: If $P\left(\frac{A}{B}\right)=P\left(\frac{B}{P}\right)$. A and B are two, non-mutually exclusive events, then

(a) A and B are necessarily same events

(b) $P(A) = P(B)$

(c) $P(A \cap B) = P(A)P(B)$ (d) All of these

Solution: **(b)** $P\left(\frac{A}{B}\right)=P\left(\frac{B}{P}\right) \Rightarrow \frac{P(A\cap B)}{P(B)}=\frac{P(A\cap B)}{P(A)} \Rightarrow P(A)=P(B)$

Example 6: 10 different books and 2 different pens are given to 3 boys so that each gets equal number of things. The probability that the same boy does not receive both the pens, is

(a) 5/11 (b) 7/11

(c) 2/3 (d) 6/11

Solution: **(a)** $n(S) = {}^{12}C_4 \times {}^8C_4 \times {}^4C_4$

$n(E) = n(S)$ – The number of ways in which one boy gets both the pens

$= n(S) - {}^{10}C_2 \times {}^8C_4 \times {}^4C_4 \times (3!)$

$$\therefore \qquad P(E)=1-\frac{{}^{10}C_2\times{}^8C_4\times{}^4C_4\times(3!)}{{}^{12}C_4\times{}^8C_4\times{}^4C_4}=1-\frac{6}{11}=\frac{5}{11}$$

Example 7: Two distinct numbers are selected at random from the first twelve natural numbers. The probability that the sum will be divisible by 3, is

(a) 1/3 (b) 23/66

(c) 1/2 (d) None of these

Solution: **(a)** Let E_3 = The event of the sum being 3.

Similarly, E_6, E_9, E_{12}, E_{15}, E_{18}, E_{21}

$n(S) = {}^{12}C_2$

$n(E_3) = 1$, $n(E_6) = 2$, $n(E_9) = 4$, $n(E_{12}) = 5$, $n(E_{15}) = 5$, $n(E_{18}) = 3$, $n(E_{21}) = 2$

$\therefore \qquad P(E) = P(E_3) + ... + P(E_{21})$

$$=\frac{1+2+4+5+5+3+2}{{}^{12}C_2}=\frac{22\times 2}{12\times 11}=\frac{1}{3}$$

Aliter: The sum of the numbers for every selection is divisible by 3 or leaves the remainder 1 or leaves the remainder 2. These are equally probable. So, the required probability = 1/3, because the sum of the three probabilities is 1.

Example 8: Four numbers are multiplied together. Then the probability that the product will be divisible by 5 or 10 is

(a) 369/625 (b) 399/625

(c) 123/625 (d) 133/625

Solution: **(a)** The divisibility of the product of four numbers depends upon the value of the last digit of each ten digits 0, 1, 2, ..., 9. So, the total number of ways of selecting last digits of four numbers is $10 \times 10 \times 10 \times 10 = 10^4$. If the product of the 4 numbers is not divisible by 5 or 10, then the number of choices for last digit of each number is 8 (excluding 0 or 5). So, favourable number of ways = 8^4. Therefore, the probability that the product is not divisible by 5 or 10 is $(8/10)^4$.

Hence, required probability = $1 - \left(\frac{8}{10}\right)^4 = \frac{369}{625}$

Example 9: A team of 8 couples, (husband and wife) attend a lucky draw in which 4 persons picked up for a prize. Then, the probability that there is at least one couple is

(a) 11/39 (b) 12/39

(c) 14/39 (d) 15/39

Solution: **(d)** Probability of selecting at least one couple for the prize

= 1 – Probability of not selecting any couple for the prize

$$= 1 - \frac{({}^{16}C_1 \times {}^{14}C_1 \times {}^{12}C_1 \times {}^{10}C_1)4!}{{}^{16}C_4}$$

$$= 1 - \frac{16 \times 14 \times 12 \times 10}{16 \times 15 \times 14 \times 13} = \frac{15}{39}$$

Example 10: A bag contains 16 coins of which two are counterfeit with heads on both sides. The rest are fair coins. One is selected at random from the bag and tossed. The probability of getting a head is

(a) 9/16 (b) 11/16

(c) 5/9 (d) None of these

Solution: **(a)** Let A be the event of selecting a counterfeit coin and B be the event of getting head.

Then, required probability = $P(A \cap B) \cup (\bar{A} \cap B)$

$$= P(A \cap B) + (\bar{A} \cap B)$$

$$= P(A)P\left(\frac{B}{A}\right) + P(\bar{A})P\left(\frac{B}{\bar{A}}\right)$$

$$= \frac{2}{16} \times 1 + \frac{14}{16} \times \frac{1}{2} = \frac{9}{16}$$

Example 11: A fair coin is tossed 99 times. If X is the number of times heads occur, then $P(X = r)$ is maximum when r is

(a) 49 (b) 52

(c) 51 (d) None of these

Solution: **(a)** Here, $n = 99$, $p = 1/2$. Therefore, $(n + 1)p = 50$.
So, $P(X = r)$ is maximum when $r = 50$ or $r = 49$

Example 12: A bag contains $n + 1$ coins. It is known that one of these coins shows heads on both sides, whereas the other coins are fair. One coin is selected at random and tossed. If the probability that toss results in heads is 7/12, then the value of n is

(a) 3 (b) 4

(c) 5 (d) None of these

Solution: **(c)** Let E_1, denotes the event "a coin with two heads" is selected and E_2, denote the event "a fair coin is selected". Let A be the event "the toss results in heads". Then,

$$P(E_1) = \frac{1}{n+1}, P(E_2) = \frac{n}{n+1}.P\left(\frac{A}{E_1}\right) = 1 \text{ and } P\left(\frac{A}{E_2}\right) = \frac{1}{2}$$

$$\therefore \quad P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right)$$

$$\Rightarrow \quad \frac{7}{12} = \frac{1}{n+1} \times 1 + \frac{n}{n+1} \times \frac{1}{2} \qquad \left[\because P(A) = \frac{7}{12}\right]$$

$\Rightarrow$ $12 + 6n + 7n + 7$ $\Rightarrow$ $n = 5$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Classical Probability

Q.1 Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1, r_2 and r_3 are the numbers obtained on the die, then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is
(a) 1/18 (b) 1/9
(c) 2/9 (d) 1/36

Q.2 If A and B are independent events of a random experiment such that $P(A \cap B)$ = and $\frac{1}{2}$ $P(\overline{A} \cap \overline{B}) = \frac{1}{3}$, then $P(A)$ is equal to
(a) 1/4 (b) 1/3
(c) 5/7 (d) 2/3

Q.3 If $P(A) = \frac{1}{12}$, $P(B) = \frac{5}{12}$ and $P\left(\frac{B}{A}\right) = \frac{1}{15}$, then $P(A \cup B)$ is equal to
(a) 89/180 (b) 90/180
(c) 91/180 (d) 92/180

2. Addition & Subtraction Law of Probability

Q.4 One ticket is selected at random from 50 tickets numbered 00, 01, 02, ..., 49. Then, the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero equals
(a) 1/14 (b) 1/7
(c) 5/14 (d) 1/50

Q.5 A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is
(a) 1/729 (b) 8/9
(c) 8/729 (d) 8/243

Q.6 The probability that a certain kind of component will survive a given shock test is 3/4. The probability that exactly 2 of the next 4 components tested survive is
(a) 9/41 (b) 25/128
(c) 1/5 (d) 27/128

Q.7 A box contains 9 tickets numbered 1 to 9 inclusive. If 3 tickets are drawn from the box one at a time, the probability that they are alternatively either {odd, even, odd} or {even, odd, even} is
(a) 5/17 (b) 4/17
(c) 5/16 (d) 5/18

Q.8 A committee of five is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all, is
(a) 1/2 (b) 5/9
(c) 4/9 (d) 2/3

Q.9 A biased coin with probability p, $0 < p < 1$ of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is 2/5, then p equals
(a) 1/3 (b) 2/3
(c) 2/5 (d) 3/5

Q.10 An urn contains 3 red and 5 blue balls. The probability that two balls are drawn in which 2nd ball drawn is blue without replacement, is
(a) 5/16 (b) 5/56
(c) 5/8 (d) 20/56

3. Independent & Conditional Probability

Q.11 An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is
(a) 2, 4 or 8 (b) 3, 6 or 9
(c) 4 or 8 (d) 5 or 10

Q.12 Let E^c denote the complement of an event E. Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then, $P(E^c \cap F^c \mid G)$ equals
(a) $P(E^c) + P(F^c)$ (b) $P(E^c) - P(F^c)$
(c) $P(E^c) - P(F)$ (d) $P(E) - P(F^c)$

Q.13 One Indian and four American men and their wives are to be seated randomly around a circular table. Then, the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife, is
(a) 1/2 (b) 1/3
(c) 2/5 (d) 1/5

Q.14 An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random without replacement from the urn. The probability that the three balls have different colours, is
(a) 1/3 (b) 2/7
(c) 1/21 (d) 2/23

Q.15 It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P\left(\frac{A}{B}\right) = \frac{1}{2}$ and $P\left(\frac{B}{A}\right) = \frac{2}{3}$. Then $P(B)$ is
(a) 1/6 (b) 1/3
(c) 2/3 (d) 1/2

4. Total Probability & Baye's Theorem

Q.16 A signal which can be green or red with probability 4/5 and 1/5 respectively, is receive by station A and then transmitted to station B. The probability of each station receiving the signal correctly is 3/4. If the signal received at station B is green, then the probability that the original signal green is
(a) 5.3 (b) 6/7
(c) 20/23 (d) 9/20

Q.17 A die is thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then, $P(A \cup B)$ is
(a) 3/5 (b) 0
(c) 1 (d) 2/5

Q.18 For $k = 1, 2, 3$ the box B_k contains k red balls and $(k + 1)$ white balls. Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$. A box is selected at random and a ball is drawn from it. If a red ball is drawn, then the probability that it has come from box B_2, is
(a) 35/78 (b) 14/39
(c) 10/13 (d) 12/13

5. Probability distribution & Binomial Distribution

Q.19 In a binomial distribution $B\left(n, p = \frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to 9/10, then n is greater than
(a) $\frac{1}{\log_{10} 4 - \log_{10} 3}$ (b) $\frac{1}{\log_{10} 4 + \log_{10} 3}$
(c) $\frac{9}{\log_{10} 4 - \log_{10} 3}$ (d) $\frac{4}{\log_{10} 4 - \log_{10} 3}$

Q.20 At a telephone enquiry system the number of phone calls regarding relevant enquiry follow poisson distribution with an average of 5 phone calls during 10 min time intervals. The probability that there is at the most one phone call during a 10 min time period, is
(a) 5/6 (b) 6/55
(c) $6/e^5$ (d) $6/5^e$

Q.21 The probability that the same number appear on throwing three dice simultaneously, is
(a) 1/36 (b) 5/36
(c) 1/6 (d) 4/13

Q.22 A coin is tossed n times. The probability of getting head at least one is greater than 0.8, then the least value of n is
(a) 2 (b) 3
(c) 5 (d) 4

Q.23 A fair die has four blank faces and two faces marked 3. The change of getting a total of 12 in 5 throws is
(a) ${}^5C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)$ (b) ${}^5C_4\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^4$
(c) ${}^5C_4\left(\frac{1}{6}\right)^4$ (d) ${}^5C_4\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right)$

Q.24 If X is a Poisson varies such that $P(X = 1) = P(X = 2)$, then $P(X = 4)$ is equal to
(a) 1/3 (b) 1/2
(c) 2/3 (d) 3/4

Q.25 A fiar die is tossed eight times. The probability that a third six is observed on the eight throw is
(a) $\frac{{}^7C_2 \times 5^5}{6^7}$ (b) $\frac{{}^7C_2 \times 5^5}{6^8}$
(c) $\frac{{}^7C_2 \times 5^5}{6^6}$ (d) None of these

Section-B
(Previous Year Question)

Q.1 Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for the complement of hte event A. Then the events A and B are
(a) equally likely but not indendent
(b) independent but not equally likely
(c) independent and equally likely
(d) mutually excusive and independent

Q.2 A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is

(a) $\frac{10}{3^5}$ (b) $\frac{17}{3^5}$

(c) $\frac{13}{3^5}$ (d) $\frac{11}{3^5}$

Q.3 Three numbers are chosen at random without replacement from {1, 2, 3, ...,8}. The probability that their minimum is 3, given that their maximum is 6, is

(a) $\frac{1}{4}$ (b) $\frac{2}{5}$

(c) $\frac{3}{8}$ (d) $\frac{1}{5}$

Q.4 If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is

(a) $P(C/D) < P(C)$ (b) $P(C/D) = \frac{P(D)}{P(C)}$

(c) $P(C/D) = P(C)$ (d) $P(C/D) \geq P(C)$

Q.5 Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval

(a) $\left[0, \frac{1}{2}\right]$ (b) $\left(\frac{11}{12}, 1\right]$

(c) $\left(\frac{1}{2}, \frac{3}{4}\right]$ (d) $\left(\frac{3}{4}, \frac{11}{12}\right]$

Q.6 An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawan at random without replacement from the urn. The probability that the three balls have different colours is
(a) 1 / 3 (b) 2 / 7
(c) 1 / 21 (d) 2 / 23

Q.7 Four numbers are chosen at random (without replacement from the set {1, 2, 3, ---, 20}.
Statement- 1 : The probabilitthat the chosen numbers when arranged in some order will form an A.P. is $\frac{1}{85}$
Statement-2: If the four chosen numbers from an A.P., then the set of all possible values of common difference is $\{\pm1, \pm2, \pm3, \pm4, \pm5\}$.
(a) Statement-1 is true, Statement-2 is true; Statement-2 is true; Statement -2 is not a correct explanation for Statement-1.
(b) Statement-1 is true, Statement -2 is true, Statement-2 is not a correct explanation for Statement-1.
(c) Statement- 1 is true, STatement-2 is false.
(d) Statement -1 is false, Statement-2 is true.

Q.8 In a binomial distribution $B\left(n, p = \frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then n is greater than

(a) $\frac{1}{\log_{10} 4 + \log_{10} 3}$ (b) $\frac{9}{\log_{10} 4 - \log_{10} 3}$

(c) $\frac{4}{\log_{10} 4 - \log_{10} 3}$ (d) $\frac{1}{\log_{10} 4 - \log_{10} 3}$

Q.9 One ticket is selected at random from 50 tickets numbered 00, 01, 02,...., 49. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals

(a) $\frac{1}{7}$ (b) $\frac{5}{14}$

(c) $\frac{1}{50}$ (d) $\frac{1}{14}$

Q.10 A die is thrown Let A be the event that the number obtained is greater 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) 0 (d) 1

Q.11 It is given that the events A and B are such that $P(A)=\frac{1}{4}, P(A \setminus B)=\frac{1}{2}$. Then P(B) is

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Q.12 A pair of fair dice is thrown independtly three times. The probability of getting a score of exactly 9 twice is

(a) 8 / 729 (b) 8 / 243

(c) 1 / 729 (d) 8 / 9

Q.13 Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is

(a) 0.2 (b) 0.7

(c) 0.06 (d) 0.14

Q.14 At a telephone enquiry system the number of phone calls regarding relevent enquiry follow Poisson distribution with a average of 5 phone calls during 10-mintue time intervals. The probability that there is at the most one phone call during a 10-minute time period is

(a) $\frac{6}{5^e}$ (b) $\frac{5}{6}$

(c) $\frac{6}{55}$ (d) $\frac{6}{e^5}$

Q.15 A random variable X has Poisson distribution with mean 2. The P (x > 1.5) equals

(a) 0 (b) $2/e^2$

(c) $3/e^2$ (d) $1-\frac{3}{e^2}$

Q.16 Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without counsulting others. The probability that all the three apply for the same house is

(a) 1/9 (b) 2/9

(c) 7/9 (d) 8/9

Q.17 Let A and B be two events such that $P(\overline{A \cup B})=\frac{1}{6}$, $P(A \cup B)=\frac{1}{4}$ and $P(\overline{A})=\frac{1}{4}$, where $\overline{A}$ stands for complement of event A. Then events A and B are

(a) equally likely but not independent

(b) equally likely and mutually exclusive

(c) mutually exclusive and independent

(d) independent but not equally likely

Q.18 THe mean and the variance of a binomial distribution are 4 and 2 respectively. then the probability of 2 successes is:

(a) 128 / 256 (b) 219 / 256

(c) 37 / 256 (d) 28 / 256

Q.19 A random variable X has the probility distribution:

X:	1	2	3	4	5	6	7	8
$P(X)$	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events $E=$ {X is a prime number } and F = {X < 4}, the probability $P(E \cup F)$ is:

(a) 0.35 (b) 0.77

(c) 0.87 (d) 0.50

Q.20 The probability that A speaks truth is 4/5, while htis probability for B is 3/4. The probability that they contradict each other when asked to speak on a fact is

(a) 7 / 30 (b) 1 / 5

(c) 3 / 20 (d) 4 / 5

Q.21 The mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively, then P (X = 1) is

(a) 1 / 16 (b) 1 / 8

(c) 1 / 4 (d) 1 / 32

Q.22 Five horses are in race. Mr A selectes two of the horses at random and bets on them. The probaility that Mr. A selected the winning horse is

(a) 3 / 5 (b) 1 / 5

(c) 2 / 5 (d) 4 / 5

Q.23 Events A, B, C are mutually exclusive events such that $P(A)=\frac{3x+1}{3}, P(B)=\frac{1-x}{4}$ and $P(C)=\frac{1-2x}{2}$. Then set of possible values of x are in the interval

(a) $\left[\frac{1}{3},\frac{2}{3}\right]$ (b) $\left[\frac{1}{3},\frac{13}{3}\right]$

(c) $[0,1]$ (d) $\left[\frac{1}{3},\frac{1}{2}\right]$

Q.24 A die is tossed 5 times. Getting an odd number is considered a success. Then the variance of distribution of success is

(a) 8 / 3 (b) 3 / 8

(c) 4 / 5 (d) 5 / 4

Q.25 A and B are events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$, $P(\overline{A}) = 2/3$ then $P(\overline{A} \cap B)$ is

(a) 5 / 12 (b) 3 / 8
(c) 5 / 8 (d) 1 / 4

Q.26 A problem in mathematics is given to htree students A, B, C and their respective probability of solving the problem is 1/ 2, 1 / 3 and 1 / 4. Probability that the problem is solved is

(a) 3 / 4 (b) 1 / 2
(c) 2 / 3 (d) 1 / 3

Section-C
(Board pattern)

Q.1 A man is known to speak truth 3 out of 4 times. He throws a die and reprots that it is a six. Find the probability that it is actually a six.

Q.2 The contents of urns I, II, III are as follows :
Urn I : 1 white, 2 black and 3 red balls.
Urn II : 2 white, 1 black and 1 red balls.
Urn III : 4 white, 5 black and 3 red balls.
One urn is chosen at random and two balls are drawn. They happen to be white and red. What is the probability that they come from Urns I, II, III?

Q.3 A bag *A* contains 2 white and 3 red balls and a bag *B* contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag *B*.

Q.4 Two groups are contesting for the positions of the Board of Directors of a Corporation. The probabilities that the first and the second groups will win are 0.6 nad 0.4 respectively. Further, if the first group wins, the probability of introducing a new product is 0.7 and the carresponding probability is 0.3 if the second group wins. Find the probability that the new product introduced by the second group?

Q.5 A factory has three machines *X, Y* and *Z* producing 1000, 2000 and 3000 bolts per day represectively. The machine *X* produces 1% defective bolts, *Y* produces 1.5% and *Z* produces 2% defective bolts. At the end of a day, a bolt is drawn at random and is found to be defective. What is the probabilitiy that this defective bolt has been produced by machine *X*?

Q.6 An insurance company issued 3000 scooters, 4000 and 5000 trucks. The probabilities of the accident Involving a scooter, a car and a truck are 0.02, 0.03 and 0.04 respectively. One of the insured vehicles meet with an accident. Find the probability that it is a (a) scooter (ii) car (iii) truck.

Q.7 A company has two plants to manufacture bicycles. The first plant manufactures 60% of the bicycles and the second plant 40%. Out of that 80% of the bicycles are rated of standard quality at the first plant and 90% of standard quality at the second plant. A bicycle is picked up at random and found to be standard quality. Find the probability that it comes from the second plant.

Q.8 Three urns *A, B* and *C* contain 6 red and 4 white; 2 red and 6 white; and 1 red and 5 white balls respectively. An urn is chosen at random and a ball is drawn. If the ball drawn is found to be red, find the probability that the ball was drawn from urn *A*.

Q.9 A is known to speak truth 3 times out of 5 times. He throws a die and reports that it is 1. Find the probability that it is actually 1.

Q.10 A factory has three machines *A, B* and *C*, which produce 100, 200 and 300 items of a particular type daily. The machines produce 2%, 3% and 5% defective items respectively. One day when the production was over, an item was picked up randomly and it was found to be defective. Find the probability that it was produced by machine *A*.

Q.11 A bag contains 1 white and 6 red balls, and a second bag contains 4 white and 3 red balls. One of the bags is picked up at random and a ball is randomly drawn from it, and is found to be white in colour. Find the probabiity that the drawn ball was from the first bag.

Q.12 A speaks the turth 8 times out of 10 times. A die is tossed. He reprots that it was 5. What is the probability that it was actually 5?

Q.13 For *A, B* and *C* the chances of being selected as the manager of a firm are in the ratio 4 : 1 : 2 respectively. The respectively probabilities for them to introduce a radical change in marketing startegy are 0.3, 0.8, 0.5. If the change does take place, find the probability that it is due to the appointment of *B* or *C*.

Q.14 An insurance company insured 2000 scooters and 3000 motorcycles. The probability of an accident involving a scooter is 0.01 and that of a motorcycle is 0.02. An insured vehicle met with an accident. Find the probability that the accidented vehicle was a motorcycle.

Q.15 There are three coins. One is two headed coin, another is a biased coin that comes up head 75% of the time and third is an unbiased coin. One of the three coins is choosen at random and tossed, it shows heads, what is the probability that it was the two headed coin?

ANSWER KEY

CHAPTER-01 (VECTOR ALGEBRA)

SECTION-A (Straight objective type questions)

1. (b) 2. (a) 3. (a) 4. (a) 5. (b) 6. (c) 7. (c)
8. (a) 9. (b) 10. (b) 11. (a) 12. (a) 13. (d) 14. (d)
15. (d) 16. (c) 17. (c) 18. (b) 19. (b) 20. (c) 21. (a)
22. (c) 23. (d) 24. (d) 25. (c) 26. (d) 27. (c) 28. (d)
29. (b) 30. (a)

SECTION-B (Previous year Question)

1. (c) 2. (d) 3. (a) 4. (d) 5. (c) 6. (b) 7. (a)
8. (a) 9. (d) 10. (a) 11. (a) 12. (b) 13. (a) 14. (d)
15. (a) 16. (b) 17. (c) 18. (d) 19. (d) 20. (d) 21. (a)
22. (a) 23. (c) 24. (d) 25. (a) 26. (b) 27. (c) 28. (a)
29. (c) 30. (c)

SECTION-C (School/Board level type questions)

1. $\hat{a}=\frac{3}{7}\hat{i}-\frac{2}{7}\hat{j}+\frac{6}{7}\hat{k}$ 2. $\frac{3}{7}\hat{i}+\frac{6}{7}\hat{j}+\frac{6}{7}\hat{k}$ 3. $\frac{2}{3}\hat{i}+\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}$

4. $\hat{i}+2\hat{j}+\hat{k}$ 5. $6\hat{i}+2\hat{k}, -\hat{i}-2\hat{j}+3\hat{k}$ 6. $\vec{b}$ is any vector

8. $\pi/3$ 9. $\cos^{-1}(-1/3)$ 10. $5/2$ 11. $8/7$ 12. $2/3$

CHAPTER-02 (THREE DIMENSIONAL GEOMETRY)

SECTION-A (Straight objective type questions)

1. (c) 2. (c) 3. (d) 4. (c) 5. (d) 6. (d) 7. (b)
8. (c) 9. (a) 10. (d) 11. (c) 12. (c) 13. (c) 14. (a)
15. (d) 16. (a) 17. (c) 18. (d) 19. (a) 20. (d) 21. (c)
22. (b) 23. (b) 24. (d)

SECTION-C (Previous Year questions)

1. (d) 2. (d) 3. (d) 4. (b) 5. (c) 6. (a) 7. (c)
8. (c) 9. (a) 10. (b) 11. (b) 12. (d) 13. (c) 14. (b)
15. (b) 16. (d) 17. (a) 18. (a) 19. (b) 20. (a) 21. (b)
22. (a) 23. (a) 24. (c) 25. (b) 26. (c)

SECTION-D (School/Board level type questions)

1. (3, 4, 5) 2. $\vec{r}=\left(\frac{1}{3}\hat{i}-\frac{1}{3}\hat{j}+\hat{k}\right)+\lambda(\hat{i}+2\hat{j}+3\hat{k})$ 3. $\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$

4. $\frac{x-1}{-2}=\frac{y-2}{14}=\frac{z-3}{3}; \vec{r}=\hat{i}+2\hat{j}+3\hat{k}+\lambda(-27\hat{i}+14\hat{j}+3\hat{k})$

5. $\left(-\frac{1}{3},\frac{1}{3},1\right); 2,1,-6; \vec{r}=-\frac{1}{3}\hat{i}+\frac{1}{3}\hat{j}+\hat{k}+\lambda(2\hat{i}+\hat{j}-6\hat{k})$ 6. $\frac{x+1}{2}=\frac{y-3}{7}=\frac{z+2}{4}$ 7. $-\frac{10}{7}$

8. 1 9. (–1, –1, –1) 11. $2\sqrt{6}, (3,-4,-2)$ 12. $2\sqrt{29}$

13. $\frac{x}{p}+\frac{y}{z}=\frac{z}{r}=3$ 14. $x - y + 3z - 2 = 0$; $\frac{2}{\sqrt{11}}$ 15. $17x + 2y - 7z = 26$

16. $5x + 9y + 11z - 8 = 0$ 17. $2x - 4y + 3z - 8 = 0$ 18. $x + y - 2z + 4 = 0$

19. $51x + 16y - 50z + 173 = 0$ 20. $\vec{r}.(2\hat{i} - 13\hat{j} + 3\hat{k}) = 0$ 21. $8x - 13y + 15z + 13 = 0$ 22. 1

23. $\frac{13}{12}\sqrt{6}; \left(-\frac{1}{12}, \frac{25}{12}, -\frac{2}{12}\right)$ 24. (1, 0, 7)

CHAPTER-03 (PROBABILITY)

SECTION-A (Straight objective type questions)

1. (c)	2. (b)	3. (a)	4. (a)	5. (d)	6. (d)	7. (d)
8. (c)	9. (a)	10. (c)	11. (d)	12. (c)	13. (c)	14. (b)
15. (b)	16. (c)	17. (c)	18. (b)	19. (a)	20. (c)	21. (a)
22. (b)	23. (a)	24. (c)	25. (b)			

SECTION-B (Previous year questions)

1. (b)	2. (d)	3. (d)	4. (d)	5. (a)	6. (b)	7. (c)
8. (d)	9. (d)	10. (d)	11. (c)	12. (b)	13. (d)	14. (d)
15. (d)	16. (a)	17. (d)	18. (d)	19. (b)	20. (a)	21. (d)
22. (c)	23. (d)	24. (d)	25. (a)	26. (a)		